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Perceptions of Implementation from Higher Education Participants in Online Course Design Professional Development

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CAPSTONE

Submitted in Partial Fulfillment of the Requirements

For the Degree of Doctor of Education, Ed.D.
Interdisciplinary Leadership, Higher Education Concentration

Governors State University
University Park, IL

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ABSTRACT

Many higher education instructors treat online teaching as a modern form of correspondence, believing students learn on their own when content is placed in a learning management system, but learning also takes place through other forms of interaction beyond solely with content. To show best practices for the online modality, instructors at a Midwest regional mid-sized higher education institution participated in professional development using the Improving Your Online Course (IYOC) workshop from Quality Matters™ (QM), a globally recognized authority in best practices for online course design. Level 3 of Kirkpatrick's Four Levels of Training Evaluation was used to evaluate the transfer of training from the IYOC workshop to the professional practice of participants. The purpose of this qualitative study was to examine perceptions of implementation after training, and determine if transference had taken place. Results show that participants have applied training concepts to multiple courses and modalities, not just online.

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CHAPTER I

INTRODUCTION

Many instructors feel that online teaching is a modern form of correspondence, simply believing that students can learn on their own when content is placed in a learning management system. This is important because many instructors take this approach when designing and teaching online courses, but student learning also takes place through other forms of interaction beyond solely interacting with the content. To show instructors best practices for online course design and delivery, professional development is necessary. The purpose of this qualitative study is to examine perceptions of implementation from higher education instructors after participating in professional development for best practices in online course design and delivery. The goal is to see if and how transference has taken place from professional development into regular practice.

Statement of the Research Problem

The topic of this study is the effectiveness of professional development in online course design best practices. This study needs to be conducted because many higher education instructors feel that online learning is a form of modern correspondence, believing that by solely placing content in a learning management system, students can learn on their own (Joosten & Cusatis, 2019; Robinson et al., 2017). When instructors take this approach, they unfortunately neglect that learning also happens through other forms of interaction. Best practices for online course design and delivery include not only student-content interaction but also student-student and student-instructor interaction (Kendall & Kendall, 2014; King, 2003; Meyer & Marsick, 2003; Nilson, 2003). To introduce higher education faculty to the best practices for the online modality, professional development is necessary (Garrison, 2017; Hodges & Fowler, 2020; Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Li et al., 2023; Luongo,

2018; Nilson, 2003; Van Wart et al., 2020; Webster, 2025). The purpose of this qualitative study is to examine perceptions of implementation from higher education instructors after participating in professional development for best practices in online course design and delivery. The goal is to see if and how transference has taken place from professional development into regular practice.

This study shows the implications for higher education faculty and leadership. Teaching pedagogy and the usage of technology are both constantly evolving, so to stay up to date, regular professional development is expected by educators (Burridge & Carpenter, 2013; Danielson, 2007; Davies & Preston, 2002; King, 2003; Shagrir, 2012). The purpose of instructors participating in professional development should be for the benefit of students (Darling-Hammond et al., 2017; Guskey, 1986; Hodges & Fowler, 2020; King, 2003). A goal of professional development is the transference of what was learned to regular practice (Borko, 2004; Darling-Hammond et al., 2017; Judd, 2015; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016). Perceptions of what was learned in professional development and then what was implemented can influence the student experience (Budzick, 2014; Cahapay, 2021; Darling-Hammond et al., 2017; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016). When students learn effectively, they are better prepared for future learning and experiences beyond the classroom (Burridge & Carpenter, 2013; Davies & Preston, 2002; Gregory et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Zimmerman et al., 2020). Implementation of what is learned in professional development can justify to leadership the value of the investment in training and provide feedback on effectiveness (Borko, 2004; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016).

The implications of this study show the importance of continual professional development in online course design and delivery. As a result of social restrictions from the COVID-19 pandemic, there is increased student and instructor demand for flexibility in the modality of course delivery (Bozkurt et al., 2022; Hodges & Fowler, 2020; Hodges et al., 2020). Therefore, instructors must continue to receive professional development for best practices in online course design. Once professional development is delivered, it is necessary to determine what the results of that professional development are to determine effectiveness. Professional development for educators in general, in addition to being specifically for effective online course design, can lead to improved student performance in addition to increased positive attitudes toward the online learning modality by instructors (Burridge & Carpenter, 2013; Davies & Preston, 2002; Gregory et al., 2020; Joosten & Cusatis, 2019; Zimmerman et al., 2020). The purpose of this study is to determine if and to what degree transference of knowledge from training to practice took place.

Background

Training in various industries includes not only those mandated by legal and workplace policies, but also professional development specific to a given profession to gain or improve skills. Professional development in the education profession is necessary and expected for learning new and evolving technologies, and to improve current practices for content design and delivery in all learning modalities (Alsalamah & Callinan, 2021; Cahapay, 2021; Kendall & Kendall, 2014; Kennedy et al., 2013; King, 2003; Kurt, 2016; Meyer & Marsick, 2003; Nilson, 2003). As much as students in higher education courses are adult learners, so are higher education instructors who participate in professional development (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; King & Lawler, 2003; Kurt, 2016). Therefore, trainers

need to be familiar with adult learning principles in addition to the needs of the organization (Meyer & Marsick, 2003).

Professional development in education is an ongoing and systematic process involving discussion, reflection, experimentation with new methods, expansion of knowledge, and acquiring new skills to implement change that improves practice (Alsalamah & Callinan, 2021; Cahapay, 2021; Guskey, 1986; Kennedy et al., 2013; Kurt, 2016; Shagrir, 2012). Educators, at all levels, are expected to be lifelong learners, so professional development, including reflection on practices, is encouraged and expected (Burridge & Carpenter, 2013; Cahapay, 2021; Danielson, 2007; Davies & Preston, 2002; Kennedy et al., 2013; King, 2003; Shagrir, 2012).

Higher education faculty, because their formal content-area training usually does not include teaching pedagogy, may be more likely to participate in professional development only in their content area (Brancato, 2003; Cranton & King, 2003; King, 2003; Zimmerman et al., 2020). PreK-12 teachers typically receive training in teaching pedagogy as part of their post-secondary or graduate-level higher education degree, in addition to their possible subject area (Davies & Preston, 2002). Higher education instructors are less likely to have any formal training in how to be teachers, which is why pedagogical professional development is so valuable for them (Brancato, 2003; Cranton & King, 2003; King, 2003; Simunich, 2023; Zimmerman et al., 2020). Professional development to better understand teaching pedagogy and effective use of technology in realistic teaching situations would be beneficial for higher education instructors in all teaching modalities (Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Lowell & Yang, 2023; Luongo, 2018).

Context of the Problem

Online learning is a course delivery modality that has been growing in popularity since the wide availability of the internet to the public in the 1990s (Nilson, 2003), especially considering the circumstances of the COVID-19 pandemic (Bozkurt et al., 2022). The mindset of online learning has shifted from the focus of correspondence to mimicking the in-person experience by taking advantage of the interactive possibilities the technology tools allow for (Joosten & Cusatis, 2019; Kurt, 2016; Robinson et al., 2017).

Online learning has its roots in the distance learning techniques developed for radio and then television in the early and middle parts of the 20th century, which were dependent mainly on content delivery and then correspondence between instructor and student for learning activities (Joosten & Cusatis, 2019). As early as the late 1980s, online education already had the characteristics of having a physical separation of instructors and students using a computer network for communication and interaction to distinguish it from traditional in-person education and using an educational organization to distinguish it from self-study or private tutoring (Johnson et al., 2019). The current formal definition of online learning is electronically mediated asynchronous and synchronous communication that utilizes interaction and collaboration beyond just content delivery (Garrison, 2017). As a result, higher education faculty who wish to be effective instructors in the online modality need specific professional development, in not only the technology tools but also the pedagogy for purposeful course design to provide a quality learning experience in online courses for adult learners (Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Luongo, 2018; Nilson, 2003).

Online learning tools allow for the introduction of course content in non-traditional ways, such as through communication groups and multimedia, to help meet the needs of more students, especially when they may be expected to participate in more on-your-own asynchronous

activities (Joosten & Cusatis, 2019; Kurt, 2016; Nilson, 2003), and historically professional development focused on the technology tools but not necessarily on teaching pedagogy with those tools (Johnson et al., 2019; Kurt, 2016; Robinson et al., 2017). So, because of the interactive nature of available learning tools, online learning is distinct from both traditional classroom learning and historical distance and correspondence learning, and therefore, professional development has been developed and is needed so instructors can rethink their approach to designing courses for the online modality (Garrison, 2017; Joosten & Cusatis, 2019; Kurt, 2016).

Purpose of the Study

The purpose of this study is to evaluate if and to what degree transference from professional development to regular practice has taken place for higher education instructors. Participants in this study were higher education faculty at a Midwest regional mid-sized institution who participated in the Improving Your Online Course (IYOC) workshop, as developed by Quality Matters™ (QM), an international standards organization in online course design and delivery best practices. Professional development for educators in general, in addition to being specifically for effective online course design, can lead to improved student performance in addition to increased positive attitudes toward the online learning modality by instructors (Burridge & Carpenter, 2013; Darling-Hammond et al., 2017; Davies & Preston, 2002; Gregory et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Zimmerman et al., 2020). Professional development tools and standards organizations are available to provide pedagogical training and evaluation tools for online course designers and instructors, including Quality Matters™ (QM), among others (Johnson et al., 2019; Zimmerman et al., 2020). QM is a sound, research-based professional development provider to increase interaction in online

courses (Joosten & Cusatis, 2019), and increased interaction and collaboration within online courses not only mimics but also enhances in-person course experiences as well (Gregory et al., 2020; Joosten & Cusatis, 2019; Robinson et al., 2017). Implementation of what was learned in IYOC is what will be evaluated.

Conceptual Framework

Kirkpatrick's (2006) model of the Four Levels of Training Evaluation will be used as the conceptual framework of this study. Resulting from Donald Kirkpatrick's 1959 dissertation with the initial intention of evaluating corporate training programs, Kirkpatrick's (2006) Four Levels can be used in any industry to evaluate training and professional development, including training educators at all levels (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). The Kirkpatrick model describes three specific reasons to evaluate training programs:

- To justify the existence and budget of the training program by showing how it contributes to the organization's objectives and goals.
- To decide whether to continue or discontinue the training program.
- To gain information on how to improve future training programs.

The Kirkpatrick model consists of four levels: reaction, learning, behavior, and results. Level 1 gauges participants' reactions during the training process: what they like most or least, and what feelings they have, positive or negative, toward the experience. Level 2 evaluates the learning that takes place through data gathering about the principles, facts, or techniques attained by the participants through the course activities. Level 3 identifies the behavioral changes that can happen after the training through self-rating scales, observations, interviews, and questionnaires. Level 4 consists of the results of the evaluation, gauging the overarching

effectiveness of the training program in creating sustainable change in the trainee and his or her organization and its culture. Level 3 will be the primary focus used in this study.

Refection and participant perceptions are key components of Level 3 (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). Reflection is considered part of the standard instructional design process through which better teaching practices are achieved for any modality (Hodges & Fowler, 2020), and effective professional development utilizes critical self-reflection to help instructors decide how or why to implement what was learned during training (Cranton & King, 2003; Meyer & Marsick, 2003). Instructor reflections on professional development can help guide educator practice when embracing new pedagogical ideas and available technology tools for online course design and delivery, and help correct previous misconceptions, making new knowledge meaningful to their own experiences (Luongo, 2018; Merseth, 1994; Picciano, 2017; Robinson et al., 2017; York et al., 2016). Reflection is an important part of an educator's active engagement in their growth process to determine what worked and how to improve (Cranton & King, 2003; Danielson, 2007; Garrison, 2017; Guskey, 1986). Reflection is actively used in QM training workshops for participants to determine what has or hasn't worked previously and how they will implement concepts from the training into their future practice. The Self Review activity in the IYOC workshop uses the online Self Review tool, a digital version of the QM Rubric, in which instructors and course designers can also add comments and reflections on what is working and what might need improvement in their online course design. Results from this study may include whether transference of what was learned occurred to other learning modalities beyond just online.

Research Questions

This study uses two phases to answer each of the following research questions:

1) How do higher education instructors perceive changes, if any, to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop?
2) How did participants perceive what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery?

Research Question 1 will be answered through interview questions conducted individually with previous IYOC participants. Questions consist of practices used in courses related to content within the training, such as forms of student interaction used, alignment of content, activities, and assessments to learning objectives, and technology tools used. Research Question 2 will be answered by comparing the responses of participants to interview questions with comments made within the Self Review activity during the IYOC training, as well as possibly comparing items and tools used within course sections within the learning management system from before training and then after training.

Operational Definitions

- **Administration** – Synonymous with institutional leadership. For this study, the deans and division chairs within individual colleges and/or institution-wide leadership at a Midwest regional mid-sized institution.
- **Adult Learners** – Any learner at the higher education or professional level, including undergraduate and graduate students as well as faculty participants in professional development.

- **Asynchronous** – For online courses, this means there are no scheduled virtual meeting times. For activities in any modality, this means students interact with them outside of scheduled meetings, and they can be synonymous with homework.
- **Center for Teaching and Learning (CTL)** – The researcher's department that provides professional development to faculty in both teaching pedagogy and technology tools for education. The researcher is the Senior Online Instruction Specialist in this department.
- **Correspondence** – A communication method in distance learning in which students submit assignments or messages, and then instructors reply and/or provide feedback. The dominant form of student-instructor and student-content interactions in distance learning courses was before the widespread use of the internet.
- **Distance Learning** – A modality of learning that allows students to complete courses remotely. This modality was dominated by correspondence through most of the 20th century and now includes online courses.
- **Emergency Remote Teaching (ERT)** – At the start of the COVID pandemic, instructors were forced to adapt their in-person courses to an online format for an unknown period due to social restrictions. Courses during this time may or may not have been purposefully designed with online best practices.
- **Higher Education (HE, higher-ed)** – The level of education beyond PreK-12, which consists of adult learners.
- **Hybrid** – A modality of learning that combines the traditional teaching modality with some online features.
- **Interaction** – Ways in which a student interacts with components of a course, any of which can be synchronous or asynchronous. In this study, changes to interactions within

courses after professional development will help determine if transference to professional practices has taken place.

- ***Student-Content Interaction*** – Students interact with the course content directly, either synchronously through virtual or in-class presentations, or asynchronously through readings in textbooks, articles, and/or instructor-provided notes, and/or interacting with multimedia such as videos or other technology tools.
- ***Student-Instructor Interaction*** – Students interact with the instructor directly, either synchronously through virtual or in-person meetings, or asynchronously through correspondence.
- ***Student-Student Interaction*** – Students interact with other students directly, either synchronously through virtual or in-person meetings, or asynchronously through designed activities.

- **Implementation** – The behavior of applying knowledge learned in training to professional practice, which will be analyzed in this study using Kirkpatrick's Level 3.
- **Kirkpatrick's Four Levels of Training Evaluation** – Initially developed to evaluate components of corporate training programs, but can be adapted for any industry, including Education.
  - ***Reaction*** – Level 1 from Kirkpatrick's model, where motivation to participate in professional development is directly analyzed. Motivation is not relevant to this study because the training has already been completed, so the application of what was learned to professional practice is what will be analyzed.
  - ***Learning*** – Level 2 from Kirkpatrick's model, where the training sessions and their content are directly analyzed. Because the pool of participants consists of faculty

who have already completed the training, learning had to have taken place for a participant to have earned a certificate of completion. Level 2 is not relevant to this study because the learning itself is not being analyzed, but the application of what was learned after training was completed will be.

- ***Behavior*** – Level 3 from Kirkpatrick's model, where transference of knowledge from training to professional practice is directly analyzed. Level 3 will be the primary focus of this study.
  - *Implementation* – The behavior of applying knowledge learned in training to professional practice, which will be analyzed in this study using Kirkpatrick's Level 3.
  - *Transfer of Knowledge* – The desired behavior of participating in professional development is the transfer of what was supposed to be learned to practical application and active practice, which is analyzed in Kirkpatrick's Level 3.
- ***Results*** – Level 4 from Kirkpatrick's model, where the data from all previous levels are gathered, organized, analyzed, and reported to the administration. The analysis from Level 3 in this study can be a factor in the overall results, but because Levels 1 and 2 are not being analyzed for this study, Level 4 leaves potential for future studies.

- **Online** – A modality of distance learning that utilizes the internet and related technology tools, allowing students to work remotely. Online courses can include synchronous and asynchronous interactions.
- **Participant –** An adult learner who participates in professional development. For this study, 12 higher education instructors out of a pool of 21 potential participants (see Appendix B) completed the IYOC workshop through the CTL.

- **Pedagogy** – In education, the best practices for teaching. QM workshops focus on pedagogical training for the online teaching modality, but some of those best practices can also be applied to other teaching modalities.
- **PreK-12** – An abbreviation of Pre-kindergarten through twelfth grade (the senior year of high school), which consists of non-adult learners.
- **Professional Development** – Synonymous with training, activities, or courses that improve competence and skill within a given profession. This study will analyze professional development for higher education instructors in best practices for online course design and delivery.
- **Quality Matters™ (QM)** – A non-profit organization and globally recognized standards authority for best practices in online course design and delivery for the field of education.
    - ***Applying the QM Rubric (APPQMR)*** – QM's flagship workshop that covers all 42 specific review standards of the QM Rubric.
    - ***Improving Your Online Course (IYOC)*** – This QM professional development workshop covers the 23 essential (required) specific review standards of the QM Rubric. Participation in IYOC will be the setting for analysis in this study.
    - ***QM Higher Education Rubric, 6th Edition (QM Rubric)*** – A set of 8 general standards categories consisting of 42 specific review standards used for reviewing online courses for best practices in course design and delivery.
        - *General Standards* – The 42 specific review standards on the QM Rubric are divided into 8 general categories.
        - *Specific Review Standards (SRS)* – There are 42 specific review standards with three levels of point values: essential SRSs are worth 3 points each, very

important SRSs are worth 2 points each, and important standards are worth 1 point each.

    - *Essential Standards* – May also be referred to as the required standards. The 23 essential SRSs on the QM Rubric are required to meet an official course review. The essential standards are also required for participants to evaluate in the IYOC workshop using the Self Review tool.

  - ***Self Review Tool*** – An online, interactive version of the annotated QM Rubric. Allows any user at a QM-subscriber institution to perform a self-review of any course, including entering numeric scores and commentary for each SRS, at any time. Performing a self-review of a course previously taught is a required activity in the IYOC workshop.

- **Synchronous** – For any course modality, this means there are specific scheduled virtual or in-person meeting times. For activities in any modality, this means students interact with them during scheduled virtual or in-person meetings.
- **Technology Tools** – In education, applications and/or websites are used to administer or enhance online courses. They can include websites, learning management systems and optional plugins, other web-based resources, and other applications that might not require internet access to utilize.
- **Traditional** – Sometimes referred to as in-person or face-to-face (which itself is sometimes abbreviated as f2f), a modality of learning that takes place in a classroom at scheduled meeting times.

- **Training** – Synonymous with professional development in the field of education. This study will analyze the transfer of knowledge from training to professional practice for participants in QM's IYOC workshop.
- **Transfer of Knowledge** – The desired behavior of participating in professional development is the transfer of what was supposed to be learned to practical application and active practice, which is analyzed in Kirkpatrick's Level 3.

Assumptions, Limitations, and Delimitations

The following section will outline the assumptions, limitations, and delimitations of this study. Some assumptions are based on suggestions from Kirkpatrick's Four Levels of Training Evaluation, which serves as the theoretical model for this study. Other assumptions, limitations, and delimitations are based on details about the pool of potential study participants.

Assumptions

According to Kirkpatrick and Kirkpatrick (2016), it can be assumed that an institution's administration is not concerned with Level 2 data, which can be part of the motivation for participants to take the training. For this study, the training has already taken place, so the motivation for taking the training is irrelevant. What is now relevant is whether transference of learning has taken place from that already-completed training into the professional practice it was intended to help improve. Because the administration has already recommended, funded, and approved participants taking the training, the administration is assuming that what was learned is going to be implemented, which is why Level 3 will be used to evaluate the implementation of what was supposed to be learned by participants in the training.

Also, according to Kirkpatrick and Kirkpatrick (2016), it can be assumed that contributions to practical application on the job take place on the job, and personal responsibility,

motivation, and reflection go hand in hand with efforts to the implementation of concepts learned in training. This means that the purpose of training is to be able to implement what was learned in training into professional practice. This study assumes that the pool of previous IYOC participants took training specifically to implement it, regardless of any other motivations. This is why Level 3 of Kirkpatrick's model was chosen to evaluate if and to what degree implementation took place.

A significant number of the IYOC participants overall were College of Education faculty, and it can be assumed that this group has had formal training in teaching pedagogy already. Even though the IYOC workshop focuses on the pedagogical aspects of online course design and delivery, some of the IYOC subject matter might be considered a review for this group. It can, therefore, be assumed that this group may already have been implementing best teaching practices in their courses, regardless of modality, so this group was eliminated as potential participants in this study.

Because participants volunteered for and then completed the training, full effort in the IYOC workshop was assumed, and therefore, the intention to implement what was learned is also assumed. As noted previously, motivation to participate in the workshop was irrelevant because the training had already concluded, and the participant pool is made up of those who successfully completed the training. Because completion of the training was successful, learning had to have taken place. What is assumed is that the results of this study will yield those successful participants who made the effort to implement what they learned.

Scope and Delimitations

The researcher is the primary facilitator of official QM training workshops at a Midwest regional mid-sized institution, including sessions of the IYOC workshop that are the basis of this

study. The researcher is also a certified QM course reviewer. QM workshops are scripted and only change when the rubric editions are updated, so the IYOC content during the time frame of this study, March 2021 to June 2023, was consistently delivered. Due to the scripted nature of the workshop, the facilitator was only allowed to customize content or feedback as it pertained to an individual participant's questions or responses during activities. The researcher was an "insider" involved in the training process being evaluated and therefore has an emic perspective (Berkovic, 2023). The researcher was actively encouraging the application of the material presented during training and, therefore, must practice reflexivity to critically examine assumptions and biases when gathering and analyzing data in this study (Berkovic, 2023; Bogdan & Biklen, 2006; Burnam, 2025; Hurst, 2023; McLeod, 2024; Merriam, 1998; Stake, 1995; Turner, 2020; Yin, 2018). Reflexive measures taken include a research journal (Hurst, 2023; Turner, 2020), peer debriefing (McLeod, 2024), and member-checking (McKim, 2023).

The sample pool will be previous IYOC participants employed at a Midwest regional mid-sized institution from March 2021 to June 2023. February 2020 is when this institution first started subscribing to QM, and afterward, the researcher was trained as an official QM workshop facilitator and course reviewer. March 2021 was the first IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. June 2023 was the last IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. After July 2023, the QM Higher Education Rubric was updated to the 7th Edition, and terminology and numbering were changed from the 6th Edition (Quality Matters™, 2023). March 2021 through June 2023 was the time frame chosen for this study for the sake of consistency, using the same QM Rubric and Self Review tool used in the IYOC workshops, and as a result, the participant pool will be from those same IYOC workshops previous to July 2023. See Appendix A for the

non-annotated 6th Edition of the QM Higher Education Rubric, which the Self Review tool is based on.

Only the 23 essential standards will be analyzed because those are the standards required in the IYOC Self Review activity. There are 42 total standards on the QM 6th Edition Higher Education Rubric, but only the 23 essential standards, which are considered required for meeting an official QM course review, were covered by the IYOC workshop. In Appendix A, the specific review standards worth 3 points are considered the essential (required) standards, and only these specific review standards need to be completed during the Self Review activity of the IYOC workshop (Quality Matters™, 2023). Commentary by participants on these specific review standards will be used for comparison to interview responses to show potential transference from training to practice.

12 of 21 total potential participants in this sample pool participated in this study, broken down by the three colleges and associated divisions in which they teach, as of June 6, 2023. Because faculty in this pool may teach in more than one of the 16 out of 22 potential programs represented and, as a result, may teach in more than one college or division, the college and division totals may differ from the actual total participants. The breakdown of participants will be discussed further in the methodology chapter.

Refer to Appendix B for the breakdown of previous IYOC participants at this institution that make up this pool of potential participants. Appendix B also shows the breakdown by program within each division. This organizational structure can also act as a potential breakdown for analysis, possibly providing an embedded context (Yin, 2018). The methodology and results analysis will be thoroughly described in future chapters. Participants from this pool were still employed at this institution and were asked to volunteer to participate in this study.

Participants from the overall pool no longer employed at this institution were not asked to participate in this study. Volunteers' requests to remain anonymous other than college, department, and subject area will be honored.

Only Level 3: Behavior from Kirkpatrick's Model for Training Evaluation will be tested for this study because it is the most relevant to the research questions. Level 1: Reaction is related to motivation and satisfaction, which are not relevant to the research questions. Level 2: Learning is relevant to the topic of this study, but the fact that learning may or may not have taken place may not mean that transference of learning has occurred into the participants' professional practice afterward. Level 3: Behavior is most relevant to the research questions because this is where the transference of learning is directly analyzed. Results from Level 3 in this study can contribute to Level 4: Results if reported to administration, and could indicate the potential for further studies, but not all of the overall results with potential to be reported within Level 4, which include the effects for the institution and students in addition to the faculty participants, are relevant to the research questions for this study.

Limitations

Potential weaknesses are the limitations of the sample pool. The 21 faculty in the sample pool represent only 8.71% of the total 241 full-time faculty population at this institution (Univstats, 2023). Therefore, this sample pool might not entirely represent the total faculty population, but the methods to be used in this study could be used in future studies to build upon the potential results from this study, which could include cross-case analysis of implementation levels across different participant groups.

Also, 15 training participants were not included in the potential sample pool for this study because they are College of Education faculty. The researcher is also a practicing instructional

designer and has had formal pedagogical training and experience as a K-12 teacher and, as such, is aware of the pedagogical background those specific faculty have. Because of this, it is assumed that the College of Education faculty participants may have had pedagogical training in teaching as well as designing courses for one or more modalities. However, QM assumes participants in QM workshops have not had training in how to be course designers or teachers (Simunich, 2022), so it can also be assumed that higher education faculty in non-education disciplines might never have been trained in how to be teachers, regardless of modality (Budzick, 2014), hence the exclusion of College of Education faculty. A potential for future studies can compare the College of Education faculty's results based on this assumption using these faculty as a control group, as defined by Kirkpatrick (2006) for Level 3, to compare to the results for non-education faculty as a whole, or by individual subject areas.

Another group of potential participants (4 out of 21, 19.05%) is College of Business (COB) faculty. Many of their peers were participants in QM workshops independently before this institution started subscribing to QM in February 2020. Although this group in the participant pool for this study did not have any previous QM training themselves, through their peers they may have already been aware of some of the pedagogical aspects of QM training. This awareness may affect their perceptions, but only one faculty member participated in this study whose COB courses are cross-listed with courses in the computer science program, which is in the College of Arts and Sciences. It is assumed this faculty member's COB affiliation does not influence their perceptions enough to affect the results of this study. More details about data analysis will be discussed in Chapter 4.

Other case studies, which will be discussed further in the literature review, have analyzed Kirkpatrick's model for QM's Applying the QM Rubric (APPQMR) workshop, a more rigorous

workshop than IYOC that covers all specific review standards of the QM Higher Education Rubric (Quality Matters™, 2023). This study will analyze IYOC, which only covers the 23 essential standards. This limitation is based on the availability of results from IYOC workshops because APPQMR has not been facilitated at this institution.

Organization of the Study

This study is a single-case retrospective case study design. Retrospective case studies, sometimes called historical case studies, try to determine changes over a period of time using existing data or documentation to analyze patterns, causes, or lessons learned (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998; Ozcan et al., 2017), and this existing data can be compared to interviews to interpret findings (Goyes & Sandberg, 2024). Because the Self Review tool used in the IYOC workshop also allows for reflection and justification by the participant and reviewer at the time of training, this qualitative information can be used for comparison to interview questions during this study. Potential changes in teaching practices for online courses can be shown through opportunities for instructor reflection (Ali & Wright, 2017). Improvements in courses because of that change can have benefits for student performance, which can also be beneficial for individual departments as well as the institution. This study could be used as a model for future studies of the transfer of knowledge from professional development both at this and other institutions.

The practical implications are also related to the ability to understand how faculty participants are influenced by the IYOC workshop as well as QM as a provider of professional development. The purpose of the IYOC training was to introduce participants to the concepts of the three forms of student interaction: student-content, student-instructor, and student-student. Participants were also to reflect on how these concepts may or may not have already been

implemented, and how they would approach implementing these concepts into future courses. If the training proves to be successful in attaining knowledge about online course design best practices, then it can be offered again knowing it is meaningful. Furthermore, if the training relates to a perceived value change for this institution because of participation in QM, then the institution may continue to use and possibly even scale up this initiative. If QM does not demonstrate a value, then the institution may wish to focus on other training efforts for this pedagogical need.

As related to Research Question 2, transference of knowledge might not be limited to online courses. Transference of knowledge to professional practice may also have occurred in other teaching modalities by the participants. Even though the IYOC workshop focuses on best teaching practices for the online modality, many of the pedagogical elements can be applied to in-person or hybrid courses as well. The results may show implementation beyond just the online modality, which can show the potential benefits of professional development efforts for students, educators, and institutions alike.

Summary

In this chapter, the need to evaluate the effectiveness of professional development for designing and delivering online courses in higher education was introduced. A brief overview of the literature regarding the analysis of training effectiveness was provided, and how Kirkpatrick's Level 3 from the Four Levels of Training Evaluation will be utilized in this study was described. As a qualitative case study, the two research questions will be answered by gathering interview data from previous professional development participants, and then comparing their current perceptions to comments made within training. The next chapter will be an in-depth review of the literature about professional development and training, including

expectations for professional development in education, the history of online education, and the need for professional development, Quality Matters™ as a provider of professional development for online course design and delivery, and evaluation of the effectiveness of professional development.

CHAPTER II

REVIEW OF THE LITERATURE

Despite the wide availability and use of online courses in higher education, many instructors still need professional development in best practices for online course design and delivery. Many instructors see teaching in the online modality as easy because they believe all they need to do is place content in a learning management system, and students can be expected to learn on their own. This mentality results in little interaction for the student beyond the content itself and minimal feedback from the instructor. This would mean online learning is no more than a modern version of correspondence courses that were once accomplished by mail, a course modality that would have been considered innovative in the early 20th century, but with the telecommunications technology available today, correspondence learning is considered archaic (Gibson, 1998; Joosten & Cusatis, 2019; Robinson et al., 2017). There are many technology tools widely available to instructors that can allow courses to be more interactive with not only the content but also the instructor and other students. Technology tools are available that allow for both synchronous and asynchronous interactions over great distances to meet the demand for convenience. For instructors to learn best practices for their online courses to become more interactive, professional development is necessary. Perceived changes to practice by participants after training, compared to actual effectiveness after training, is how the implementation of that professional development can be measured.

The purpose of this qualitative case study is to examine the perceived effectiveness of the implementation of professional development for best practices in online course design by higher education instructors to determine if transference of knowledge has taken place from training to professional practice. Participants in this study were previous participants in the official Quality Matters™ (QM) Improving Your Online Course (IYOC) workshop as offered through the Center

for Teaching and Learning (CTL) at a Midwest regional mid-sized institution. The researcher was the facilitator of the IYOC workshops and is also a certified QM course reviewer.

Themes

Professional development, also referred to as training, is necessary for various career fields. Participating in training allows professionals to gain new skills or to improve current practices (Alsalamah & Callinan, 2021; Borko, 2004; Cahapay, 2021; Darling-Hammond et al., 2017; Kendall & Kendall, 2014; Kennedy et al., 2013; King, 2003; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016; Meyer & Marsick, 2003; Nilson, 2003). Adult learners are any student or professional development participant in courses or training outside of PreK-12 education. As much as students in higher education courses are adult learners, so are higher education instructors who participate in professional development (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; King & Lawler, 2003; Kurt, 2016). In education, at all levels, professional development is encouraged and expected (Danielson, 2007).

Professional development for educators is especially necessary for the online learning modality in both best teaching practices and the use of technology tools. Online learning is a course delivery modality that is growing in popularity, especially considering the circumstances of the COVID-19 pandemic (Bozkurt et al., 2022). As a result, higher education faculty who wish to be instructors in the online modality need professional development in not only the technology tools but also the pedagogy for purposeful course design of online courses for adult learners (Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Luongo, 2018; Nilson, 2003).

The institution where this study takes place has adopted Quality Matters™ (QM) as a standards system for course design evaluation and professional development delivery. Professional development tools and standards organizations are available to provide pedagogical training for online course designers and instructors (Johnson et al., 2019; Zimmerman et al., 2020). Instructor reflections on their professional development can help guide practice when embracing new pedagogical ideas and available technology tools for online course design and delivery (Luongo, 2018; Merseth, 1994; Picciano, 2017; Robinson et al., 2017; York et al., 2016). QM workshops emphasize reflection throughout their activities. This study will evaluate participant reflections both within and after professional development to determine if transference has taken place from training to professional practice.

Definition and History of Professional Development in General

Training is necessary and expected for professionals in several career fields to gain new skills. Improving current practices or learning new and evolving technologies are important goals of professional development in specific industries (Kendall & Kendall, 2014; King, 2003; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Meyer & Marsick, 2003; Nilson, 2003), including education (Alsalamah & Callinan, 2021; Borko, 2004; Cahapay, 2021; Darling-Hammond et al., 2017; Kennedy et al., 2013; Kurt, 2016).

This institution adopted QM because they are an internationally recognized standards organization in online course design and delivery. Training on QM concepts is provided by the Center for Teaching and Learning (CTL) at this Midwest regional mid-sized institution, the department of which this researcher is a member. Professional development can be provided by either training departments within an organization or be provided for by outside organizations

considered to be authoritative in the field in which training is required (Burridge & Carpenter, 2013).

As the facilitator of the QM Improving Your Online Course (IYOC) workshops that provide the basis of this study, this researcher has followed an apprenticeship style of facilitation. As the facilitator, this researcher was considered a mentor for participants' learning and applying a possibly new set of concepts. The apprenticeship teaching orientation sees teachers or trainers as demonstrating best practices that participants can mimic in practical application (Daley, 2003). Trainers need to be familiar not only with adult learning principles but also with the needs of the organization (Meyer & Marsick, 2003). Trainers and other innovation leaders deliver training by modeling, usually by peers and sometimes by early adopters of an innovation, to potential future adopters via organized demonstrations (King & Lawler, 2003; Rogers, 2003). Just as training would need to be ready to implement in any professional field, professional development for educators needs to be hands-on and ready to implement in the classroom (Alsalamah & Callinan, 2021; Borko, 2004; Cahapay, 2021; Darling-Hammond et al., 2017; Kennedy et al., 2013; Kurt, 2016; Marceau, 2003).

Definition, History, and Challenges of Professional Development in Education

Professional development in education may take different forms at various levels and within multiple disciplines. Professional development for best teaching practices in PreK-12 is expected and regularly offered, but in higher education, training tends to be specific to subject areas, climate and culture, or compliance training, and not always on best teaching practices. Regardless of level, professional development in education is an ongoing and systematic process involving discussion, reflection, experimentation with new methods, expansion of knowledge, and acquiring new skills to implement change that improves practice (Alsalamah & Callinan,

2021; Borko, 2004; Cahapay, 2021; Darling-Hammond et al., 2017; Guskey, 1986; Kennedy et al., 2013; Kurt, 2016; Shagrir, 2012). Educators, at all levels, are expected to be lifelong learners, so professional development, including reflection on practices, is encouraged and expected (Burridge & Carpenter, 2013; Danielson, 2007; Davies & Preston, 2002; King, 2003; Shagrir, 2012).

Educators at all levels have a variety of methods and opportunities to participate in professional development. Instructors may be expected to seek out professional development from any available sources, including but not limited to the training department at their institution, training resources at other institutions, books, articles, online, or their peers (Danielson, 2007; King & Lawler, 2003; Nilson, 2003; Zimmerman et al., 2020). Professional development and training departments have become increasingly common on higher education campuses since the 20th century due to the increased changes in technology as well as pedagogical methods for teaching with the goal of improving educational practice (Guskey, 1986; Hodges et al., 2020; King, 2003; King & Lawler, 2003; Nilson, 2003).

Educators at all levels experience challenges to participating in training opportunities. Challenges to pursuing professional development include availability for the time commitment due to teaching load and other obligations (Brancato, 2003; Cahapay, 2021; Luongo, 2018), the perception that professional development is uninspired or ineffective (Guskey, 1986), lack of familiarity with technology tools and comfort level with technology in general (Kurt, 2016; Luongo, 2018), as well as the perceived burden of having to add yet another task to an instructor's plate with or without compensation (Cahapay, 2021; Luongo, 2018; Johnson et al., 2019). This is why professional development for educators must provide participants with the confidence to be successful when implementing changes to their practice. Participating in

training must be worthwhile not only for instructors' benefit but, most importantly, for students' benefit (Borko, 2004; Darling-Hammond et al., 2017; Guskey, 1986; Hodges & Fowler, 2020; King, 2003).

PreK-12 teachers typically receive training in teaching pedagogy as part of their post-secondary or graduate-level higher education degree, in addition to their possible subject area (Davies & Preston, 2002). Professional development can lead to significant changes in subject matter knowledge, understanding student thinking, and instructional strategies (Borko, 2004). Higher education faculty are more likely to participate in professional development only in their content area because their formal content-area training might not include teaching pedagogy. As a result, higher education instructors are less likely to have any formal training in how to be teachers, which is why pedagogical professional development is so valuable for them (Brancato, 2003; Cranton & King, 2003; King, 2003; Zimmerman et al., 2020). Higher education instructors' teaching experience is based on practice: repetition, possibly self-reflection on what worked and what didn't, and maybe adaptation as necessary (Cranton & King, 2003). Professional development to better understand teaching pedagogy and effective use of technology in realistic teaching situations would be beneficial for higher education instructors in all teaching modalities (Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Lowell & Yang, 2023; Luongo, 2018).

An additional challenge for educators includes using technology tools for both online courses and traditional in-person courses, in addition to the pedagogy behind when and how to use technology when teaching. Educators are increasingly expected to utilize technology tools, even though they and their students may not have access, the experience, or training to use them (Brancato, 2003; Johnson et al., 2019; Kurt, 2016). Educators sometimes feel the training itself

is either not available or is of poor quality, so they may be reluctant to participate (Johnson et al., 2019), but educators who do participate in technology-related professional development perceive that the experience positively impacts their teaching practice and helps create a culture of support (Crews & Wilkinson, 2015; Gregory et al., 2020; Hollowell et al., 2017; Li et al., 2023; Shagrir, 2012).

Adult Learners in Higher Education

Behaviorism describes education's purpose as producing a desired behavior change, and this certainly is relevant to the professional development of higher education instructors to adopt best practices for course design and delivery (Daley, 2003). When working with adult learners, how the instruction is designed will affect how the learner will interact with the content and delivery (York et al., 2016). K-12 learners, especially at younger grades, can be seen as "blank slates" due to a lack of life experience, but adult learners register for courses or participate in training already coming in with life experiences, which affect their educational goals, whether good or bad, relevant to the course material or teaching methods or not (Lawler, 2003). As much as students in higher education courses are adult learners, so are higher education instructors who participate in professional development (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; King & Lawler, 2003; Kurt, 2016). In addition, trainers of higher education instructors are also adult learners because they have to learn the material to be trainers of it (King & Lawler, 2003), and should also be looking at teachers of adults as adult learners themselves (Lawler, 2003).

Professional Development for Higher Education Instructors of Online Courses

Online learning is a course delivery modality that has been growing in popularity since the wide availability of the internet to the public in the 1990s (Kurt, 2016; Nilson, 2003),

especially considering the circumstances of the COVID-19 pandemic (Bozkurt et al., 2022). Before the pandemic, the pool of faculty seeking professional development for online teaching was relatively small but has greatly increased since (Hodges et al., 2022). The emergency remote teaching scenario that arose out of the pandemic has shown that institutions need to scale up their approach to online learning to meet the needs of students, and that based on those student needs each institution may need to scale up based on its own unique needs, so it is therefore essential for future professional development in online teaching to be guided by evidence-based needs (Li et al., 2023).

The mindset of online learning has shifted from the focus of correspondence to mimicking the in-person experience by taking advantage of the interactive possibilities the technology tools allow for (Joosten & Cusatis, 2019; Robinson et al., 2017), and unfortunately, different institutions have different definitions of distance, online and blended learning, which means that different institutions handle training for and assignments of faculty to online courses differently (Hodges et al., 2020). As a result, higher education faculty who wish to be effective instructors in the online modality need specific professional development, in not only the technology tools but also the pedagogy and best practices for purposeful course design to provide a quality learning experience in online courses for adult learners (Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Luongo, 2018; Nilson, 2003).

Differences between Online and Traditional In-person Modalities

In his analysis of various learning theories and frameworks regarding online education to create an integrated model, Picciano (2017) determined that online learning should be considered a subset of learning in general, not just another form of distance learning. Online learning is rooted in the distance learning techniques developed first for radio and then television in the

early and middle of the 20th century, which was dependent mainly on content delivery followed by correspondence between instructor and student for learning activities (Joosten & Cusatis, 2019). As computers and the internet evolved, so did the delivery methods for emerging online courses, but the correspondence mindset and methods remained unchanged for many higher education instructors into the early 21st century (Joosten & Cusatis, 2019; Robinson et al., 2017). This mindset contributed to many higher education instructors traditionally believing that online learning was of lower quality (Hodges et al., 2020) and that online course design could simply mean posting syllabi and reading assignments in a learning management system, communicating only via email. As a result, online courses were historically focused only on content delivery and independent learning (Garrison, 2017; Gibson, 1998; Joosten & Cusatis, 2019; Nilson, 2003; Robinson et al., 2017).

As early as the late 1980s, online education already had the characteristics of having a physical separation of instructors and students using a computer network for communication and interaction to distinguish it from traditional in-person education and using an educational organization to distinguish it from self-study or private tutoring (Johnson et al., 2019). By the turn of the century Graham, Cagiltay, Lim, Craner, and Duffy, in their 2001 study of evaluating online courses using Chickering and Gameson's *Seven Principles of Good Teaching* developed a list of lessons learned for the online environment: there should be clear guidelines for interaction with students, activities that foster meaningful collaboration between students, challenging tasks for students with regular feedback, and the opportunity for students to choose projects that incorporate diverse views based on student experiences (Crews & Wilkinson, 2015). The current formal definition of online learning is electronically mediated asynchronous and synchronous

communication that utilizes interaction and collaboration beyond just content delivery (Garrison, 2017).

Online learning tools allow for the introduction of course content in non-traditional ways, such as through communication groups and multimedia, to help meet the needs of more students, especially when they may be expected to participate in more on-your-own asynchronous activities (Joosten & Cusatis, 2019; Nilson, 2003), and historically professional development focused on the technology tools but not necessarily on teaching pedagogy with those tools (Johnson et al., 2019; Robinson et al., 2017). Due to the abilities of online learning tools evolving to be more interactive and the transactional nature of learning activities within the online modality, teachers can become facilitators where they can also be students, and students can potentially be teachers as well, sharing their own relevant experiences and knowledge within courses (Garrison, 2017).

Modern technology tools provide opportunities for synchronous and asynchronous interactions that make correspondence learning obsolete. The traditional correspondence approach in the online modality is ineffective because more engagement is necessary for effective learning, especially when students are not always going to be self-driven (Gibson, 1998). The ability to be interactive and utilize multiple content delivery methods, including multimedia, is a potential strength in designing a course for the online modality (Branch, 2009; Gibson, 1998; Meyer & Marsick, 2003; et al., 2011; Nilson, 2003; Robinson et al., 2017). The Community of Inquiry and Constructivist theories view online learning as interactive collaborations between instructor and student and students and each other, where both can act as instructor and student simultaneously (Garrison, 2017; Robinson et al., 2017). So, because of the interactive nature of available learning tools, online learning is distinct from both traditional

classroom learning and historical distance and correspondence learning, and therefore, professional development has been developed and is needed so instructors can rethink their approach to designing courses for the online modality (Garrison, 2017; Joosten & Cusatis, 2019).

Purposeful Online Course Design and Delivery vs. Emergency Remote Teaching

Prior to the COVID-19 pandemic, many educators were not fully aware of the influence the lack of effective purposeful online learning was already having on education; online learning was already on the rise and predicted to become mainstream before the pandemic (Picciano, 2017; Simunich, 2022; Zimmerman et al., 2020), and the number of faculty teaching online was already increasing but many did not have proper training or preparation to do so (Johnson et al., 2019). Many higher education instructors have never been online students and therefore have never experienced what best practices in online course design look like from a student's point of view (Simunich, 2022). The development time for a well-designed online course can take up to a full semester before being delivered, but due to reflection and adjustment, as well as becoming more familiar with technology tools, instructors may not be fully comfortable teaching online until the second or third time delivering a course (Hodges et al., 2020).

In Robinson, Kilgore, and Warren's (2017) study, higher education instructors expressed their perceptions of online learning in terms of the in-person modality with which they were most familiar, although they acknowledged they had to make accommodations to distinguish between the intentional course design and delivery methods associated with each modality. Garrison (2017) surmised that online courses would fail if they were merely add-ons to or the repackaging of current educational designs. Therefore, online course designers and instructors pre-pandemic already needed to be prepared to rethink current approaches and be purposeful in

how learning experiences are designed to take advantage of the interactive possibilities within the online modality (Garrison, 2017; Johnson et al., 2019).

What happened because of the pandemic was essentially a worldwide mandate to move into an emergency remote teaching mindset: moving instructional materials and potential interactions that were meant for traditional in-person courses into an online delivery medium, however limiting that may be based on experience and comfort level of the instructors and students, for the sake of just getting through the current disaster, however long it may last (Hodges et al., 2020; Van Wart et al., 2020). Hodges and Fowler (2020) suggested that after emergency remote teaching ended after Spring 2020, preparation of courses should take place to be ready for fully online delivery in case another emergency occurs, so that the new normal can be delivered in either in-person or online modality should it need to change mid-semester. The question thus becomes what will happen to instructors and students who may wish to continue teaching and learning in the online modality now that the pandemic seems to be subsiding, which is why professional development is necessary to help instructors and course designers to move out of the emergency remote mindset into a purposeful design mindset (Hodges & Fowler, 2020; Hodges et al., 2020; Van Wart et al., 2020).

Professional Development for Higher Education Online Learning

Professional development for educators in general, in addition to being specifically for effective online course design, can lead to improved student performance in addition to increased positive attitudes toward the online learning modality by instructors (Borko, 2004; Burridge & Carpenter, 2013; Darling-Hammond et al., 2017; Davies & Preston, 2002; Gregory et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Zimmerman et al., 2020). Professional development tools and standards organizations are available to provide pedagogical training and

evaluation tools for online course designers and instructors, including Quality Matters™ (QM), the Program for Online Teaching (POT), the Association of Educational Communications and Technology (AECT), the Instructional Design Standards for Distance Learning, and the Online Learning Consortium (OLC) Scorecard (Johnson et al., 2019; Zimmerman et al., 2020).

Quality Matters™ is a sound, research-based professional development provider to increase interaction in online courses (Joosten & Cusatis, 2019), and increased interaction and collaboration within online courses not only mimics but also enhances in-person course experiences as well (Gregory et al., 2020; Joosten & Cusatis, 2019; Robinson et al., 2017). Quality Matters™ course review process serves as an excellent example of the implementation of best practices for online course design and delivery (Bogle et al., 2009; Gregory et al., 2020; Hodges & Fowler, 2020) as well as an example of best practices for professional development for adult learners (Lawler, 2003; Wlodkowski, 2003). Training workshops provided by QM can also be evaluated using the Kirkpatrick Model (Budzick, 2014; Chisum, 2019; Oberfoell et al., 2019; Simunich, 2023).

Instructor Reflections on Professional Development for Online Learning

Reflection is considered part of the standard instructional design process through which better teaching practices are achieved for any modality (Hodges & Fowler, 2020), and effective professional development utilizes critical self-reflection to help instructors decide how or why to implement what was learned during training (Alsalamah & Callinan, 2021; Cahapay, 2021; Cranton & King, 2003; Kennedy et al., 2013; Kurt, 2016; Meyer & Marsick, 2003). Instructor reflections on professional development can help guide educator practice when embracing new pedagogical ideas and available technology tools for online course design and delivery and help correct previous misconceptions, making new knowledge meaningful to their own experiences

(Darling-Hammond et al., 2017; Kurt, 2016; Luongo, 2018; Merseth, 1994; Picciano, 2017; Robinson et al., 2017; York et al., 2016).

Reflection is an important part of an educator's active engagement in their own growth process to determine what worked and how to improve (Alsalamah & Callinan, 2021; Branch, 2009; Cahapay, 2021; Cranton & King, 2003; Danielson, 2007; Darling-Hammond et al., 2017; Garrison, 2017; Guskey, 1986; Kennedy et al., 2013; Kurt, 2016). Transformative Learning Theory is defined as adaptation based on experiences: what happens once is expected to happen again, but when it doesn't, we question why and what we did to cause the change or what we can do to affect the next result (Cranton & King, 2003; Luongo, 2018). Formative evaluation of instruction, including reflection, is a component of Dick, Carey, and Carey's (2015) Systems Approach Model of Instructional Design because the purpose is to help create and improve instructional processes. The ADDIE approach, as defined by Branch (2009), consisting of analyze, design, develop, implement, and evaluate, can be a fundamental approach to creating learning experiences, including professional development. Reflection, as part of the evaluation component in the ADDIE process of instructional design, is a form of evaluation for the effectiveness of what was learned in professional development when applied to an instructor's course design and delivery (Branch, 2009; Hodges & Fowler, 2020). Reflection is actively used in the Quality Matters™ Course Review Management System, a digital version of the Quality Matters™ Rubric, in which instructors and course designers can use a Self-Review tool to comment on what is working and what might need improvement in their online course design.

Transfer of Knowledge from Professional Development to Practice

Effective professional development is not a one-off event, but requires sufficient time for learning, application, reflection, and iterative improvement (Darling-Hammond et al., 2017).

The Kirkpatrick Model can be used to evaluate professional development, and consists of four levels: reaction, learning, behavior, and results (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). Level 1 gauges participants' reactions during the training process: what they like most or least, and what feelings they have, positive or negative, toward the experience. Level 2 evaluates the learning that takes place through data gathering about the principles, facts, or techniques attained by the candidates through the course activities. Level 3 identifies the behavioral changes that happen after the training through self-rating scales, observations, interviews, and questionnaires. Level 4 consists of the results of the evaluation, gauging the overarching effectiveness of the training program in creating sustainable change in the trainee and his or her organization and its culture. Level 3 will be the primary focus used in this study because that is where the transfer of training takes place.

Level 3 of the Kirkpatrick Model specifically evaluates the effectiveness of training by analyzing changes in behavior after training. Although Kirkpatrick's model was developed for evaluating training in business and industrial professions, it can be used to evaluate training for any industry, including higher education (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). Per recommendation from QM Vice President Bethany Simunich (2023), Level 3 from Kirkpatrick's model can be used to evaluate the transfer of knowledge to practice to meet the goals of this study. Evaluation of training involves assessing a participant's change in behavior (Alsalamah & Callinan, 2021; Cahapay, 2021; Judd, 2015; Kennedy et al., 2013; Kurt, 2016). Several studies note that data from Levels 1 and 2 are easily gathered but do not provide much information beyond motivation to participate, satisfaction with the delivery modality or the facilitator,

assessment of activity completion for participants, and the possible intention of implementing what was learned (Budzick, 2014; Cahapay, 2021; Chisum, 2019; Judd, 2015; Kennedy et al., 2013; Maina et al., 2022; Oberfoell et al., 2019; Sowcik, 2018). As a result, there is a call for more studies that specifically address changes in behavior and the potential to contribute more thoroughly to the overall results that can benefit an institution (Alsalamah & Callinan, 2021; Budzik, 2014; Cahapay, 2021; Chisum, 2019; Kennedy et al., 2013; Kurt, 2016; Maina et al., 2022; Oberfoell et al., 2019).

Transfer of training, sometimes referred to as transfer of knowledge, to professional practice is the change in behavior that this study will evaluate. Level 3 of the Kirkpatrick Model analyzes changes in behavior after training, and part of that behavior is the transfer of training to regular practice. It can be assumed that up to 70% of all learning that contributes to job performance takes place on the job, which is why evaluating behavior after training has been completed is the most important part of evaluating training (Kirkpatrick & Kirkpatrick, 2016). Evaluating changes in behavior can be done with reflective comments by participants, using pre- and post-tests, and using control groups for comparison (Kirkpatrick & Kirkpatrick, 2006).

Practice Model Guiding This Study

Kirkpatrick's (2006; 2016) model of Four Levels of Training Evaluation has been used to evaluate training programs in the professional world in various modalities, including online, and can be applied to professional development in education. Donald Kirkpatrick was a doctoral student at the University of Wisconsin-Madison when he developed his four-level approach to evaluating management training programs for factory workers in his 1954 dissertation (Kirkpatrick Partners, LLC, 2023). Those concepts were published in a series of articles in the *US Training and Development Journal* in 1959, and then published again in the first edition of

his book *Evaluating Training Programs* in 1994 (Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016). Kirkpatrick was a former president of the American Society for Talent Development (ASTD) and Professor Emeritus at the Management Institute at the University of Wisconsin-Madison until his death in 2014 (Kirkpatrick Partners, LLC, 2023).

Kirkpatrick's model for evaluating training is broken down into four levels (Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016). Level 1: Reaction measures the quality of training design based on participant satisfaction; this can be described as customer satisfaction based on whether participants feel they have gotten value for their time or financial investment, and a positive reaction may not ensure learning, but a negative reaction almost certainly reduces the possibility of. Level 2: Learning measures knowledge and skills gained during training; learning has taken place if there are changes in attitude, increase in knowledge, or improvements in skill. Level 3: Behavior measures the degree to which the new knowledge and skills introduced in training are implemented into professional practices; this change in behavior can be defined as transfer of training, and this needs to be analyzed after some time has passed after training was completed. Level 4: Results analyzes how training results affect not only the individual participants but also the effects for the institution; results can be reported for individual participants, their respective departments, and the entire institution, both performance and financial, and how those results can be aligned to achieving an institution's mission and goals.

Kirkpatrick (2006; 2016) states that evaluation of learning is necessary because if learning hasn't taken place, there will not likely be a change in behavior. Levels 2 and 3 both can utilize pre- and post-test data and participant reflections to determine if learning has taken place and if there has been a transfer of training to professional practice. Control groups can also

be used for comparative analysis to determine the degree of both learning and implementation. Administrators might not be interested in whether learning has taken place because they might assume that by participating in training, participants have learned (Cahapay, 2021; Kirkpatrick & Kirkpatrick, 2016), but leadership may also be interested in the results of Level 3 as reported in Level 4 to evaluate the value of the training (Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2016). Transfer of training requires more than just evaluation of the learning within the training, but also participants' reflections on how they applied what they learned (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016).

Kirkpatrick (2006) specifically states that evaluating e-learning is the same as evaluating learning in any modality because it is the learning outcomes that are being evaluated, not the modality. Different modalities have different techniques and tools, but the change in behavior as a result of training in e-learning can still be evaluated the same way as in traditional classroom learning (Horton, 2001; Kirkpatrick & Kirkpatrick, Evaluating training programs : The four levels, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). E-learning, because of easier access to information and tools on demand through the internet, can provide an easier process for participants to partake in training (Horton, 2001; Kirkpatrick & Kirkpatrick, Evaluating training programs : The four levels, 2006; Kurt, 2016). QM's IYOC workshop to be evaluated in this study was delivered in an e-learning modality and can be evaluated using the Kirkpatrick model. More on the methodology will be introduced in the next chapter.

Summary and Conclusions from the Literature

Training, called professional development in the education profession, is necessary and expected for professionals in several career fields to gain new skills, including learning new and evolving technologies or improving current practices for individuals and for specific industries

(Kendall & Kendall, 2014; King, 2003; Kurt, 2016; Meyer & Marsick, 2003; Nilson, 2003). As much as students in higher education courses are adult learners, so are higher education instructors who participate in professional development (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; King & Lawler, 2003; Kurt, 2016), therefore trainers need to be familiar with adult learning principles in addition to the needs of the organization (Meyer & Marsick, 2003).

Professional development in education is an ongoing and systematic process involving discussion, reflection, experimentation with new methods, expansion of knowledge, and acquiring new skills to implement change that improves practice (Guskey, 1986; Kurt, 2016; Shagrir, 2012). Educators, at all levels, are expected to be lifelong learners, so professional development, including reflection on practices, is encouraged and expected (Burridge & Carpenter, 2013; Cahapay, 2021; Danielson, 2007; Darling-Hammond et al., 2017; Davies & Preston, 2002; Kennedy et al., 2013; King, 2003; Shagrir, 2012).

PreK-12 teachers typically receive training in teaching pedagogy as part of their post-secondary or graduate-level higher education degree, in addition to their possible subject area (Davies & Preston, 2002). Higher education faculty, because their formal content-area training usually does not include teaching pedagogy, are more likely to participate in professional development only in their content area, which means they are less likely to have any formal training in how to be teachers, which is why pedagogical professional development is so valuable for them (Brancato, 2003; Cranton & King, 2003; King, 2003; Zimmerman et al., 2020). Professional development to better understand teaching pedagogy and effective use of technology in realistic teaching situations would be beneficial for educators in all teaching

modalities (Borko, 2004; Darling-Hammond et al., 2017; Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Lowell & Yang, 2023; Luongo, 2018).

Online learning is a course delivery modality that has been growing in popularity since the wide availability of the internet to the public in the 1990s (Kurt, 2016; Nilson, 2003), especially considering the circumstances of the COVID-19 pandemic (Bozkurt et al., 2022). The mindset of online learning has shifted from the focus of correspondence to mimicking the in-person experience by taking advantage of the interactive possibilities the technology tools allow for (Joosten & Cusatis, 2019; Kurt, 2016; Robinson et al., 2017). As a result, higher education faculty who wish to be effective instructors in the online modality need specific professional development, in not only the technology tools but also the pedagogy for purposeful course design to provide a quality learning experience in online courses for adult learners (Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Luongo, 2018; Nilson, 2003).

The current formal definition of online learning is electronically mediated asynchronous and synchronous communication that utilizes interaction and collaboration beyond just content delivery (Garrison, 2017). Online learning tools allow for the introduction of course content in non-traditional ways, such as through communication groups and multimedia, to help meet the needs of more students, especially when they may be expected to participate in more on-your-own asynchronous activities (Joosten & Cusatis, 2019; Kurt, 2016; Nilson, 2003), and historically professional development focused on the technology tools but not necessarily on teaching pedagogy with those tools (Johnson et al., 2019; Robinson et al., 2017). So, because of the interactive nature of available learning tools, online learning is distinct from both traditional classroom learning and historical distance and correspondence learning, and therefore,

professional development has been developed and is needed so instructors can rethink their approach to designing courses for the online modality (Garrison, 2017; Joosten & Cusatis, 2019).

Technology is an ongoing challenge for users, both learners and educators, and professional development is there to help address this challenge (Kurt, 2016; Lawler & King, 2003). Because using technology can be a challenge for educators who may wish to pursue designing courses and teaching in the online modality, learning best practices for utilizing technology in teaching may also be a challenge. Institutions may subscribe to outside organizations to provide training resources for their professional development departments or instructors and designers directly, and one of the providers of standards for online course design and delivery is Quality Matters™, among others (Johnson et al., 2019; Zimmerman et al., 2020). QM is a sound, research-based professional development provider to increase interaction in online courses (Joosten & Cusatis, 2019), and increased interaction and collaboration within online courses not only mimics but also enhances in-person course experiences as well (Gregory et al., 2020; Joosten & Cusatis, 2019; Kurt, 2016; Robinson et al., 2017).

Professional development designers and course instructors are both challenged to reflect, seek opportunities for action and research, and actively question what is and isn't working for the benefit of both the instructors and their students (Borko, 2004; Darling-Hammond et al., 2017; Hodges & Fowler, 2020; Lawler & King, 2003). Reflection by professional development participants can help determine if they perceived the professional development to be valuable and applicable to their teaching practices, and if measurable evaluation of those teaching practices before and after professional development matches participant perceptions. Reflection is utilized in training evaluation with the Kirkpatrick model (Alsalamah & Callinan, 2021;

Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016), and is actively used in QM training workshops.

Kirkpatrick's (2006) model of Four Levels of Training Evaluation can be used in any industry to evaluate training and professional development, including training educators at all levels (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). Level 3 identifies the behavioral changes after training is completed – reflections and participant perceptions are key components of this (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). The behavior of implementation of what was learned into the participants' professional practice can occur as a result of training, which is why Level 3 will be the primary focus used in this study.

There are gaps in the research that help justify performing this study. When evaluating professional development in higher education, many studies have struggled to establish a clear link between professional development and behavioral change for participants (Cahapay, 2021), which Level 3 helps evaluate. This is because Levels 1 and 2 have been overused when applying the Kirkpatrick model to evaluating training in higher education (Cahapay, 2021), and other factors, such as time after training for implementation and administration goals, were not taken into consideration (Alsalamah & Callinan, 2021; Cahapay, 2021). Self-assessments by participants, especially at the lower levels, tend to be biased to show completion of the training, and do not reflect any implementation over time after training (Cahapay, 2021; Kennedy et al., 2013). Changes in behavior after training require evaluation at higher levels of Kirkpatrick's model (Cahapay, 2021; Kennedy et al., 2013).

These gaps in the research support answering the following research questions:

1) How do higher education instructors perceive changes, if any, to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop?
2) How did participants perceive what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery?

The intention of this study is to determine the effectiveness of professional development delivered through the official QM Improving Your Online Course (IYOC) workshop as delivered to faculty participants at a Midwest regional mid-sized institution. The following chapter will describe this study's methodology in greater detail.

CHAPTER III

METHODS

The purpose of this study is to examine the perceived effectiveness of implementing professional development for best practices in online course design by higher education instructors to determine if transference of knowledge has occurred between training and professional practice. Participants in this study were previous participants in the official Quality Matters™ (QM) Improving Your Online Course (IYOC) workshop as offered through the Center for Teaching and Learning (CTL) at a Midwest regional mid-sized institution. The researcher was the facilitator of the IYOC workshop and is also a certified QM course reviewer. This study will answer the following research questions:

1) How do higher education instructors perceive changes, if any, to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop?
2) How did participants perceive what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery?

The intention of this study is to determine the effectiveness of professional development delivered through the official QM Improving Your Online Course (IYOC) workshop as delivered to faculty participants at this institution.

Theoretical Perspectives Guiding this Study

Kirkpatrick's (2006) model of Four Levels of Training Evaluation can be used in any industry to evaluate training and professional development, including training educators at all levels (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). Resulting from Donald Kirkpatrick's 1959 dissertation on evaluating management training programs for factory

workers, the Kirkpatrick model consists of four levels of training evaluation: reaction, learning, behavior, and results. Level 3 identifies the behavioral changes after training is completed – reflections and participant perceptions are key components of this level (Alsalamah & Callinan, 2021; Cahapay, 2021; Kennedy et al., 2013; Kirkpatrick & Kirkpatrick, 2016; Kurt, 2016). Because the behavior of implementing what was learned into the participants' professional practice can occur due to training, Level 3 will be the primary focus used in this study.

Teaching pedagogy and the usage of technology are both constantly evolving, so to stay up to date, participating in regular professional development is expected by educators (Burridge & Carpenter, 2013; Danielson, 2007; Davies & Preston, 2002; King, 2003; Kurt, 2016; Shagrir, 2012). The purpose of instructors participating in professional development should be for the benefit of students (Guskey, 1986; Hodges & Fowler, 2020; King, 2003). A goal of professional development is the transference of what was learned to regular practice (Horton, 2001; Judd, 2015; Kirkpatrick & Kirkpatrick, Evaluating training programs : The four levels, 2006; Kirkpatrick & Kirkpatrick, 2016). Perceptions of what was learned in professional development and what was implemented can influence the student experience (Budzick, 2014; Horton, 2001; Kirkpatrick & Kirkpatrick, Evaluating training programs : The four levels, 2006; Kirkpatrick & Kirkpatrick, 2016). When students learn effectively, they are better prepared for future learning and experiences beyond the classroom (Burridge & Carpenter, 2013; Davies & Preston, 2002; Gregory et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Zimmerman et al., 2020).

It is the responsibility of higher education leadership to be aware of ongoing trends in technology and then provide professional development for instructors accordingly. Because teaching pedagogy and the usage of technology are both constantly evolving, regular professional development is expected to be offered by the administration to educators (Burridge

& Carpenter, 2013; Danielson, 2007; Davies & Preston, 2002; King, 2003; Shagrir, 2012). Professional development providers within institutions, because of the need to support changes in technology and best teaching practices, are expected to display the leadership competencies of the "knowledge of technology," "knowledge of instructional strategies," and "change management" (Ritzhaupt et al., 2021). It is then necessary for education leaders to evaluate the effectiveness of any professional development offered.

Qualitative Approach to Inquiry

Qualitative methods have been used to evaluate professional development. Previous qualitative studies involved input from students as well as instructors. Studies with student data have included feedback during and after courses, usually related to satisfaction. Studies with instructor data have traditionally included satisfaction with training immediately after, but in recent years, faculty evaluations are increasingly noting either participation in professional development or evaluation of implementation (Burridge & Carpenter, 2013; Johnson et al., 2019). Participants in this study received a certificate of completion after professional development that they can choose to add to their institutional evaluation materials.

Implementation of concepts after professional development at both the PreK-12 and higher education levels has been evaluated using qualitative methods, using both instructor and student perspectives. When evaluating higher education instructor experiences with a third-party training package, participants found value in the professional development provided because the outsourced material gave a sense of validity to the material, and the training was trusted to be pedagogically sound (Zhao et al., 2024). PreK-12 student perceptions of instructor implementation of technology tools noted that instructor ability with the technology tools improved over time as usage increased (Chang et al., 2014). Student perceptions toward learning

objectives implemented by instructors after professional development had a positive effect on student achievement (Burridge & Carpenter, 2013). However, instructors and students alike noted that training on technology tools alone is not enough, and without pedagogically sound course design, technology should not be used for technology's sake (Gravel, 2014). Professional development should provide more than just an understanding of how a piece of technology works but also how to use it for instructional purposes effectively (Barton, 2020; Gravel, 2014). Lessons need to be properly designed first; then, technology tools can be integrated to help meet learning objectives (Barton, 2020; Gravel, 2014). It is also possible for learning activities to have multiple different technologies to fulfill the same instructional needs (Gravel, 2014). Participants in professional development stated the importance of integrating practical uses of technology tools into the professional development of all subject areas to ease negative feelings and increase confidence toward using them in teaching practices (Barton, 2020). It is assumed that because participation in the training was voluntary and because participants successfully completed the training, participants for this study intended to implement practical applications for what was learned in training. This study seeks to evaluate that implementation.

Qualitative studies have been used to evaluate professional development effectiveness. Taking professional development does not always lead to implementation (Ali & Wright, 2017), but critical reflection by instructors can lead to lasting changes in their teaching approach because reflection can lead to participants being motivated to implement what was learned into their teaching practices (Ali & Wright, 2017; Burridge & Carpenter, 2013; Gregory et al., 2020). Li et al. (2023), using interviews with administrators in several professional sectors, including education, determined that participants want effective professional development with evidence-based, proven methods that can be readily implemented and will only participate in training if

that is the case, unless otherwise required to. Perceived requirements might motivate some instructors to implement what was learned in training, but others might critically reflect on their practices as a result of professional development and continually try to improve over time (Ali & Wright, 2017). Participants in this study were either requested, not required, by their respective administrators to participate in professional development, or volunteered, and this gathers reflections on the implementation of what was learned after training. Instructor reflections and perceptions of student performance will be used to evaluate student interactions designed by instructors, which was one of the primary learning goals of the IYOC workshop.

This study uses a single-case retrospective case study design. Retrospective case studies, sometimes called historical case studies, try to determine changes over a period of time using existing data or documentation to analyze patterns, causes, or lessons learned (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998; Ozcan et al., 2017), and this existing data can be compared to interviews to interpret findings (Goyes & Sandberg, 2024). In this study, the existing data would be the comments and responses made on the Self Review activity by faculty participants in the IYOC workshop as administered by this researcher. These types of studies can compare historical data to new data to triangulate findings in observed behaviors, attitudes, or outcomes (Bogdan & Biklen, 2006; Merriam, 1998). In this study, changes in behavior have been documented and analyzed based on participant responses to interview questions compared to their responses from within the IYOC training, which would be after one or more semesters. Retrospective studies can suggest causal relationships by showing how variables change in response to interventions (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998). This study may determine if there was a causal relationship between participating in professional development and changes to course design and delivery.

Participants in this study were faculty participants in the QM IYOC workshops administered by the researcher. Participants first answered a series of reflection questions during individual interviews using Level 3 of the Kirkpatrick Model as a guide. Because participants have been able to teach one or more courses since participating in professional development, reflection questions asked what kinds of activities are being used in their course design, how they implemented the types of student interactions introduced during training, and if any changes were made to their professional practice after training. To measure the perceptions of the effectiveness of implementation, participant submissions during training will be compared to the responses from the interviews. Analysis of these comments may help determine if transference has taken place between what was learned in professional development and what may have become regular practice in participants designing and teaching online courses.

Suggested future studies include analyzing student performance and perceptions of their learning experiences from both the students themselves and their instructors. Transfer of training to performance is one of the most critical concerns for trainers and administrators (Dick, Carey, & Carey, 2015; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016). Suggested future studies suggest there should be feedback from instructors about their actual practices to see how professional development has been implemented (Barton, 2020; Dick, Carey, & Carey, 2015; Luongo, 2018) in addition to student data, satisfaction, and perceptions. Interviews can be used to evaluate instructors' implementation of concepts during training into professional practice for qualitative case studies (Dick, Carey, & Carey, 2015). This study uses interviews with participants compared to reflections within the IYOC workshop to evaluate if and to what degree training was implemented into professional practice.

Description of the Site

Participants in this study are current faculty members at a Midwest regional mid-sized institution. This institution was founded in 1969 as an upper-division institution with a primary focus on serving community college graduates and undergraduate transfer students pursuing baccalaureate and graduate degrees (Governors State University, 2021). Starting in 2014, this institution started allowing undergraduate students at all levels, and since then, it serves approximately 6,000 students annually in 85 different bachelor's, master's, doctoral, and certificate programs (Governors State University, 2021).

Participants

The pool of potential study participants consists of previous IYOC participants employed at a Midwest regional mid-sized institution from March 2021 to June 2023. February 2020 is when this institution first started subscribing to QM. After the researcher was trained as an official QM workshop facilitator and course reviewer, March 2021 was the first IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. June 2023 was the last IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. After July 2023, the QM Higher Education Rubric was updated to the 7th Edition, and some terminology and numbering were changed from the 6th Edition (Quality Matters™, 2023). March 2021 through June 2023 was the time frame chosen for this study for the sake of consistency, using the same QM Rubric and Self Review tool used in the IYOC workshops.

This sample pool has 21 total potential participants, broken down by the three colleges and divisions in which they taught as of June 2023. This would be considered a purposeful sample due to their unique shared experience, as well as a convenience sampling due to location and availability (Merriam, 1998). All potential participants can be grouped together into one pool because the QM workshops are scripted and only change when the rubric editions are

updated, so the IYOC content during the time frame of March 2021 to June 2023 was consistently delivered. Due to the scripted nature of the workshop, the facilitator was only allowed to customize content or feedback as it pertained to an individual participant's questions or responses during activities. Refer to Appendix B for a more detailed breakdown of the potential faculty participants in this study, but because the 21 potential participants in this pool may teach in more than one of the 22 programs represented and may also teach in more than one college or division, the college and division totals may differ from the actual total of potential participants:

- College of Arts & Sciences – 11 faculty across 13 programs
    - Division of Arts and Letters – 8 faculty
    - Division of Science, Mathematics and Technology – 3 faculty
- College of Business – 4 faculty across 3 programs
    - Division of Accounting, Finance, Management, Information Systems, and Economics – 3 faculty
    - Division of Management, Marketing, and Entrepreneurship – 1 faculty
- College of Health and Human Services – 7 faculty across 6 programs
    - Division of Health Administration – 1 faculty
    - Division of Nursing – 4 faculty
    - Division of Physical Therapy – 1 faculty
    - Division of Social Work – 1 faculty

If still employed at this institution, participants from this pool were asked to volunteer to participate in this study. If participants from this pool were no longer employed at this institution, they were not asked to participate in this study. Volunteers were anonymized in

results; only the college, department, and subject area were identified, along with a number as part of the coding process. Refer to Appendix B for a more detailed breakdown of the potential faculty participants in this study, but because the 12 faculty who actually participated may teach in more than one of the 16 programs represented and may also teach in more than one college or division, the college and division totals may differ from the actual total of participants:

- College of Arts & Sciences – 9 faculty across 10 programs
  - Division of Arts and Letters – 7 faculty
  - Division of Science, Mathematics and Technology – 3 faculty
- College of Business – 1 faculty from 1 program
  - Division of Accounting, Finance, Management, Information Systems, and Economics – 1 faculty
- College of Health and Human Services – 4 faculty across 4 programs
  - Division of Nursing – 3 faculty
  - Division of Physical Therapy – 1 faculty

Recruitment was via direct outreach using email. Direct outreach is an effective practice for soliciting participation in research (Moss, 2025). Because potential participants are peers of the researcher, the researcher verified if they were still employed at this institution before sending the email. Follow-up to the initial email was done through a phone conversation, meeting in person, or follow-up emails. Direct person-to-person contact with prospective participants can be effective for building rapport and addressing specific concerns before formal research begins (Berry, et al., 2013). It was expected that some of the participants may not want to or may not be able to participate in this study. Some members did not respond to any contact attempts.

The projected number of participants was 8 to 10 from this pool. Single-case retrospective studies typically focus on a group of participants such as a community of educators or an organization such as an educational institution (Bogdan & Biklen, 2006; Kratochwill et al., 2010; Merriam, 1998). While there is no universal threshold, studies often define small groups as 3 to 10 participants (Manolov et al., 2021). Small numbers of participants are frequently used in psychology and education, single-case studies to establish patterns and maintain depth of analysis (Lakens, 2022; Kratochwill et al., 2010; Manolov et al., 2021). 8 to 10 participants would be between about one-third and just under one-half of the 21 potential participants within the available pool. Of the 15 respondents to the initial email, there were 12 actual participants in this study. Three respondents opted out due to availability.

Upon approval of this study, ethical steps were taken before proceeding. The first step was to obtain approval via the Institutional Review Board (IRB). Next, after recruitment through email, an informed consent form was provided to potential participants. Using an informed consent form is integral to the research study to ensure the safety of participants (Smith & Nizza, 2022). Because of the small sample size as well as to protect the identity of the participants, identifiers of faculty will only be their respective program and a number, because there may be more than one participant in the study from that same subject area. Any names of participants, faculty peers, or students were blocked out in the transcription of recorded interviews as well as in previous training comments. Using the internet as both a tool and a way to facilitate research requires considering data storage (Buchanan & Zimmer, 2021). Interviews were recorded digitally, stored within the researcher's password-encrypted campus Panopto video hosting account, and set to private. All transcriptions of videos and previous training documents

containing comments were stored on the researcher's password-protected campus laptop, and all cloud backups are also password-protected.

The Researcher

The researcher has been an instructional designer in the Center for Teaching and Learning (CTL) at this Midwest regional mid-sized institution since July 2019. Primary duties of CTL include providing professional development for faculty in best practices for course design and delivery for all learning modalities, and using educational technology tools. The researcher has 10 years of teaching experience in K-12, specifically in middle school technology and visual arts subject areas, and 15 years of teaching experience at the higher education level in computer science, specifically in office applications, web development, and multimedia software. The researcher has taught and designed curriculum for both K-12 and higher education courses for in-person, online-only, and hybrid modalities. Before changing careers to education, the researcher was a graphic designer in the printing industry and a freelance web developer for 10 years. In February of 2020, upon this institution beginning its subscription to Quality Matters™, the researcher began pursuing official QM training facilitator certifications. In March 2021, the researcher started offering official QM workshops, including the IYOC to be analyzed as part of this study, as part of the CTL department's catalog of professional development opportunities for faculty.

It is important to acknowledge potential bias on the part of this researcher because this study is based on professional development that this researcher was the facilitator of. As such, the researcher is not independent of the potential qualitative data – the material learned during training was delivered by the researcher, and the reflections and comments by the participants within that training may be directly affected by the researcher's participation. It is also possible

that the participants made the comments they did during training to “please the teacher”, but they may also be willing to participate in this study because they are professional peers of the researcher. It is important, then, that this researcher practice reflexivity. Active reflection during the research process can help a researcher continually question their thought process and consider how their own perspective may influence the conclusions drawn (Burnam, 2025). It is important during this process for the researcher to regularly reflect on interactions with participants and their comments made in findings as objectively as possible, because this researcher has been involved directly in the learning process being analyzed.

This researcher used a research journal to reflect on the process and help mitigate bias. A research reflective journal is a systematic record where researchers document their thoughts, decisions, experiences, and insights throughout the research process (Hurst, 2023; Lincoln & Guba, 1982). A research journal can capture both the practical aspects of conducting research and the intellectual and emotional responses to the process (Turner, 2020; Lincoln & Guba, 1982). Especially because this researcher was the facilitator of the professional development being evaluated in this study, using a journal helps recognize how background, assumptions, and biases might influence the research, which is essential for maintaining research integrity and transparency (Hurst, 2023; Turner, 2020; Lincoln & Guba, 1982). The following four guidelines are recommended for a reflexive research journal: keep a log of evolving perceptions, a log of day-to-day procedures, a log of methodological decision points, and a log of day-to-day personal introspections (Lincoln & Guba, 1982). This was done with a written log after each interview, as well as notating in the transcripts of each interview where these items may have taken place.

Peer debriefing was also used to help the researcher practice reflexivity. Peer debriefing is the process of engaging with colleagues or experts to review research results and minimize

personal biases (McLeod, 2024). Peer debriefing was used by consulting with other members of the CTL team, who are familiar with this researcher and this study, on participant responses and conclusions drawn to help with the reflection process. This was done by sharing the research journal and a draft of the conclusions before being shared with the participants for final review during the final member-checking process.

Member-checking was used to enhance the credibility, accuracy, and trustworthiness of the qualitative data in this study. Member-checking uses the techniques of formative feedback, informant reviews, and internal validity to establish validity in qualitative research (Bogdan & Biklen, 2006; McKim, 2023; Merriam, 1998; Stake, 1995; Yin, 2018). During interviews, collaborative dialogue can take place between the researcher and participants. Participant responses during the interviews were checked against tools used and student activities created within the learning management system before and after training to validate and allow for further comment, as necessary. Considering that the researcher was also the facilitator of the training being evaluated, including perceptions of the participants during interviews can help mitigate researcher bias by recognizing participants as co-constructors of knowledge because the responses are grounded in the participants' realities (Merriam, 1998; McKim, 2023). Sharing drafts of findings with participants before completing the final draft of the study allows participants to participate in the review process to validate the researcher's interpretation of interview data (Bogdan & Biklen, 2006; McKim, 2023; Stake, 1995; Yin, 2018). This structured approach to member-checking, which also includes the consistency of interview questions and comments within the training to compare the responses to, helps strengthen the validity of the study (McKim, 2023).

Data Collection

The purpose of this qualitative case study is to examine the perceived effectiveness of the implementation of professional development for best practices in online course design by higher education instructors to determine if transference of knowledge has taken place from training to professional practice. Participants in this study were previous participants in the official Quality Matters™ (QM) Improving Your Online Course (IYOC) workshop as offered through the Center for Teaching and Learning (CTL) at a Midwest regional mid-sized institution. The researcher was the facilitator of the IYOC workshops and is also a certified QM course reviewer.

This study utilizes a single-case retrospective case study design. Retrospective case studies, sometimes called historical case studies, try to determine changes over a period of time using existing data or documentation to analyze patterns, causes, or lessons learned (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998; Ozcan et al., 2017), and this existing data can be compared to interviews to interpret findings (Goyes & Sandberg, 2024). Participants first answered a series of reflection questions in individual interviews. See Appendix C for interview questions. Next, potential themes were identified in the researcher's journal and then quotes were identified in participant interview transcripts using ATLAS.ti software. ATLAS.ti software was used because it is a valid evaluation tool for the organization and analysis of themes (Deterding & Waters, 2021; Saldaña, 2016). Next, commentary from the IYOC Self Review activity was gathered and compared to interview responses, also using ATLAS.ti software; because the Self Review tool used in the IYOC workshop also allows for reflection and justification by the participant and reviewer at the time of training, this qualitative information can be used for comparison to interview questions during this study. Potential changes in teaching practices for online courses can be shown through opportunities for instructor reflection (Ali & Wright, 2017). Improvements to student interactions were the focus of the IYOC

workshop. Implementation of those potential improvements in courses, as noted in the interview responses, will help answer Research Question 1. Comparison to intended improvements noted in the IYOC Self Review activity will indicate if and to what degree transference from training to practice has taken place, helping answer Research Question 2. Comparing direct responses to previous responses is a form of methodological triangulation for validation of research (Stake, 1995).

Also related to Research Question 2, transference of knowledge might not be limited to online course design and delivery. Transference of knowledge to professional practice may also have occurred in other teaching modalities by the participants. Even though the IYOC workshop focuses on best teaching practices for the online modality, many pedagogical elements can also be applied to in-person or hybrid courses. Results will show implementation beyond just the online modality, which can show the benefits of future professional development efforts for students, educators, and institutions alike. More about analysis of the data will be shown in the following chapter.

The practical implications are also related to the ability to understand how the faculty participants are influenced by the IYOC workshop as well as QM as a provider of professional development. This study could be considered heuristic because it can help explain whether an innovation worked or failed to work (Merriam, 1998). This institution's administration decided to begin a subscription to Quality Matters™ in February 2020, before there was even a thought of a possible forced nationwide shutdown due to COVID. Administration already saw a potential value in utilizing QM as a training resource to help improve faculty understanding of best practices for teaching online courses. Through the evaluation of QM training at this institution with this study, if the training proves to be successful in faculty attaining better

knowledge about online course design best practices, then the administration will likely continue to subscribe to QM and encourage future offerings of QM training at this institution, knowing it is meaningful. In this study, Kirkpatrick's Level 3 helps evaluate faculty perceptions of student performance and satisfaction. Furthermore, if the training relates to a perceived value change for this institution in terms of student performance and satisfaction because of faculty participation in QM training, then the administration may possibly even scale up the QM training initiative. If QM does not demonstrate a value, then administration may wish to pursue other training efforts for this pedagogical need. Future studies could be conducted utilizing Kirkpatrick's Level 4 to show any possible relationship between QM training for faculty and actual student performance in online courses such as course grades and evaluations on individual assignments. More cross-case analyses or studies involving larger groups could show more institution-wide results and trends.

Procedures

This study is a single-case retrospective case study. Participants in this study were faculty participants in the QM IYOC workshops administered by the researcher at a Midwest regional mid-sized institution. Participants from this pool were asked to volunteer for this study. Participants first answered a series of reflection questions during individual interviews using Level 3 of the Kirkpatrick Model as a guide.

The culminating activity in the IYOC workshop is for participants to perform a self-review of a course they have previously taught and identify ways to improve based on the best practices for the three student interactions introduced during training: student-student, student-content, and student-instructor. See Table 1 for a list of the QM-specific review standards specifically addressing student interactions.

Table 1

The following QM Specific Review Standards introduced in Module 6 of the QM IYOC Workshop specifically address student interactions.

Specific Review Standard 2.4	*The relationship between learning objectives or competencies and learning activities is clearly stated.*
Specific Review Standard 4.2	*The relationship between the use of instructional materials in the course and completing learning activities is clearly explained.*
Specific Review Standard 5.2	*Learning activities provide opportunities for interaction that support active learning.*
Specific Review Standard 5.3	*The instructor's plan for interacting with learners during the course is clearly stated.*
Specific Review Standard 6.2	*Course tools promote learner engagement and active learning.*

Because participants have been able to teach one or more courses since participating in professional development, reflection questions asked what kinds of activities are being used in their course design and how they implemented each type of student interaction. Questions also asked how that has changed since before taking the IYOC workshop. See Appendix C for interview questions.

To measure the perceptions of the effectiveness of implementation, the results from the Self Review from the IYOC workshop were compared to the responses to the reflection questions from the survey and/or interviews. Analysis of these comments may help determine if transference has taken place between what was learned in professional development and what may have become regular practice in participants designing and teaching online courses.

Data Analysis

This is a retrospective case study, evaluating changes to behavior over time by comparing interview responses to previous responses made during a professional development activity. The IYOC workshop specifically asks participants to identify courses that could be changed and reflect on those potential changes. This study focuses specifically on changes in interaction within courses after professional development. The reflections of participants in this study could

determine if and how any changes to interaction took place in the courses identified during training or in any other courses taught since.

As was noted in the previous chapter, some assumptions can be made regarding the participants in this study and their previous use of best practices before participating in professional development. Given the correspondence-like nature and self-teaching mindset some instructors might have about how courses should be designed for students in distance courses, at the very least, student-content interaction is assumed to be present. This form of interaction was still evaluated during this study, but after participating in the IYOC workshop, it was assumed that the most potential implementations after training would likely be in the student-student and student-instructor interactions. It is also assumed that instructors in distance courses, at the very least, provide feedback on assignments and are available for communication in some way, so this study assumes that student-instructor interaction is already present but also has the potential for improvement. Given the possibilities because of the technology tools available and demonstrations through the training, it is assumed that student-student interaction will be the area of greatest implementation.

Strategies for Validating the Findings

As noted in Table 1, QM Specific Review Standards 2.4, 4.2, 5.2, 5.3, and 6.2 are specifically related to a given course's activities and their potential student interactions. The interview questions for this study are specifically written to answer these questions because changes to participants' course design will be centered around the implementation of improved student interactions. In the IYOC Self Review activity, participants were asked to comment on what they were doing at that time to meet these standards or how they intended to improve this. The IYOC comments were used for comparison to the interview questions. Comparing

interviews to previous non-interview comments can enhance the credibility of findings by cross-verifying information from different perspectives (Alamri, 2019; Kaman & Othman, 2016; Vogl et al., 2018; Yin, 2018). Non-interview comments, such as those made during training, offer spontaneous reflections (Vogl et al., 2018) that may not be affected by recall bias after time has passed (Alamri, 2019). Because retrospective studies can identify potential changes made over time, comparing interview responses to pre-existing data can help researchers understand shifts in behavior (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998).

Triangulation was also used to validate findings. Triangulation is the practice of validating results across multiple sources of evidence (Bogdan & Biklen, 2006; Merriam, 1998; Stake, 1995; Yin, 2018). This researcher, as part of the support team for online learning tools at this institution, has access to Blackboard, the learning management system used to house all online learning materials for courses in all modalities. As part of the permissions granted by participants in this study and as part of this researcher's job duties, application of what was learned in training was identified in the course sections mentioned, not only in the interviews of this study, but also in the comments made during training that will be compared to. As part of the IYOC training, participants noted the courses they intend to improve, so sections within Blackboard before and after the training could be compared for tool usage and types of assignments, checking for differences in student interactions. Interview responses dictated if such comparisons are necessary. More analysis details will be discussed in the following chapter.

This study is a single case in that the participants were from a pool of faculty from the same institution who participated in the same training. The data will determine if the analysis is a holistic or embedded context. Holistic single-case studies have a single unit of analysis, and

embedded single-case studies have multiple units of analysis (Yin, 2018). If there is a small sample size, there might not be enough participants or variety within responses to be considered more than a single unit of analysis, so the data may prove to be holistic. But depending on the variety of colleges, divisions, programs, and subject areas of the participants, and the richness of the data provided, there might be multiple units of analysis to be considered embedded (Yin, 2018).

Because of the variety of subject areas represented by the participants willing to participate in this study, the entire group could be analyzed as a whole, making this study a holistic context, but there are specific standout sub-groups that could allow for a potential embedded context in a future study. With the assumption that many higher education faculty do not receive training on how to be teachers, the topic of interactivity from the IYOC training is considered a review for the College of Education (COE) faculty within the faculty pool. This eliminated the COE faculty from being part of this study and allowed for analysis of the other subject areas represented. Multiple sources of data, which in this study would be the variety of subject areas represented based on participants who volunteered, enhances the reliability and validity of the research and reduce bias (Kaman & Othman, 2016; Quintao & Andrade, 2020). Eliminating the education group, because they are likely already familiar with the topics presented during training, which this study analyzes, is necessary to reduce bias and not skew the results (Alexander et al., 2014; Forbes, 2013).

For Level 3: Behavior to be evaluated, critical behaviors need to first be identified (Kirkpatrick & Kirkpatrick, 2016). These critical behaviors were the potential themes determined in the researcher's journal. Coding of themes was used to organize findings from the interviews and training comment data. Coding can be used to classify interviews or forms of

qualitative data to make them more appropriately retrievable at any time (Bogdan & Biklen, 2006; Merriam, 1998; Stake, 1995). Following the strategies provided by Saldaña (2014), the first step would be to foresee a possible organization. Knowing the subject matter as the facilitator of the professional development, the themes of student interactions already provide an organization for participant interview responses, which will be further broken down into types of activities implemented within their course designs. Motivations for using those activities could also be themes of analysis.

Next steps included collecting the interview data and to analyze for themes using axial coding (Deterding & Waters, 2021; Saldaña, 2014), verifying if what was foreseen was valid and if new themes emerged (Yin, 2018). Saturation is where new information no longer brings fresh insights to research, meaning that enough data has been collected to confidently understand patterns and themes (Webster, 2025). Because of the themes determined, selective coding, a more focused approach for identifying core themes and discarding irrelevant or redundant themes, ensured that this study retains only meaningful trends (Saldaña, 2014) and helped address saturation (Deterding & Waters, 2021; Saldaña, 2014; Webster, 2025). Because this study is retrospective, temporal coding was used to track when themes may have developed or faded over time (Saldaña, 2014) when comparing interview responses to in-training comments. ATLAS.ti software was used because it is a valid evaluation tool for the organization and analysis of themes (Deterding & Waters, 2021; Saldaña, 2016). More details about analysis will be discussed in the following chapter.

Summary

Many instructors feel that online teaching is a modern form of correspondence, believing that students can learn independently when content is placed in a learning management system.

This is important because many instructors take this approach when designing and teaching online courses, but student learning also occurs through other forms of interaction beyond solely interacting with the content. Showing instructors the best practices for online course design through the delivery of professional development is necessary. This study examines the perceived effectiveness and implementation of training after higher education instructors have participated in professional development for best practices in online course design and delivery. Professional development has been delivered using resources from Quality Matters™ (QM), a globally recognized authority on online course design best practices. QM is aligned with Level 3 of Kirkpatrick's model of Four Levels of Training Evaluation, which evaluates the transfer of knowledge from training to practice and provides the foundation for this study. Participants were previous faculty participants in official QM Improving Your Online Course (IYOC) workshops previously facilitated by the researcher at the same institution. Participant perceptions of what was learned in professional development and what was implemented after can influence how courses are taught and, therefore, the student experience as a result. The goal of professional development is to help participants implement what was taught into regular practice. The following chapter will provide an analysis evaluating whether that transference has taken place.

CHAPTER IV

RESULTS

Online learning is a course delivery modality that has been growing in popularity since the wide availability of the internet to the public in the 1990s (Nilson, 2003), especially considering the circumstances of the COVID-19 pandemic (Bozkurt et al., 2022). The mindset of online learning has shifted from the focus of correspondence to mimicking the in-person experience by taking advantage of the interactive possibilities the technology tools allow for (Joosten & Cusatis, 2019; Kurt, 2016; Robinson et al., 2017). The current formal definition of online learning is electronically mediated asynchronous and synchronous communication that utilizes interaction and collaboration beyond just content delivery (Garrison, 2017). As a result, higher education faculty who wish to be effective instructors in the online modality need specific professional development, in not only the technology tools but also the pedagogy for purposeful course design to provide a quality learning experience in online courses for adult learners (Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Luongo, 2018; Nilson, 2003).

The purpose of this study is to evaluate if and to what degree transference from professional development to regular practice has taken place for higher education instructors. Participants in this study were higher education faculty at a Midwest regional mid-sized institution who participated in the Improving Your Online Course (IYOC) workshop, as developed by Quality Matters™ (QM), an international standards organization in online course design and delivery best practices. Professional development for educators in general, in addition to being specifically for effective online course design, can lead to improved student performance in addition to increased positive attitudes toward the online learning modality by instructors (Burridge & Carpenter, 2013; Darling-Hammond et al., 2017; Davies & Preston,

2002; Gregory et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Zimmerman et al., 2020). Professional development tools and standards organizations are available to provide pedagogical training and evaluation tools for online course designers and instructors, including Quality Matters™ (QM), among others (Johnson et al., 2019; Zimmerman et al., 2020). QM is a sound, research-based professional development provider to increase interaction in online courses (Joosten & Cusatis, 2019), and increased interaction and collaboration within online courses not only mimics but also enhances in-person course experiences as well (Gregory et al., 2020; Joosten & Cusatis, 2019; Robinson et al., 2017). Implementation of what was learned in IYOC is what will be evaluated.

This study uses two phases to answer each of the following research questions:

3) How do higher education instructors perceive changes, if any, to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop?
4) How did participants perceive what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery?

Research Question 1 will be answered through interview questions conducted individually with previous IYOC participants. Questions consist of practices used in courses related to content within the training, such as forms of student interaction used, alignment of content, activities, and assessments to learning objectives, and technology tools used. Research Question 2 will be answered by comparing the responses of participants to interview questions with comments made within the Self Review activity during the IYOC training, as well as comparing items and tools used within course sections within the learning management system Blackboard from before training and then after training.

Summary of the Methodology

Participants in this study are current faculty members at a Midwest regional mid-sized institution. The pool of potential study participants consists of previous IYOC participants employed at a Midwest regional mid-sized institution from March 2021 to June 2023. February 2020 is when this institution first started subscribing to Quality Matters™ (QM). After the researcher was trained as an official QM workshop facilitator and course reviewer, March 2021 was the first IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. June 2023 was the last IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. After July 2023, the QM Higher Education Rubric was updated to the 7th Edition, and some terminology and numbering were changed from the 6th Edition (Quality Matters™, 2023). The time frame chosen for this study, from March 2021 to June 2023, was selected for the sake of consistency, using the same QM Rubric and Self Review tool employed in the IYOC workshops. The researcher was the facilitator of the IYOC workshops and is also a certified QM course reviewer.

This sample pool has 21 total potential participants, broken down by the three colleges and divisions in which they taught as of June 2023. 12 participants from this pool volunteered to participate in this study. Volunteers were anonymized in the results; only the subject area and course number were identified, along with a number as part of the coding process:

- Communications (COMM) – 1 faculty
- Computer Science or Information Technology (CPSC) – 3 faculty
- English and Literature (ENGL) – 2 faculty
- History (HIST) – 1 faculty
- Interdisciplinary Studies (IDSS) – 1 faculty

- Nursing (NURS) – 2 faculty
- Physical Therapy (PHYT) – 1 faculty
- Sociology (SOC) – 1 faculty

Refer to Appendix B for a more detailed breakdown of the faculty participants for this study.

This study utilizes a single-case retrospective case study design. Retrospective case studies, sometimes called historical case studies, try to determine changes over a period of time using existing data or documentation to analyze patterns, causes, or lessons learned (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998; Ozcan et al., 2017), and this existing data can be compared to interviews to interpret findings (Goyes & Sandberg, 2024). Participants first answered a series of reflection questions in individual interviews, which were recorded in Microsoft Teams, and then transcripts were downloaded for analysis. See Appendix C for interview questions.

Next, potential themes based on critical behaviors were identified and coded in the researcher's journal. For Level 3: Behavior to be evaluated, critical behaviors need to first be identified (Kirkpatrick & Kirkpatrick, 2016). Developing a coding system involves searching through qualitative data for regularities and patterns, writing down, highlighting and color-coding key words and phrases (Bogdan & Biklen, 2006). Themes determined from the researcher's journal were then then used to identify participant quotes using ATLAS.ti software. ATLAS.ti software was used because it is a valid evaluation tool for the organization and analysis of themes (Deterding & Waters, 2021; Saldaña, 2016). Next, commentary from the IYOC Self Review activity was gathered and compared to interview responses, also using ATLAS.ti software; because the Self Review tool used in the IYOC workshop also allows for

reflection and justification by the participant and reviewer at the time of training, this qualitative information can be used for comparison to interview questions during this study.

Potential changes in teaching practices for online courses can be shown through opportunities for instructor reflection (Ali & Wright, 2017). Improvements to student interactions were the focus of the IYOC workshop. Implementation of those potential improvements in courses, as noted in the interview responses, will help answer Research Question 1. Comparison to intended improvements noted in the IYOC Self Review activity will indicate if and to what degree transference from training to practice has taken place, helping answer Research Question 2.

The culminating activity in the IYOC workshop is for participants to perform a self-review of a course they have previously taught and identify ways to improve based on the best practices for the three student interactions introduced during training: student-student, student-content, and student-instructor. See Table 1 for a list of the QM-specific review standards specifically addressing student interactions.

Because participants have been able to teach one or more courses since participating in professional development, interview questions asked what kinds of activities are being used in their course design and how they implemented each type of student interaction. Questions also asked how that has changed since before taking the IYOC workshop. See Appendix C for interview questions.

To measure the perceptions of the effectiveness of implementation, the results from the Self Review from the IYOC workshop were compared to the responses to the reflection questions from the survey and/or interviews. Analysis of these comments may help determine if transference has taken place between what was learned in professional development and what

may have become regular practice in participants designing and teaching online courses. Comparing direct responses to previous responses is a form of methodological triangulation for the validation of research (Stake, 1995). See Table D1 for a summary of the perceptions of the participants in both training and in this study.

Peer debriefing was used to help the researcher practice reflexivity and mitigate bias. Peer debriefing is the process of engaging with colleagues or experts to review research results and minimize personal biases (McLeod, 2024). Peer debriefing was used by consulting with other members of the CTL team, who are familiar with this researcher and the training being evaluated, on participant responses and conclusions drawn to help with the reflection process. This was done by sharing the research journal and a draft of the findings and conclusions before being shared with the participants for final review during the final member-checking process.

Member-checking was also used to mitigate bias and enhance the credibility, accuracy, and trustworthiness of the qualitative data in this study. Member-checking uses the techniques of formative feedback, informant reviews, and internal validity to establish validity in qualitative research (Bogdan & Biklen, 2006; McKim, 2023; Merriam, 1998; Stake, 1995; Yin, 2018). During interviews, collaborative dialogue can take place between the researcher and participants. Participant responses during the interviews were checked against tools used and student activities created within the learning management system before and after training to validate and allow for further comment, as necessary. Considering that the researcher was also the facilitator of the training being evaluated, including perceptions of the participants during interviews can help mitigate researcher bias by recognizing participants as co-constructors of knowledge because the responses are grounded in the participants' realities (Merriam, 1998; McKim, 2023). Sharing drafts of findings with participants before completing the final draft of the study allowed

participants to participate in the review process to validate the researcher's interpretation of interview data (Bogdan & Biklen, 2006; McKim, 2023; Stake, 1995; Yin, 2018). This structured approach to member-checking, which also includes the consistency of interview questions and comments within the training to compare the responses to, helps strengthen the validity of the study (McKim, 2023).

It should also be noted that many of the interviews became support sessions after the interview questions had completed. This research was not only the facilitator of the training, but is also a member of the support department for online learning tools and best practices at this institution. As noted in the researcher's journal, offering support to instructors was purposely not done during interview questioning to allow participants to provide honest reflections of their perceived implementations of training without interference. It was after the questions were answered that questions about troubleshooting tool usage, possibly improving current methods, or implementing upcoming software updates and alternatives were addressed. These issues included future software implementations at this institution, including a campus-wide upgrade to the Blackboard learning management system one year from the time of this study, deprecation of several multimedia platforms and adoption of updated alternatives, and addressing the legal requirement for all resources at public institutions nationwide to meet accessibility guidelines before the end of the next semester as of the time of this study (U.S. Department of Justice, Civil Rights Division, Disability Rights Section, 2024). Participants in this study regularly consult the researcher's department for troubleshooting or training needs, so it was a natural progression for interviews to progress into discussions involving specific questions and concerns to meet the needs of individual participants.

Kirkpatrick's Level 3: Evaluating Behavior

Kirkpatrick (2006) suggests the following guidelines for evaluating behavior after training:

1. Use a control group if practical.
2. Allow time for behavior change to take place.
3. Evaluate both before and after the program if practical
4. Survey and/or interview one or more of the following: trainees, their immediate supervisor, their subordinates, and others who often observe their behavior.
5. Get 100 percent response or a sampling.
6. Repeat the evaluation at appropriate times.
7. Consider cost versus benefits.

For number 1, using a control group was not practical for this study due to the low number of potential participants and the background of the possible control group. A significant number of the IYOC participants overall were College of Education faculty, and it can be assumed that this group has had formal training in teaching pedagogy already. Even though the IYOC workshop focuses on the pedagogical aspects of online course design and delivery, some of the IYOC subject matter might be considered a review for this group. It can, therefore, be assumed that this group may already have been implementing best teaching practices in their courses, regardless of modality, so this group was eliminated as potential participants in this study. Asking for volunteers from the remaining pool of non-education faculty would be considered what Merriam (1998) calls purposeful sampling, because of the attempt to gain insight from their new learning experience, which might be considered review for the College of Education faculty. A potential for a future study could use the College of Education faculty as a control group to compare to participants in this study or future studies.

The purpose of this study was specifically designed to address items 2 and 3. The IYOC trainings as part of this study took place between March 2021 and October 2023, between two and four-and-a-half years before this study, which is a significant period of time to allow changes in behavior to take place. Evaluation specifically compared qualitative data from before and after training. The Self Review responses were the culminating activity in the IYOC workshop, and those responses were compared to the interviews of this study, along with verification in the Blackboard learning management system of course sections from before and after the training took place. If any Self Review responses during training were left blank, this would have been because the participant believed they were already implementing those specific QM standards in their current practices at that time.

For number 4, the trainees were interviewed. Also, according to Kirkpatrick (2007), interviews should consist of structured questions that can be administered to query the degree to which new behaviors are being applied on the job. See Appendix C for interview questions, the structure of which was based on the QM Specific Review Standards being taught in the IYOC workshop: 2.4 (presentation of learning objectives), 4.2 (alignment to learning objectives), 5.2 (forms of student interaction), 5.3 (policies for student interactions), and 6.2 (variety of technology). See Appendix A for the QM Rubric and details about each of the Specific Review Standards.

The job in question here is faculty designing and delivering online courses, and the IYOC training was intended to help those instructors improve their online courses. Follow-up questions to action plans are an excellent way to administer interviews (Kirkpatrick & Kirkpatrick, 2007). The action plans being followed up on would be the Self Review responses provided by participants during the IYOC workshop. Tracing changes over time in case studies

is part of what Kratochwill (1978) refers to as time series analysis. Data analysis for this type of case study consists of tracing actual events over time while giving close attention to their chronological sequence (Kratochwill, 1978; Yin, 2018).

Although the trainees were interviewed, supervisors and subordinates were not. The supervisors in this case would be program coordinators or division chairs, those who would perform performance evaluations of the faculty participants via direct observation. Subordinates would be the students. Potential for future studies could include interviews with supervisors compared to instructor perceptions, interviews with students compared to instructor perceptions, or evaluation of student performance data for a mixed-methods comparison of grade data to instructor perceptions.

For number 5, getting 100 percent response was not possible, but 12 out of 21 potential participants is a significant sampling, 57.143%. Only 8-10 participants were expected. See Appendix B for a detailed breakdown of the participants in this study. A potential future study could pursue the remaining potential participants and repeat this study's methodology. Repeating this study is item number 6, and the potential future study ideas described within this study could fulfill that item. Additional future studies could also include trainings that took place after the time frame of this study. This and future studies could help address item number 7 and provide a justification for this institution's investment in QM. More on the conclusions and implications for institutional leadership will be discussed in the next chapter.

Methodological Decision Points

This researcher used a research journal to reflect on the process and help mitigate bias. A research reflective journal is a systematic record where researchers document their thoughts, decisions, experiences, and insights throughout the research process (Hurst, 2023; Lincoln &

Guba, 1982). A research journal can capture both the practical aspects of conducting research and the intellectual and emotional responses to the process (Turner, 2020; Lincoln & Guba, 1982). Especially because this researcher was the facilitator of the professional development being evaluated in this study, using a journal helps recognize how background, assumptions, and biases might influence the research, which is essential for maintaining research integrity and transparency (Hurst, 2023; Turner, 2020; Lincoln & Guba, 1982). The following four guidelines are recommended for a reflexive research journal: keep a log of evolving perceptions, a log of day-to-day procedures, a log of methodological decision points, and a log of day-to-day personal introspections (Lincoln & Guba, 1982). This was done with a written log after each interview, as well as notating in the transcripts of each interview where these items may have taken place. Significant to the interview process in this study were several methodological decision points. See Table 2 for the coding for instructors and corresponding subject areas.

Table 2

Coding for the 12 instructor participants and their corresponding subject areas.

Instructor Codes	Subject Area
COMM-1	Communications
CPSC-1 *CPSC-2* *CPSC-3*	Computer Science or Information Technology
ENGL-1 *ENGL-2*	English and Literature
HIST-1	History
IDSS-1	Interdisciplinary Studies
NURS-1 *NURS-2*	Nursing
PHYT-1	Physical Therapy

SOC-1	Sociology

Courses in Training vs. Courses in Interviews

Because some time has passed since the training was completed, for some participants going all the way back to the middle of the COVID lockdown in March 2021, the courses discussed during training were not consistent with all interview responses. It was assumed that the courses utilized during training would still be taught in some form, either online or another modality. In the one instance of CPSC-3, the course utilized in training was intended to be developed, but was never actually taught. Other deviations included courses taught online during COVID as emergency-remote but are now back to in-person, and courses that are now taught in multiple modalities by the same instructor, sometimes taught concurrently, but sometimes modalities alternate semesters. In all these instances, the interviews needed to be adapted to individual instructor scenarios.

Some courses utilized in training are now taught in different modalities. In the instances of HIST-1, NURS-1, and COMM-1, courses utilized in training were taught during COVID as emergency-remote online but have now returned to in-person. In the case of HIST-1, he intends to convert his in-person First Year Seminar course to online, but it is currently only offered in-person. Sometimes sections with different modalities are taught concurrently, sometimes modalities switch each semester. In the instances of CPSC-2, IDSS-1, and ENGL-1, these instructors teach different online and in-person sections of the same course concurrently each semester. In the instance of IDSS-1, she strongly recommends that her online students attend in-person sections. In the instance of ENGL-1, he has unofficially converted his in-person sections to hybrid providing students the option to attend synchronous sessions who cannot otherwise be

present in the classroom, which online students can also attend. More on the themes of student interactions, intended future improvements, and transference to other modalities will be discussed in the next section.

Helping Guide Participants with Interview Questions

There were some specific instances where participants needed to be guided through questions or where clarification was necessary. This was either due to some individual participants' foreign accents and using English as a second language, or based on their level of tech-savviness, some specific aspects from their responses needed clarification. See Appendix C for interview questions.

There were two instances where participants had a heavy foreign accent. In the case of NURS-2, her initial responses seemed passionate and thorough, but then she had to be reminded about other parts of the questions. Consciously trying not to lead her, her own statements in previous responses were used as reminders of something she might have wanted to talk about when she got off topic. In the case of CPSC-3, for clarity on both the participant's and the researcher's part, either questions or answers needed to be repeated due to the language barrier.

There was one instance where a participant was required to provide clarification based on their use of technology. In the case of SOC-1, there were technical aspects of her approach to narrating PowerPoint slides. Her current approach is to insert an audio file on each slide, which requires students to download the PowerPoint, then click the audio icon on each slide to hear the narration. After the interview, support was provided on how to turn these presentations into video files, easing the burden on students to hear the narration of the presentation.

Interviews Becoming Support Sessions

Based on upcoming future technology deprecations or upgrades at this institution, several interviews became support sessions after the interview questions had been completed. It should be noted that not only was the researcher the facilitator of the training being evaluated in this study, but the researcher is also a member of the department that provides professional development and support for instructional technology and best practices at this institution. Conscious efforts were made by the researcher to refrain from interjecting supportive comments during the interview, in order not to affect the validity of the responses. Based on specific instructor responses, support comments were provided after the interviews, sometimes at the participants' request, as a response to any additional questions or comments they may have had at the end. See Appendix C for the interview questions.

Some of these additional responses and comments contributed to the theme of intended future improvements. In the instance of HIST-1, the instructor asked about how to start using cloud-based sharing tools in both in-person and online courses for asynchronous collaborative group projects. In the instances of PHYT-1, NURS-1, and NURS-2, they all had questions about specific technologies that will either be deprecated due to the institution not renewing their subscription or other specific technology upgrades. In the instance of ENGL-2, the instructor asked about different approaches to presenting alignment of learning objectives to students. The theme of intended future improvements will be discussed in the next section.

It should also be noted that the researcher is also an adjunct instructor who teaches computer science courses, including web development coding. Some of the after-interview dialogue was between colleagues comparing different techniques and tools used for student interactions. In the cases of all three CPSC instructors, online tutorials and coding tools that students are required to download in courses taught by both the instructors and the researcher

were compared and discussed. The theme of purposeful use of technology will also be discussed in the next section.

Presentation of Findings

Following the strategies provided by Saldaña (2014), the first step would be to foresee a possible organization. The organization would be based on identifying themes that help answer the research questions. Themes identified in interviews would be what Merriam (1998) calls narrative analysis, simultaneously coding the raw data and constructing categories based on the characteristics of each transcript's contents. Those themes would be based on what Kirkpatrick (2016) calls critical behaviors. The critical behaviors identified in Level 3: Behavior will have the most influence on Level 4: Results (Kirkpatrick & Kirkpatrick, 2016). Results and conclusions based on the data will be discussed in greater detail in Chapter 5.

In order for Level 3: Behavior to be evaluated, critical behaviors need to be identified (Kirkpatrick & Kirkpatrick, 2016). These critical behaviors were the potential themes that were first determined in the researcher's journal, and then matching quotes were identified in participant interview transcripts using ATLAS.ti software. Themes were identified by what Yin (2018) defines as working with the data from the ground up, poring through the data until useful concepts are found, and possibly suggesting additional relationships. Coding of themes was used to organize findings from the interviews and training comment data. Coding can be used to classify interviews or forms of qualitative data to make them more appropriately retrievable at any time (Bogdan & Biklen, 2006; Merriam, 1998; Stake, 1995). Next steps included collecting the interview data and analyzing for themes using axial coding (Deterding & Waters, 2021; Saldaña, 2014), verifying if what was foreseen was valid, and if new themes emerged. Saturation is where new information no longer brings fresh insights to research, meaning that

enough data has been collected to confidently understand patterns and themes (Webster, 2025). Because of the themes determined, selective coding, a more focused approach for identifying core themes and discarding irrelevant or redundant themes, ensured that this study retains only meaningful trends (Saldaña, 2014) and helped address saturation (Deterding & Waters, 2021; Saldaña, 2014; Webster, 2025). Because this study is retrospective, temporal coding was used to track when themes may have developed or faded over time (Saldaña, 2014) when comparing interview responses to in-training comments.

Strategy codes were used to identify themes. Strategies refer to the tactics, methods, and techniques instructors use to influence student behavior (Bogdan & Biklen, 2006). Themes should reflect the purpose of the research (Merriam, 1998). From a holistic standpoint as defined by Yin (2018), the following themes help answer Research Question 1: the organization of content within the learning management system, the use of discussion activities, the use of all three forms of student interaction, and the purposeful use of technology. The following themes help answer Research Question 2: the presentation of and alignment with learning objectives, intended future improvements to course design and delivery, addressing accessibility, and the transfer of training concepts to other course modalities. Even though there were only 12 participants in this study out of a possible 21 who completed IYOC training and are still employed at this institution, there were three distinct sub-groups of participants based on subject area that allow for a potential embedded context as defined by Yin (2018).

Holistic Themes for Research Question 1

Research Question 1 asks how higher education instructors perceive any changes to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop. Responses to interview questions answering this question yielded the

following themes: organization of content within the learning management system, utilization of discussion activities, implementation of all three forms of student interaction, and purposeful use of technology.

Organization of Content within the Learning Management System

Because one of the main goals of the IYOC training was the improvement of student interactions, organization of content is reflected in the QM Specific Review Standards (SRSs) of 2.4, 4.2, and 5.3. See Appendix A for the descriptions of the QM standards. SRS 2.4 expects that the relationship between learning objectives, learning activities, and assessments is made clear. SRS 4.2 expects the relationship between instructional materials and completion of activities and assessments is clearly explained. SRS 5.3 expects that the instructor's plan for substantive student interaction is clearly presented. Meeting each of these standards is helped by providing a clear organization of content within the learning management system. Instructors had many approaches to organizing content within their Blackboard shells based on their personal teaching styles and the learning objectives for their respective programs of study.

According to the interview with ENGL-1, organizing content was one of the key takeaways from the training when he stated,

> "Up until (the training) I'd sort of followed the format of like the Blackboard shell as it kind of appears when you first get it, and all the material ended up going into Course Content (folder). And now I start each semester by getting rid of any sections (in the Blackboard navigation) I'm not gonna use. I just set up in the left hand column the START HERE (area) and include necessary information like learning outcomes and stuff like that from the syllabus … and then having unit folders and weekly unit folders … based on like say for example in Shakespeare that there's four plays, so I'll do 4 units

> rather than so that all the readings for a particular play will be available in that one folder and stuff so. Having like the unit learning outcomes and stuff like that in each of those folders, along with the links to reading materials, and submission folders and stuff like that. So yeah, the sort of structure of my Blackboard shells are very much influenced by having done that (training) course. It has very much so in terms of pedagogy from doing that."

These intentions were identified in his training comments in the Self Review activity. He mentioned the benefit of organizing learning materials in his response to SRS 1.2, "This information could be provided in a more accessible way than just putting it all in a syllabus document." Although he has not created a curriculum map yet, his intention to address organization was also stated in his response to SRS 2.4, "I think having a visual map of the course including learning activities and linking them to objectives would help in an online course," and in his response to SRS 4.2,

> "While in broad terms students are made to understand why we are reading things like major texts, I have tended to assume students will get why they are asked to read certain materials or why they're given particular links and resources. This is probably not helpful because it can feel like an info dump. When including links, I usually also explain what's in them, but this could be more directly related to how it will help them complete particular assignments or learning objectives. A resources section of a visual map would probably help too."

According to the interview for CPSC-3, modules are also used for the organization of content into manageable chunks for students, and also demonstrating alignment to learning objectives. This was also verified in her Blackboard shells. She stated,

> "So every course, like every lecture, I will break that into modules. So, for example, if a lecture has three modules in the Blackboard, there will be 3 separate modules, so students don't get overwhelmed with all the materials posted, you know, so they're going to be categorized into 3 modules. And at the top of each module, I will list out all the learning objectives of this particular module and how they, you know, match the learning objectives of. The entire course, you know, there's like a simple mapping, maybe just three to four objectives there for each module, and then they map to the learning objective of the entire course at the top of each module."

According to her Self Review comments for SRSs 2.4, 4.2 and 5.3 during training, CPSC-3 appears to have already been doing this.

CPSC-1 organizes his Blackboard shells to match the environment provided by the external content provider. As stated in his interview,

> "(Cisco) provides a similar learning environment (to Blackboard), where it tracks the users. Everything's broken down. It almost simulates (Blackboard) as far as breaking everything down into sessions, breaking down the different topics, having labs built in, and everything else. So even when I'm doing in-person, we're using online resources to access our materials that I have in a book to access our labs and everything else. So the standard practices that I learned in there, luckily, Cisco, this is where developers seem to apply the same standard practices. So it might be whatever I do online, or when I'm on the ground, we use online materials that mirror that (organizational approach). … I give (students) the link (in Blackboard), and they go to a separate site."

According to his Self Review comments for SRSs 2.4, 4.2 and 5.3 during training, CPSC-1 appears to have already been doing this.

Although all but one of the participants are using modules for content organization, all participants are using an orientation area to introduce students to course policies and other key items from the syllabus. But there is one participant, ENGL-2, who believes the use of the orientation area specifically was not beneficial for students when she stated in her interview,

> "I'll tell you I tried and it didn't work … my orientation page was way too long with way too much information, and you thought the students were going to get lost in it, and that I should break it down into separate files and links, which I did for Spring '22. And, it was a disaster. What I mean by that is, I don't know if it was just clicking on the link or whatever, but students wouldn't look at everything. I had a tremendous upsurge or uptick, I guess, of students constantly emailing me for information that was right in the links. And it was, I mean, it was noticeable. It was seriously about three times the amount of email that I usually get. And so in the fall, I went back to the single document in there, and honestly, it's been fine, but I did try it … And I look at it, and I think it's better because you just click on it, you know, right where the information is. I mean, I think it's easier. It makes sense to me, but they just wouldn't click on them."

One of the participants, COMM-1, uses discussions not only for discussion activities but to organize content instead of modules. This was verified in her Blackboard shells. In her interview she stated,

> "I'll post practice activities on the discussion boards. So, like, if I put a (video) clip on there, I'll ask them to find certain elements of like audience, tone, and structure, and stuff like that, and so. It's less of a discussion question and more of an activity, but it is still on the discussion board. So I try to keep all the like activity stuff on the boards. I think it gets a little lost in the modules."

It should be noted that after participating in the IYOC training in June 2021, COMM-1 went on to pursue and pass an official Quality Matters course review in November 2024. Passing an official QM review requires meeting all required SRSs, which are covered in the IYOC training, and also achieving an overall score of 85% or better on the remaining non-required SRSs (Quality Matters™, 2025). This signifies that it doesn't matter what tool is used for organizing content, as long as the content is presented to students in a clear fashion for the benefit of their learning interactions.

Utilization of Discussion Activities

In addition to COMM-1's use of discussions as both an interaction and an organizational tool, many of the participants use the discussion tools for student interactions. Because one of the main goals of the IYOC training was the improvement of student interactions, presenting opportunities for student interactions is reflected in the QM Specific Review Standards (SRSs) of 5.2 and 5.3. See Appendix A for the descriptions of the QM standards. SRS 5.2 expects that learning activities provide opportunities for interactions that support active learning. SRS 5.3 expects that the instructor's plan for interaction during the course is clearly stated. Meeting each of these standards is helped by providing use of the discussion tool within Blackboard to foster all three forms of student interaction: student-student, student-content, and student-instructor. Instructor interactions may or may not be as prevalent as to what might be expected for students, depending on subject area or teaching style, but the policies for instructor interaction are usually within the syllabus, orientation area, or instructions for individual discussion activities.

Participants stated in their interviews that interaction within discussions requires students to post throughout each unit of study while meeting certain deadlines to ensure active participation. According to CPSC-1 during his interview,

> "A large part of the course interaction is based on the discussions. So the students have to do discussions, and they can't just do a single post. The rules are what help to make sure they are participating. So it's not only between themselves but also between (the instructor) and with the content. Also, because they have to post on three different days, so that they typically, if they don't have the rule to post on three different days, and the first one has to be by Wednesday, they (would) all post on Sunday night, and there is no interaction. Everything is just boom straight on down. They may post three times, but they didn't interact with anybody. By forcing the rules of posting first by Wednesday and then on three different days, we get a lot more interaction because students, you know, they see what somebody else post, then they can have a chance to respond. They don't have to create their own new post, they can always respond to someone else's."

Identified in his Self Review comments, CPSC-1 mentioned his intention to use this plan in his response to SRS 5.3,

> "Discussion post grading has a standard set of comments to help the students achieve the needed points for posting. This helps in the first weeks to train the learners on what is expected if they did not read the very detailed link to the discussion rule."

According to HIST-1, the participation expectations might be different depending on the content and on the age level of the students. He stated in his interview,

> "When I was teaching (First Year Seminar) online, we would, we had weekly discussion boards. Students were required to post depending on the semester, either two or three times per week. And generally I would stipulate that the 1st post needed to be a general comment on the learning materials, what they read, what they viewed that week, and then the second and third posts would need to be direct replies to fellow students. I would

> engage in those conversations periodically. I think if I were to go back to the online course, I would probably make myself more of a presence on the discussion boards. I generally monitored and just let students interact with one another. That worked okay. It works really well when I have adult learners, which I do in my upper-level history courses. So when we do discussion boards in an upper-level history course, you know, a bunch of 30 and 40-year-olds, they can interact with each other just fine, and they can have substantive, even deep conversations on the discussion forum without me."

Some discussions use multimedia tools, allowing students the option to post video or audio responses instead of or in addition to typed responses. Use of multimedia tools was verified in participants' Blackboard shells. HIST-1 continues in his interview,

> "I did a little bit with VoiceThread, having students actually listen to some kind of short lecture or set of questions from me, and then they could respond. But I never mandated that students respond with a VoiceThread (video). I allowed them to give written responses if that's what they preferred, but that was one way, you know, during the pandemic years, where students would kind of interact with me, or at least a recording of me directly on the discussions."

HIST-1 stated his intention to use VoiceThread in his response to SRS 5.2 during training,

> "VoiceThreads will occasionally replace the weekly discussion forums. The discussions promote student-student interactions. VoiceThreads promote student-content interactions. All writing assignments involve extensive instructor comments and thus promote student-content and student-instructor interactions."

NURS-2 uses virtual synchronous sessions to have live discussions, trying to mimic the in-person experience. She stated in her interview,

> "Even though it is an online learning, I want to make it lively to have the students feel that the instructor is there like in a real class scenario. How do I do that? I check my course messages and my discussions boards daily, daily without fail. That's one thing. Second, I have Collaborate Ultra sessions we use in the class and we have like a discussion. Let's say there was a topic that they don't know about, (and) that we have a discussion (about). So we have both the asynchronous and the synchronous modes of training. So that way the instructor is really live in the course, which makes my students love having an online (course) with me."

In her Self Review comments for SRS 5.3, NURS-2 listed intended criteria for discussions to be used in a student "learning agreement" and an "instructor commitments" document that students would need to sign and submit for a grade. The learning agreement document contains the following statement regarding discussions: "I will log into the course a minimum of 3 different days during the week, at a time convenient to my schedule, to complete the assigned learning activities." The instructor commitments document contains the following statements regarding discussions:

> "I commit to logging into the course a minimum of 5 times per week. In the event of circumstances that would interfere with this schedule (e.g. military obligations), notifications or a substitute facilitator will be available … I commit to maintaining an active presence throughout the course, with the exception of those that are reserved for team interactions on group projects."

Some discussions are used for peer reviews, and this can also involve using the group tool within Blackboard. This is especially helpful in writing classes like those of ENGL-1 and ENGL-2. ENGL-1 stated in his interview that students "do peer reviewing of most of their

assignments. So like most of the big assignments they'll read drafts of one another's work and and provide feedback and learn." ENGL-2 stated in her interview,

> "We do also peer review the essays on the discussion board where students are, I actually match up peer review partners. And then they peer review the, you know, the essay and give the feedback. And students often have kind of ongoing, you know, discussion dialogue about the feedback, you know, that they're getting on the peer review. And I look at those as well, so I can give them some feedback."

In her response to SRS 5.3 during training, ENGL-2 provided a sample announcement in which she states the expectation for student interaction with discussions, "You should have the reading/viewings/podcasts completed no later than Tuesday of the week, so you are ready to engage in discussion throughout the remainder of the week." She then posted her own intended interaction expectations for grading discussions, "You can expect that I will … grade discussion posts and essays within seven days of the due date."

Some discussions are open-ended, not requiring any interaction, but are optionally available if students have any questions or concerns throughout the semester. This is different than requesting one-on-one conferences due to the possibility that the questions would benefit multiple students. NURS-2 stated in her interview,

> "So I have different modes how I share certain things with my students. One is Cyber Cafe. They all love it. It's called the Cyber Chat Box. This discussion thread is helpful for information to follow professional news. Some news that is out there on Informatics and Nursing Exit for Nurse Executives, they can share that. So that is a separate discussion board on its own. They can post it throughout the semester. Let's say they have to attend a seminar or a conference. Students will post it. They're saying (to me), such and such a

> conference is held in Washington, DC. Can we participate in that webinar? They can share that information with other students in the discussion board."

Regardless of whether a discussion is synchronous, optional, or required, NURS-2's training response to SRS 5.3 show that she intended to ensure active student interaction. She stated in her Self Review comments about the intended student learning agreement,

> "I agree to actively engage in learning activities in the course and understand that failure to do so may result in a full letter grade deduction in my final course grade."

She later clarified during her interview that she advises students "to log in to Blackboard three times a week for a minimum of two hours each time." It is shown here that she intended to present students with discussion expectations by integrating them into the grading policy.

Implementing All Three Forms of Student Interaction

As we can see, participants show that discussions can offer the opportunity for all three forms of student interaction. Because one of the main goals of the IYOC training was the improvement of student interactions, presenting opportunities for student interactions is reflected in the QM Specific Review Standards (SRSs) of 5.2 and 6.2. See Appendix A for the descriptions of the QM standards. SRS 5.2 expects that learning activities provide opportunities for interactions that support active learning. SRS 6.2 expects that the tools used within the course promote learner engagement and active learning. Meeting each of these standards is helped by providing activities and tools to engage with those activities to foster all three forms of student interaction: student-student, student-content, and student-instructor. It is possible for an activity, including discussions, to meet one, two, or even all three forms of student interaction (Garrison, 2017), as COMM-1 stated in her interview,

> "I would say discussion board prompts meet all three (forms of student interaction) because I'll put a prompt up there that requires (students) to explain the content and how they think about it in relation to the question. And then they're also required to interact with one another about that and have a conversation. And then I join in to redirect … or to encourage them if they say something particularly interesting. I might ask a follow-up question, so I would say the discussion boards probably hit all three."

Going back to her comments on organizing using the discussion tool instead of modules in Blackboard, COMM-1 stated in her Self Review comments SRS 6.2,

> "I do use the Blackboard tools to promote learner engagement, but I focus mostly on the Discussion Board. I could explore some of the other tools available on Blackboard. I like to keep things pretty streamlined for students, though, so I don't want to have too many spots that they need to check in on."

But the discussion tool is not the only opportunity, either within Blackboard or with external tools, that can promote student interaction. ENGL-1 stated in his interview that he uses both the discussion tool and an external tool called FlipGrid,

> "So that would be the main way that we would speak, sort of person to person. In terms of learning, they'll use things like discussion forums, which again are built into Blackboard. I would also use FlipGrid, so they would have an alternative to change things up from traditional discussion forums. FlipGrid obviously is like they record themselves in short video bursts, and they can set up threads and reply to one another."

In his Self Review comments for SRS 6.2, ENGL-1 stated how he wished to improve discussions with a wider variety of interactive options:

> "I think keeping an online learning situation fresh probably does require a wider variety of modes of engagement than I currently provide. My assignments are pretty innovative, but the regular engagement with the class can probably be enhanced by including more of … a variety of tools within the LMS, such as Flipgrid and discussion forums, to engage with one another and with me in an active and supportive environment. There's always room for improvement on this."

During synchronous sessions, PHYT-1 uses a polling feature built into the virtual meeting platform to foster engagement with the content and to check for understanding. He stated in his interview,

> "For the imaging class, I'd sometimes pull up an image, you know, what do you see in this image? And I do like a polling question, you know, ABC or D or things like that. So it's more interactive where they can interact with the imaging. But then I can also gauge where they are with the material if 80% miss this one question."

Although his use of discussions is already acknowledged in his training comments, PHYT-1 notes that the variety of tools used in his courses can be improved per SRS 6.3. It appears that he did this by integrating the polling feature for synchronous sessions.

Group projects are used by NURS-1, IDSS-1, and ENGL-1 to foster student interactions. NURS-1 stated in her interview,

> "A group project that they work on is that they have to critique an article. Whether it's a qualitative article or a quantitative article, they work as a group to answer a number of questions, and then they're graded as a group. I grade the assignment based on student participation and how much interaction, meaning the students get an opportunity to rate their peers, because there is a form that they complete to provide feedback, as far as

> basically whether or not they felt their peers contributed to the assignment or not. So I get feedback that way, and that was based on one of the recommendations from the students at the end of (a previous) course. A number of them stated that sometimes some students don't do anything, and only a few of them do all the work. So I created a tool so that they're allowed to rate themselves, and basically, if they don't contribute anything, the assignment may be worth say 20 points, whereas everyone would not receive 20 points if they did not contribute equitably to the assignment."

According to her Self Review comments for SRSs 5.2 and 6.2 during training, NURS-1 appears to have already been doing this.

According to IDSS-1, individual student familiarity with technology tools can sometimes be a hindrance to group participation, so she provides as many tools as she can think of in her online classes so students can use whatever they feel comfortable with. She states in her interview,

> "They have all of the tools that they need to work with. So it is not one of those conundrums that says, oh boy, now I'm doing group work. Now I got to try to figure out time to meet with people and I've got to leave my home, and you know, all of that. So I've given them as many tools as possible. That makes it seamless for them to work together as a group."

According to her Self Review comments for SRSs 5.2 and 6.2 during training, IDSS-1 listed the tools she was using then for this purpose. In her interview, she mentioned some additional tools. Purposeful use of technology will be discussed in the next section.

ENGL-1 assigns groups to perform parts of Shakespeare's plays virtually for the class, but they first need to meet as groups virtually to prepare. He stated in his interview,

> "Students have shared, and in (synchronous sessions) they will often do things like small group discussions, but also like big class discussions are sort of structured around the kind of an approach where it's not just teacher-student, like it's not teacher-led, so we'll design discussion processes where they have to respond directly to the previous student who has spoken. … So, and then they'll also do assignments that are student-to-student. So like they'll do things like a performance where a group of, say, three or four students will have to choose a scene from a play, work together on it, and do a performance for the rest of the class and stuff like that."

As was identified in his comments for SRS 5.2 during training, ENGL-1 stated how he intended to improve his approach to groups,

> "Need to find better ways to facilitate group work in an online environment, especially during synchronous sessions, as breakout groups have not worked well."

Virtual synchronous sessions are not always offered in asynchronous online courses, but several participants use some form of optional virtual meeting to encourage student interactions. ENGL-2 has regularly scheduled optional office hours each week that she encourages students to make appointments for to improve their writing. She also records the meetings so student do not have to take notes and can focus on the conversation without having to be distracted to write things down. She stated in her interview,

> "I'll hop on Teams and do a private session with the students. And what I do also is to record that so that they have access to it afterwards. I found, particularly when you're doing writing instruction, I prefer that the students aren't trying to take notes while we're talking, because when they're writing and trying to listen, they can't really think. And so what I tell them is it's recorded. So you can go back, you'll have the transcript, and you

> can review it. But this way we can have a dialogue, we can talk, exchange ideas. And they seem to learn a lot more there because they're not concerned with trying to get the details written down. They know that they'll have that. And I found that the thinking, the questions that I get from them, quite frankly, are much better than if their attention is kind of split. Between me and trying to get all the information down, it's been very successful."

ENGL-2 did not make any comments for SRSs 5.2 and 6.2 during training to verify if her interview statements there were any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time.

Even though the course is asynchronous, CPSC-3 schedules optional synchronous review sessions for exams. She stated in her interview,

> "I actually do have a review session like before the exams, two exams, final and midterm, before the exams. I do have a review session that I have a couple of exercises I ask students to do, well, that's synchronous. But meanwhile, it's not enforced for student attendance because you know it's an online course. I cannot say you have to come at a certain time. But to my surprise, usually there are like two-thirds of the classes there. If they want to get a good grade, they sort of want to know, you know, before the exam, they want to know if I give them practice questions, they sort of get a sense of what's gonna be tested, you know, like what kind of content it's gonna cover."

According to her Self Review comments for SRSs 5.2 and 6.2 during training, CPSC-3 appears to have already been doing this.

In her asynchronous online courses, IDSS-1 offers her online students the option to attend scheduled sessions for her in-person sections to increase student interaction and understanding. She states in her interview,

> "The online course is 100% asynchronous, but what I do is try to alleviate that and assist the students. I offered them what I call live informational classes … They are not required, but they are highly recommended, right? And in (the online) platform, I share with them what is going to be offered in each (in-person) class. So for example, if a student is struggling with trying to understand, well, I don't understand how to research based upon one discipline, right? Then they have the ability to sit in on that particular (in-person) class where I can walk them through the process. Those (in-person sessions) are where they actually interact with me live."

IDSS-1 did not make any comments for SRSs 5.2 and 6.2 during training to verify if her interview statements indicated any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time.

In his in-person classes, ENGL-1 has created a hybrid alternative for his in-person students. In multiple semesters, there have been students who prefer to work remotely, either due to distance or availability. Along with using Blackboard for his content, assignment submission, and grading comments, this essentially creates an unofficial online option. He stated in his interview,

> "There are some circumstances where there might be a student who is unable to attend in person, and I've usually tried to accommodate them by having Blackboard Collaborate open so like, the last time I taught the Shakespeare course, I did have one student who

> occasionally had to not, who was unable to be there in person, and we would do that relatively low tech. So I'd just do it through my laptop, and I'd have the laptop camera pointing out at the rest of the class, and it worked pretty well. But then in other, I've had in other classes, not Shakespeare, I've had students who are just never able to attend in person. So we moved to a Hyflex classroom, or whatever they call that … where they were to be properly integrated into the sound and visual system of the classroom, and their face will be projected on the projector, and they will be able to hear everything that was going on in the class."

ENGL-1 did not make any comments for SRSs 5.2 and 6.2 during training to verify if his interview statements indicated any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time. Based on the interview responses, this appears to have been a new idea thought of and then implemented after training.

Although not directly part of the IYOC, the variety of technology tools used is addressed in SRS 6.3, which is also related to addressing active learning in SRS 6.2. HIST-1 tries to provide students with multiple alternatives for interacting with content to help address different learning preferences. Accessibility and inclusion will be addressed in the next section. HIST-1 stated in his interview,

> "I can draw on documentaries, short YouTube videos, there are lots of maps, like environmental justice maps that show particular environmental justice hotspots around the nation or around the world. So all kinds of interactive materials, many of which are audio-visual that I can have students engage with. So the variety of materials is, you know, obviously a benefit. Different students are going to learn in different ways and

> take more or less meaning from material that is delivered in different ways. So, that's been the easy thing, is creating variety in the student learning materials. The challenging thing is to make sure that students are engaging with all of the learning materials in a way that is substantive."

As was indicated in his comments for SRSs 5.2 and 6.2 during training, HIST-1 stated how he intended to improve his approach to variety of student interactions,

> "In the future, I want to incorporate more VoiceThread assignments and occasional short lectures to frame topics. The lectures don't necessarily promote active learning, but they will be short and will seek to provide some contextualization/summary that my weekly course announcements currently do. My hope is to vary the tools enough that students feel engaged."

As we can see from this and the other previous examples, student interactions require a variety of materials as well as a variety of technology tools to foster engagement.

Purposeful Use of Technology

Based on the previously mentioned examples, participants show that in online courses, fostering all three forms of student interaction requires a variety of materials, activities, and technology tools to interact with. Because one of the main goals of the IYOC training was the improvement of student interactions, presenting opportunities for a variety of tools is reflected in the QM Specific Review Standards (SRS) of 6.3, although the promotion of active learning using course tools is reflected in SRS 6.2, which was the SRS taught during training. See Appendix A for the descriptions of the QM standards. But variety alone is not expected; rather, educators should use technology to support student learning when it serves a pedagogical purpose (Beggs, 2018; Casa-Todd, 2025). Meeting this standard is helped by providing activities and tools to

engage with those activities to foster all three forms of student interaction: student-student, student-content, and student-instructor.

Although many online courses are asynchronous, some of the participants opted to have some synchronous components, many times optional. Various tools have been available at this institution since COVID: Collaborate Ultra as a campus-wide Blackboard plugin, Webex provided by the institution's Cisco on-campus phone system, some instructors or programs have their own Zoom subscriptions, VoiceThread for multimedia discussions, Panopto for video hosting and to replace VoiceThread as a Blackboard discussion plugin, and the campus just recently adapted Microsoft Teams to replace Collaborate Ultra and Webex. As has been previously mentioned in the above themes, several participants are using these tools in some capacity, among others, to provide and record virtual synchronous sessions or narrated lectures. In order to best accommodate student interactions with content, CPSC-3 noted in her interview,

> "The other thing I learned is that when you make online videos, don't make them too long. Like student attention can be 10 minutes, for example, or 15 minutes maximum. Don't, you know, post like hour-long videos just like, you know, when you teach in person, you teach maybe an hour, right? And then take a short break. But in videos, you know, if you do that, students will just get bored or distracted easily if you do it that way. So I just chunk my subject into smaller sections and then group them into learning modules."

CPSC-3 also mentions the tools she has tried to address captioning within recorded lectures,

> "I never, to be honest, I never used Collaborate for the meetings outside the class. I used to use Webex, then I tried to switch to Teams. … Some of my videos I uploaded to YouTube so that it can provide automatic closed caption. One of the tools in Blackboard

> also does that, but it was not as good as YouTube. … The voice recognition is better in YouTube. So I upload those videos to YouTube, but I mark them as unlisted so only my students can see them, not to the public. But to be honest, I spend a lot of time still, even with YouTube, I spend a lot of time correcting those captions because of a lot of the technology terms I use in my lecture. I did that mainly because one semester, there was a disabled student who had a hearing problem. He has to read your mouth in order to sort of understand, read your lip, you know, to understand what you're saying. But during COVID, we don't have in-person classes. So you know, in the online version, it's kind of hard for him to understand. So that's why I did that, you know, to put the closed caption in my video. And also accent because I'm a foreigner, so it's easier for a student to understand me if I, you know, they can read."

CPSC-3 did not make any comments for SRSs 6.2 and 6.3 during training to verify if her interview statements indicated any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time. More on learning preferences will be discussed in the accessibility and inclusion theme in the next section.

One of the other approaches to narrating lectures is by SOC-1, who in her interview needed to clarify how she provided narration to her PowerPoint presentations. She records audio and inserts the narration on each individual slide, and students have to download the file, then click on the audio icon to hear the narration. As she stated in her interview, "It's just my voice, lecturing. So, hitting, like, the audio record button and then they'll have the slide and then hear me explain and read and add some additional points to the lecture text itself." In her interview she expressed that she wants to streamline this process by converting her PowerPoint files to

Panopto videos, but she did not make any comments for SRSs 6.2 and 6.3 during training to verify if her interview statements indicated any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time. More on intended future improvements will be discussed in the next section.

In addition to what was mentioned for alternative submissions for discussions above, IDSS-1 allows students to submit video presentations instead of written assignments if they choose to. As stated in her interview,

> "So in my classes I sometimes give them the option of either creating a video or doing a written assignment … We were using Screencast-o-matic. We were using VoiceThread. Now we're pretty much using Panopto. They also use YouTube. So, say for example, like the meet and greet (assignment), I require it to be video so people will be able to see the individual that they are actually listening to. And so in the process that we use, many students are comfortable with using their phone to create a video, and so I leave tools to show them how to upload the video to YouTube and then bring it back into the class as a link."

As was indicated in her comments for SRS 6.2 during training, IDSS-1 listed virtual meeting tools she intended to use to promote learner engagement: "Smartphone, Collaborate Ultra for virtual discussions, videos, unlimited use of the internet, email, and student-generated video production."

CPSC-2 also has an alternative method for student work submission due to the computer science subject area and as a method to discourage submissions of AI-generated code. As stated in his interview,

> "I have them submit screenshots. At the end of (a lab project), I want a screenshot showing me that you know that you did X, Y, and Z. So that is how I grade the course project. And in that course project, I'll also ask them some questions, saying, 'You know what this is,' or 'Give me your top ten commands that every Linux user should use,' and stuff like that. But so, it's a combination of me manually putting in the grades from the NetAcad grade book and them giving me screenshots from the virtual machine, and that tests that they're (actually) doing the course project. … Everything is screenshots because to me they're showing that they're actually doing this stuff, and that is where I'm like this was your initials, so that way it's a little bit easier to determine if there's like plagiarism going on, and then everything was going with AI, then there are concerns about that. But this, you know, there's really no getting around, not doing the work. You have to do the work."

CPSC-2 did not make any comments for SRSs 6.2 and 6.3 during training to verify if his interview statements indicated any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time.

Because of dissatisfaction with the textbooks available, PHYT-1 has developed his own online content document that he shares with students and regularly updates. As stated in his interview,

> "I don't care for either of those textbooks too much, and I haven't found a good one. So I've basically written a virtual-type textbook that has embedded links and things like that from the (course) for them to go to different websites or interactive, you know, type modules or things like that. It also has, you know, specific readings that I want them to do

> for the book, but it's basically a big, long Word document that they read prior to, like, the imaging class, prior to our synchronous sessions."

PHYT-1 did not make any comments for SRSs 6.2 and 6.3 during training to verify if his interview statements indicated any perceived improvements. If no comments were made during training, the participant believed they were already implementing these QM standards at that time.

At this institution, all students are automatically enrolled in a Student Orientation course in Blackboard every semester. The purpose of this is to introduce students to the commonly used Blackboard tools, including content items, discussions, tests, multimedia tools, and grading. In order to promote familiarity with the online learning environment, and also due to the perceived age and lack of tech-savvy from her graduate-level students, NURS-1 requires taking the Student Orientation for a grade. As stated in her interview,

> "Because most of the students are older students … one of the things I had them do, they're required to complete the Blackboard training and get the certificate, so that they know how to navigate throughout Blackboard, just because some of them are not savvy when it comes to computer skills. Generally, the average student is somewhere in their 40s or 50s. We have a few younger ones that know how to navigate, but the majority of the students are older because they're already (registered nurses), and so they're just coming to gain that (Bachelor of Science) degree because they (already) have an associates degree in nursing."

NURS-1 did not make any comments for SRSs 6.2 and 6.3 during training to verify if her interview statements indicated any perceived improvements. If no comments were made during

training, the participant believed they were already implementing these QM standards at that time.

Because of specific industries some courses might be preparing students for, online courses may require specific technologies not available as a campus-wide subscription. This was true for health care subject areas and computer science courses, where specific technologies might be linked to as part of the content delivery and interaction. As stated in the interview for NURS-2,

> "This course is designed for the graduate program at the master's level where the students are anticipating to work as a director, or as a manager, or as a chief nurse executive at any of the hospitals. In order to do their work efficiently, they need to know certain technology parts of it, and that's what this course teaches … There's a lot of healthcare technologies that is out there, including artificial intelligence where they use ChatGPT, electronic health records, boss code scanning for medication administration. So this course emphasizes on that part on that aspect of technology for the nurse executives."

As stated in the interview for PHYT-1,

> "For the imaging class, some of those links take them to websites like Radiopedia, which has interactive imaging tools like (magnetic resonance imaging) and (computed tomography scanning), where they can scroll through and look through it and things to that extent. … In our synchronous sessions in the imaging class, since some of the imaging software does not, you know, you have to pay for it or things like that, I've gone through, I've showed videos of, you know, me accessing through the hospital that I work at, how I access MRI's and how I scroll through them and read them."

Neither NURS-2 nor PHYT-1 made any comments for SRSs 6.2 and 6.3 during training to verify if their interview statements indicated any perceived improvements.

Specifically true in computer science courses is purposeful use of technology because students are inherently dealing with technology as part of the course's subject area. For the three CPSC instructor participants, there happened to be some common tools used through the vendor Cisco, namely NetLabs/NetAcad. As was mentioned in his method for organizing Blackboard content mimicking the Cisco online tutorial environment, CPSC-1 mentioned in his interview,

> "We have the capability of the students not doing their own virtualization, but using a system called NetLabs, which is online cloud-based virtualization for them that gives them a playground (for ethical hacking) and locks them into a scenario where no matter what they do, they can't affect the real world or blow up their computer."

CPSC-2 also uses NetAcad in his coding course, but also uses another simulation software that is used professionally within the industry. As stated in his interview,

> "And then we do use VMware, specifically VMware Workstation, for the course projects, because when they do their course project that's not in NetAcad, that's something that they would be doing. But then I also post detailed instructions as well as a video explaining how to create their machine and how to work with it. So that would be the only additional software beyond Blackboard, NetAcad and VMware, which is all a free download anyway."

CPSC-3 also mentioned in her interview the use free coding tools, "I just asked them to use IDLE, which is like a scripting kind of tool that's part of the Python package, like when you download Python." None of the computer science instructors made any comments for SRSs 6.2

and 6.3 during training to verify if their interview statements indicated any perceived improvements, but it was verified in their Blackboard shells that these tools are being used.

As has been mentioned previously in this theme, AI is being used, for good or for bad, by students when submitting their work. Sometimes it is expected, other times it is not allowed. Each instructor at this institution has the choice whether or not to allow AI, and some departments do not want it used at all. As ENGL-2 stated in her interview,

> "Overall, the English department has been looking at students using generative AI as an absolute no-no. And the research that I did over the summer, as AI is just exploding worldwide, the professional writers are using it, everybody's using it. And we have had an explosion, particularly in the last two years of students, quite frankly, cheating with AI. You know, going ahead and allowing AI to write their papers, answer their questions, and you know, it's pretty easy to detect, but it's become extremely problematic. We had a writing retreat with the English department … where the main focus was talking about AI. And I spent a good part of the summer, I think I reviewed 18 AI books over the summer, and I tried out about 10 AI tools. What I was looking for was what would be the best as well as the most navigable for students. Particularly, the freshmen coming in, and I ended up settling on Claude AI, which is, you know, from what not just what I've read, but from my own experience, is now working with it, the best AI tool for writing, and I actually used Claude quite a bit. … The reason that I wanted to do this is that I feel that the best way to curtail unethical use of AI is to teach them how to use it properly. And so instead of, you know, hiding, you know, hiding our heads in the sand, I think we have to meet this. I mean, students are going to use it whether we want them to or not. So let's teach them how to use it the right way."

ENGL-2 did not make any comments for SRSs 6.2 and 6.3 during training to verify if her interview statements indicated any perceived improvements, but the topic of AI was not as prevalent when she took the IYOC training in November 2021.

Because of the changing nature of technology and the wide variety of tools being used by all of the participants in this study, many of the uses of technology in this theme could not be checked beyond verifying within Blackboard shells that specific tools were present. Some of the technologies mentioned during interviews or in training comments are being deprecated from the institution, either because subscriptions to some tools are being cancelled, or other tools are going through upcoming upgrades. Some of the participants mentioned the need to learn these new technologies. More on intended future improvements will be discussed in the next section.

Holistic Themes for Research Question 2

Research Question 2 asks how participants perceived what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery. Responses to interview questions answering this question yielded the following themes: presentation and alignment of learning objectives, intended future improvements to course design and delivery, addressing accessibility, and transfer of training concepts to other course modalities.

Presentation of and Alignment with Learning Objectives

Because one of the main goals of the IYOC training was the improvement of alignment to learning objectives in courses, presentation of learning objectives to students is reflected in QM Specific Review Standard (SRS) 2.4, and alignment of content, activities, and assessments is reflected in SRS 4.2. See Appendix A for the descriptions of the QM standards. SRS 2.4 expects that the relationship between learning objectives, learning activities, and assessments is

made clear. SRS 4.2 expects the relationship between instructional materials and completion of activities and assessments is clearly explained. Meeting each of these standards is helped by presenting course learning objectives, usually within the syllabus and course orientation area, and then describing alignment to students within the course content, activities, and assessments. Instructors had many approaches to demonstrating alignment within their Blackboard shells based on their personal teaching styles and the learning objectives for their respective programs of study.

According to HIST-1, organizing a content within a course has be done with alignment in mind. As stated in his interview,

> "I think the concept of alignment, of trying to align everything from, you know, broad philosophical goals to course level objectives to specific assignment level learning outcomes and, and objectives, that's been probably the most valuable part of Quality Matters. Most people, I guess, would probably assume that that's the most obvious thing, but it's also challenging to implement, right? Going through this, this process of self-evaluation, how do I align literally everything in the course, whether it's the assignments themselves or even the learning materials that students are reading or otherwise interacting with? It's obvious, but it's a huge challenge. I don't think the course that I've designed is perfect yet, but that's been, I think probably for me, the most substantial thing that has come out of Quality Matters is just trying to think consistently about alignment."

As was indicated in his comments for SRS 4.2 during training, HIST-1 believed he was already addressing alignment in his activities and assessments, but also needed to "provide explicit instructions connecting the assignments to the course (objectives) that will be relevant to an individual assignment."

One way some participants approached alignment is by creating a course map. Course maps are introduced in IYOC as a possible tool instructors can use to align course-level and module-level objectives to course content, activities, and eventually the assessments they will be graded on. NURS-1 states in her interview,

> "(Electro-Physiology student learning objectives)… I use all those mapping concepts, even in my face-to-face classes. They want us to put our objectives. So let's say through a discussion board (assignment), what objective was met? We have to put the numbers (of the objectives in the description) … mapping the course objectives with the EP SLOs and with the program objectives, which is it's tough, but the end result is good."

As was indicated in her comments for SRS 4.2 during training, NURS-1 expressed her need to improve alignment by explaining "more clearly how the instructional materials in the course and completing learning activities relationship is mapped out by creating a table or graphic display." CPSC-3 also uses alignment to organize learning modules. As stated in her interview,

> "So every course, like every lecture, I will break that into modules. So, for example, if a lecture has three modules in the Blackboard, there will be three separate modules, so students don't get overwhelmed with all the materials posted, you know, so they're going to be categorized into three modules. And at the top of each module, I will list out all the learning objectives of this particular module and how the activities, you know, match the learning objectives of. The entire course, you know, there's like a simple mapping, maybe just three to four objectives there for each module, and then they map to the learning objective of the entire course at the top of each module."

CPSC-3 goes on to talk about how objectives are introduced to students in the beginning of each module. Continuing from her interview,

> "Mapping of the learning goals or learning outcomes on each module. That was one of the things I never thought about before I took that workshop. Sort of gives you like a bigger picture of a clear idea of what you're learning today. Rather than just, you know, dive into the material directly. I first give them an overview so that they have a clear goal of what they are gonna achieve."

According to her Self Review comments for SRSs 2.4 and 4.2 during training, CPSC-3 appears to have already been doing this.

IDSS-1 also presents alignment during her synchronous sessions at the start of each module. As stated in her interview,

> "So one of the additional things that I do on our first meeting is I make sure that as I do presentations for them (at the start of each module), I share with them what is the module objective and how it relates to the course objective and they see that live in living color in the presentation."

According to her Self Review comments for SRSs 2.4 and 4.2 during training, IDSS-1 appears to have already been doing this.

Demonstrating alignment is not limited to being presented at the start of the course or the start of a module. NURS-2 has an assignment at the end of the semester asking students to reflect on how they met the course learning objectives. As stated in her interview,

> "Students are given a reflective statement (assignment) to fill out at the end of the course. They do it from their end saying we met this objective by doing this activity. Required to write the reflective statement in conjunction with the objectives."

NURS-2 did not make any comments for SRSs 2.4 and 4.2 during training to verify if her interview statements indicated any perceived improvements. If no comments were made during

training, the participant believed they were already implementing these QM standards at that time.

Presenting alignment helps students see the why behind what they are being asked to do. Regardless of alignment, what students are being asked to do may not necessarily be what they are able to do or presented in ways that they would prefer to experience the content.

Addressing Accessibility

Given the legal requirement for all public entities, including higher education institutions, to be compliant with federal Title II accessibility regulations by April 2026 (Rudra, 2025), accessibility and inclusion is a high concern for instructors. Because one of the main goals of the IYOC training was the improvement of student interactions, accessibility and inclusion, and support resources, is reflected in the QM General Standards 7 and 8. See Appendix A for the descriptions of the QM standards. Although these general standards were covered in the IYOC training, and interview questions did not directly ask participants for comments about these standards, many participants in this study were concerned and specifically commented on this theme.

The first concern for participants was the legal requirements in online vs. in-person courses. NURS-2 expressed concern over being able to tell if a student needed accommodations, and then how to address those accommodations based on the modality the student was registered in. As she stated in her interview,

> "Problem about accessibility issues. I don't know if there was a student with a disability would come to my face-to-face class. Like I said, I have synchronous and asynchronous sessions. If they were to come to my class, I have no clue how I will incorporate the accessibility piece into it. I have no idea. I need to know more about it."

HIST-1 acknowledges having to meet legal requirements when presented with official notification of required accommodations, but then started thinking about to be more inclusive, especially when having to worry about giving students extra time for assignments or the entire course. As he stated in his interview,

> "I would say that Quality Matters helped me to start thinking about student supports in ways that are probably more concrete. So prior to doing Quality Matters training, if I would get something from the (accessibility department) for students with disabilities, I'd get, you know, the accommodations letter at the beginning of the semester. I would try to make sure that I was meeting students' needs, you know, not only for ADA compliance, but just to make sure that students were able to get through the course in the way that everybody else did. But after Quality Matters, I started thinking about, okay, so what is necessary to really effectively translate all of these materials to students that might have or require different accommodations the most? The two most common things that I've seen are that students quite often will, for a variety of neat reasons, need extra time on assignments, so flexible deadlines, but also like very clear statements about the flexibility of deadlines and the way that giving students enough time to complete work actually better meets our goals than just having one single hard and fast deadline. And this is something that (the former dean) would talk to me about is, you know, the semester being 15 weeks is arbitrary. We want students to learn if they, it takes them 17 weeks to learn, what we say should be 15. That's just the randomness of saying that semesters are 15 weeks long. There's no, there's no logical reason why it's 15 rather than 17. And that stuck with me. And Quality Matters, I think that made me start thinking really more concretely about, okay, so what? What does accommodation look like when we have

> students who may not even have a documented disability or an accommodation letter when they need extra time? How do I build that into the course and then make it very explicit that extra time is okay? Asking for a little grace is fine. And actually being an advocate for yourself better helps us to meet learning outcomes. So stating that all a little bit more explicitly in syllabus or in Blackboard announcements has been, I think you know, something that I've taken from Quality Matters."

HIST-1 then went on to express concerns about the readability of his content. He continues from his interview,

> "And then learning materials themselves, like I have occasionally, students with visual impairments who need to use software that translates text to speech. And so taking PowerPoint lectures, which for me I've got like pretty visually heavy because I'll do lectures, you know, essentially off the top of my head at this point, like I don't have written notes too much. I can usually, I've taught this so many times that I can lecture without notes or with very minimal notes. So, figuring out how to take this PowerPoint that has evocative images and a few words and then turning it into something that, you know, text-to-speech software can actually translate to the students. That's been quite an undertaking. And that was prompted by Quality Matters to really think about, okay, well, how do I actually make this (content) accessible?"

COMM-1 also expressed concern with being able to check the readability of text. As she states in her interview,

> "I'm trying very hard to make sure that I'm using the right headers and you know, the right level of things so that anybody who's using a reader or something that, when reading to them, will pick it up correctly. I find that to be the hardest because I can't

always see when I've not done it. You know, like when I can't tell if the header is the right level or whatever it's supposed to be."

IDSS-1 realizes that it is not only the reactive nature of accessibility that instructors need to be concerned about, but also the proactive nature of anticipating different student learning preferences. As she stated in her interview,

> "I was very excited about when I took the (training) course to learn these new policies, and I appreciated the whole thought process around the accessibility, around the fact that students learn differently, and so … I give them an opportunity, some students are great writers, and other students express themselves differently. That does not mean that they didn't learn the content, right? And so I'm real conscious about allowing new avenues to express their learning capabilities."

Because this specific theme was not directly asked about during the interviews, and also because of the challenge of checking multiple courses for accessibility compliance and the lack of tools available for that task at this institution, this theme could not be verified within training comments or within Blackboard shells. But, given the legal importance of this issue and the impending federal deadline for compliance, this is a valid future concern that these participants acknowledge the importance of improving.

Intended Future Improvements to Course Design and Delivery

Because one of the important concepts of the IYOC training was the continual improvement of courses, participants were asked if what was learned in training was applied to other courses beyond the courses commented on during training. See Appendix C for the interview questions. Improvements made between training and when this study took place have

already been documented and verified in the previous themes. In addition to the accessibility issues, here are the future concerns from the participants.

First and foremost would be technology changes. This institution is currently undergoing an upgrade to the Blackboard learning management system which affects both the functionality and the cosmetic interface. This requires technology training in addition to best practices for most faculty outside of this study. CPSC-1 was commenting on his own subject area, but the comment also rings true for online course design as well. As he stated in his interview, technology "will change over time because, unfortunately, and in our industry, something can change overnight, and the whole screen can look different next time."

Specific technology changes at this institution include video technology. Due to changes in virtual meeting platform subscriptions across campus, Webex by Cisco is being replaced by Microsoft Teams, and the subscription to the Collaborate Ultra plugin within Blackboard is going to be replaced with a Microsoft Education plugin, which includes Teams and other interactive applications. Due to low usage campus-wide, the subscription VoiceThread, an application that allows for multimedia responses to discussions, has been cancelled. VoiceThread has been replaced by Panopto, an application that hosts videos like YouTube, allows for multimedia responses from students in all assignments, and integrates a quizzing feature with the Blackboard gradebook. Addressing having to adapt to these changes in video technology, SOC-1 commented in her interview,

> "(I've) been using Panopto more, getting more comfortable with that, and it seems, yeah, that's available … in our Blackboard. So it's, I'm getting, it's getting better. I had some technical issues with it last year, so there were some issues. So I had just switched to Microsoft PowerPoint and then MP4 files, but that was like taking up too much space (on

the Blackboard server). So I'm getting much better at it and requiring students to also only use the integrated Panopto recording."

As is important to any subject area, HIST-1 addressed concerns about applying activities and technology tools to real-world applications beyond the classroom. As stated in his interview,

"Especially if we are doing a kind of a real world project, say something like the (recent local) water crisis, if I were going to map that on to an online-only First Year Seminar, then one assignment we do in that course is a media campaign where we try to figure out a way to raise awareness about the water quality issues in (town) or relatedly, the water rate increases in (town) and other (local) communities. We develop a media campaign to address one particular problem. So I could create surveys for students to actually vote on what they think we should focus on. Now we have to vote on what it is we as a class are going to do for our media campaign. So I think surveying is something I would be that I'd be really interested in doing moving forward because the way I want to translate the course, if the second half of my course ends up being essentially a workshop. We do a media campaign, and the students are kind of in the driver's seat."

PHYT-1 has given thought to addressing IYOC training concepts in multiple modalities, but the challenge comes when courses he designs are then taught by others, or possibly co-taught. As stated in his interview,

"I have (applied training concepts to other modalities), but probably not as well. It's something I still think about and I want to do at some point, but time-wise, I haven't gotten to the point that I want to yet. Plus, many of the face-to-face classes I teach, we co-teach in our department. So sometimes it's a little challenging 'cause you're always working with somebody else to align with that way. I think I have done it better since

> taking (training), but it's something I think I do better in an online format and I have to work on better in a face-to-face aspect."

Because many of the concepts from IYOC could also apply beyond the online modality, participants were also asked if any IYOC training concepts were applied to their in-person or hybrid courses as well. See Appendix C for the interview questions.

Transfer of Training Concepts to Other Course Modalities

Transference of training to practice was not limited to online course design and delivery. Transference also occurred to other teaching modalities by all participants. Even though the IYOC workshop focuses on best teaching practices for the online modality, many pedagogical elements can also be applied to in-person or hybrid courses. As has been noted in comments about the different previous themes, participant comments noted implementation beyond just the online modality. This shows that there may be benefits of future professional development efforts for students, educators, and institutions alike, in any course modality. More about conclusions based on the data will be shown in the following chapter.

Some instructors have made some adjustments to courses in other modalities, utilizing the same setup for external tools relevant to their subject area. An example of this comes from CPSC-1, as stated in his interview,

> "I give (students) the link (in Blackboard), and they go to a separate site. … The grading of and submission of labs (from Cisco) and any papers are all in Blackboard. So Cisco provides the material. I provide the basic structure for the grading and assignments in Blackboard, and all my courses or all my sessions follow the same procedure."

The concept of alignment and mapping of learning objectives to content and activities has been applied to other modalities. As NURS-2 stated in her interview, "(the Electro Physiology student learning objectives … I use all those mapping concepts, even in my face-to-face classes."

Tools usage within Blackboard is used in multiple modalities. This has been noted in previous themes for interaction and also for grading. NURS-2 also uses tools within Blackboard for her in-person courses, continuing, "so when I do face-to-face traditional class, I use journals, I use (online and open resource) textbooks, … I use the Blackboard and teach them. I have no problems." HIST-1 notes about his discussion usage, "in the in-person course, certainly for the online courses, I think I rely … on the Blackboard discussion forum pretty heavily." And IDSS-1 comments on training concepts and usage of groups in multiple modalities from her interview,

> "I've actually applied (training concepts) to my capstone (course), and it works beautifully with the capstone. Many of the interdisciplinary courses I integrate group projects, for example, in them all. And I actually have applied the same concepts to all of the courses that I teach online. (Another) class that I actually taught, you know, was a hybrid. And I absolutely did apply those concepts to that particular course also."

Potential Embedded Analysis of Participant Groups

There was a potential for an embedded analysis in this study, but the small sample size and lack of data within subgroups did not make an embedded analysis feasible at this time. A potential breakdown for analysis will be described here. While there were only 12 participants in this study out of a possible 21 who completed IYOC training and are still employed at this institution, there were three distinct groups of participants based on subject area that could allow for a potential embedded context as defined by Yin (2018). Within this case study, there was a sub-group of three Computer Science instructors, a sub-group of two English and Literature

instructors, and a sub-group of three instructors from the health care fields of Nursing and Physical Therapy. See Table 3 for the coding for instructors and corresponding subject areas.

Table 3

Of the 12 instructor participants, these three subject-area sub-groups present an opportunity for an embedded analysis in future studies.

Instructor Codes	Subject Area
CPSC-1 *CPSC-2* *CPSC-3*	Computer Science (includes Information Technology)
ENGL-1 *ENGL-2*	English and Literature
NURS-1 *NURS-2* *PHYT-1*	Health Care (includes Nursing and Physical Therapy)

Because of the small sample size, there are only a few similar themes between members of these sub-groups worth noting, but not enough similarities between this set of sub-groups to do a full embedded analysis. The small sample size and lack of richness of data from the sub-groups do not make a true embedded case study design feasible (Yin, 2018). The breakdown of these sub-groups possibly presents an opportunity to analyze larger numbers from distinct subject areas in a future study.

Embedded Themes for Research Question 1

Research Question 1 asks how higher education instructors perceive any changes to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop. Responses to interview questions answering this question yielded the following themes: organization of content within the learning management system, utilization of discussion activities, implementation of all three forms of student interaction, and purposeful use of technology. These themes will be broken down by subject area sub-groups.

Without surprise, given the subject area, the most dominant theme of the Computer Science sub-group was the purposeful use of technology. As described in the holistic analysis, all three members noted in their interviews that they use similar setups for assignment submission, utilizing technology tools for student interactions, and linking to online supplements and labs, specifically Cisco. The only utilization of discussion was noted in the interview of CPSC-1.

Like Computer Science, the most dominant theme in the English and Literature sub-group was the purposeful use of technology. Also significant was the implementation of student interactions. ENGL-1 has unofficially converted his in-person courses to hybrid to accommodate student availability, and regularly uses group projects and discussions. ENGL-2 uses peer review groups and discussions, records virtual one-on-one conferences and had openly embraced using AI with lessons.

Also like Computer Science and English and Literature, the most dominant theme in the Health Care sub-group was the purposeful use of technology. Also significant was the implementation of student interactions. All three implement technologies that are relevant to what students will be using professionally. All three also use a variety of student interactions using both synchronous and asynchronous virtual sessions and group projects. Of note, PHYT-1 has developed his own online text supplement for all modalities, and NURS-2 has an end-of-semester reflection assignment on how students perceive meeting the learning objectives.

Embedded Themes for Research Question 2

Research Question 2 asks how participants perceived what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery. Responses to interview questions answering this question yielded the

following themes: presentation and alignment of learning objectives, intended future improvements to course design and delivery, addressing accessibility, and transfer of training concepts to other course modalities. These themes will be broken down by subject area sub-groups.

The most dominant theme of the Computer Science sub-group was the transfer to other modalities. As described in the holistic analysis, all three members noted in their interviews that they use similar setups for the organization of content within Blackboard and linking to online supplements and labs, specifically Cisco. The only future improvement theme was noted in the interview of CPSC-1, acknowledging that technology is in a constant state of change.

Like Computer Science, the most dominant theme in the Health Care sub-group was the transfer to other modalities. All three members described how they utilize similar organization, tools usage, and presentation of materials between modalities, but PHYT-1 noted in his interview about the challenges of collaborating with other instructors between modalities. NURS-1 also noted the application of alignment mapping to all modalities she teaches and future concerns about accessibility.

The English and Literature sub-group did not have any similarities in themes worth noting. Because of this and the small sample size within these sub-groups for both research questions, a full embedded analysis is not appropriate for this study. A lack of richness of data for sub-groups would not make an embedded case study design feasible (Yin, 2018), but this could be the basis for a possible future study with a larger sample pool and the potential for sub-groups with more similarities to analyze.

Summary of Findings

The purpose of this qualitative case study is to examine the perceived effectiveness of the implementation of professional development for best practices in online course design by higher education instructors to determine if transference of knowledge has taken place from training to professional practice. Participants in this study were previous participants in the official Quality Matters™ (QM) Improving Your Online Course (IYOC) workshop as offered through the Center for Teaching and Learning (CTL) at a Midwest regional mid-sized institution. The researcher was the facilitator of the IYOC workshops and is also a certified QM course reviewer.

This study utilizes a single-case retrospective case study design. Retrospective case studies, sometimes called historical case studies, try to determine changes over a period of time using existing data or documentation to analyze patterns, causes, or lessons learned (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998; Ozcan et al., 2017), and this existing data can be compared to interviews to interpret findings (Goyes & Sandberg, 2024). Participants first answered a series of reflection questions in individual interviews. See Appendix C for interview questions.

Next, potential themes were determined in the researcher's journal, and then quotes were identified in participant interview transcripts using ATLAS.ti software. ATLAS.ti software was used because it is a valid evaluation tool for the organization and analysis of themes (Deterding & Waters, 2021; Saldaña, 2016). Then, commentary from the IYOC Self Review activity was gathered and compared to interview responses, also using ATLAS.ti software; because the Self Review tool used in the IYOC workshop also allows for reflection and justification by the participant and reviewer at the time of training, this qualitative information can be used for comparison to interview questions during this study. Potential changes in teaching practices for online courses can be shown through opportunities for instructor reflection (Ali & Wright, 2017),

which the interviews and comments from the Self Review exercise during training provided. Comparing direct responses to previous responses is a form of methodological triangulation for validation of research (Stake, 1995). Improvements to student interactions were the focus of the IYOC workshop. Implementation of those potential improvements in courses, as noted in the interview responses, helped answer Research Question 1. Comparison to intended improvements noted in the IYOC Self Review activity indicates that transference from training to practice has taken place, helping answer Research Question 2. See Table D1 for a summary of the perceptions of the participants in both training and in this study.

Research Question 1 asks how higher education instructors perceive any changes to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop. Responses to interview questions answering this question yielded the following themes: organization of content within the learning management system, utilization of discussion activities, implementation of all three forms of student interaction, and purposeful use of technology. All participants used modules to organize their content in a way that made sense for their teaching styles, with the exception of one instructor, COMM-1, who uses discussions not only for interaction but also content organization. Many instructors also use discussions for student interactions, either with the Blackboard discussion tool, other multimedia tools, optional or required virtual synchronous class meetings, or virtual individual conferences. One instructor, ENGL-1, records virtual conferences in Teams so students do not have to worry about taking notes on feedback. Participants recognize that activities implemented and tools used foster up to all three forms of student interaction, and activities can have more than one. Many instructors utilize group projects organized within Blackboard. Technology is purposefully used, either using Blackboard's built-in tools or external tools, and for the computer science and health care

subject areas, this is significant because the tools used are relevant to their respective industries. NURS-1 requires her students to take the institution's Blackboard Student Orientation Course as a graded assignment.

Research Question 2 asks how participants perceived what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery. Responses to interview questions answering this question yielded the following themes: presentation and alignment of learning objectives, intended future improvements to course design and delivery, addressing accessibility, and transfer of training concepts to other course modalities. Each of the participants recognized the value in alignment of content, activities, and assessments to learning objectives within their courses, and in several cases alignment was the basis of course organization. In the case of CPSC-1, he uses the Cisco online lab environment to mimic how he organizes content within his Blackboard shell because the Cisco exercises are aligned to the course's learning objectives. Many of the participants expressed concerns over future improvements, due to either content needs or technology upgrades being implemented at the institution. These future improvements will affect their course design and student interactions. As CPSC-1 noted, when it comes to technology, changes occur frequently, and it is the responsibility of instructors to adapt accordingly. One of these adaptations expressed by many participants was the legal requirement of accessibility, which as a concept was introduced in training but is now a major concern due to federal legal requirements by public institutions by April 2026, only a semester away at the time of this study.

Another adaptation related to Research Question 2, transference of knowledge was not limited to the courses commented on during training, but also transferred to other online courses, and even other modalities. Even though the IYOC workshop focuses on best teaching practices

for the online modality, many pedagogical elements can also be applied to in-person or hybrid courses. Interview responses and verification within Blackboard shells show implementation beyond just the online modality. Many instructors use the same organization within Blackboard because they teach the same course in different modalities. In the case of ENGL-1, he implemented hybrid delivery options for students in his in-person sections based on their availability. In the case of IDSS-1, she offers the option for online students to come to the in-person section for additional interaction and support. Because this training showed benefits beyond the intended audience of online instructors, the results of this study also reflect the possible benefits of future professional development efforts in any teaching modality. This can affect instructor trainings, student achievement, and institutional leadership choices.

In this study, Kirkpatrick's Level 3 provided a model to evaluate faculty perceptions of student performance and satisfaction in this training. Furthermore, because the training relates to a perceived value change for this institution in terms of student performance and satisfaction because of faculty participation in QM training, then the administration may wish to scale up the QM training initiative. Future studies could be conducted utilizing Kirkpatrick's Level 4 to show any possible relationship between QM training for faculty and actual student performance in online courses such as course grades and evaluations on individual assignments. More cross-case analyses or studies involving larger groups could show more institution-wide results and trends.

The practical implications relate to understanding how faculty participants are influenced by the IYOC workshop, as well as QM's role as a provider of professional development. This study is considered heuristic because it explains whether an innovation worked or failed to work (Merriam, 1998). This institution's administration decided to begin a subscription to Quality

Matters™ in February 2020, before there was even a thought of a possible forced nationwide shutdown due to COVID. Administration already saw a potential value in utilizing QM as a training resource to help improve faculty understanding of best practices for teaching online courses. Through the evaluation of QM training at this institution with this study, the training has proven to be successful in faculty attaining better knowledge about online course design best practices. A recommendation for leadership will be to continue to subscribe to QM and encourage future offerings of QM training at this institution, knowing it is meaningful. More about conclusions based on the data, possible future studies, and recommendations for leadership at this institution will be shown in the following chapter.

CHAPTER V

DISCUSSION

The topic of this study is the effectiveness of professional development in online course design best practices. This study is necessary because many higher education instructors believe that online learning is a form of modern correspondence, assuming that by placing content in a learning management system, students can learn independently (Joosten & Cusatis, 2019; Robinson et al., 2017). When instructors take this approach, they unfortunately neglect the fact that learning also occurs through other forms of interaction. Best practices for online course design and delivery include not only student-content interaction but also student-student and student-instructor interaction (Kendall & Kendall, 2014; King, 2003; Meyer & Marsick, 2003; Nilson, 2003). To introduce higher education faculty to the best practices for the online modality, professional development is necessary (Garrison, 2017; Hodges & Fowler, 2020; Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Li et al., 2023; Luongo, 2018; Nilson, 2003; Van Wart et al., 2020; Webster, 2025). The goal of this qualitative study is to examine perceptions of implementation from higher education instructors after participating in professional development for best practices in online course design and delivery.

The purpose of this study is to evaluate if and to what degree transference from professional development to regular practice has taken place for higher education instructors. Participants in this study were higher education faculty at a Midwest regional mid-sized institution who participated in the Improving Your Online Course (IYOC) workshop as developed by Quality Matters™ (QM), an international standards organization in online course design and delivery best practices. Professional development for educators in general, in addition to being specifically for effective online course design, can lead to improved student performance in addition to increased positive attitudes toward the online learning modality by

instructors (Burridge & Carpenter, 2013; Darling-Hammond et al., 2017; Davies & Preston, 2002; Gregory et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Zimmerman et al., 2020). Professional development tools and standards organizations are available to provide pedagogical training and evaluation tools for online course designers and instructors, including Quality Matters™ (QM), among others (Johnson et al., 2019; Zimmerman et al., 2020). QM is a sound, research-based professional development provider to increase interaction in online courses (Joosten & Cusatis, 2019), and increased interaction and collaboration within online courses not only mimics but also enhances in-person course experiences as well (Gregory et al., 2020; Joosten & Cusatis, 2019; Robinson et al., 2017). Implementation of what was learned in IYOC is was evaluated using Level 3: Behavior of the Kirkpatrick Model for evaluating training. The researcher was the facilitator of the IYOC workshops and is also a certified QM course reviewer.

Summary of Findings

This study utilizes a single-case retrospective case study design. Retrospective case studies, sometimes called historical case studies, try to determine changes over a period of time using existing data or documentation to analyze patterns, causes, or lessons learned (Bogdan & Biklen, 2006; Hess, 2004; Merriam, 1998; Ozcan et al., 2017), and this existing data can be compared to interviews to interpret findings (Goyes & Sandberg, 2024). Participants first answered a series of reflection questions in individual interviews. See Appendix C for interview questions. Refer to Table 2 for the coding of instructors and their corresponding subject areas.

In order for Level 3: Behavior to be evaluated, critical behaviors need to be identified (Kirkpatrick & Kirkpatrick, 2016). These critical behaviors were the potential themes determined in the researcher's journal, and then quotes based on those themes were identified in

participant interview transcripts using ATLAS.ti software. ATLAS.ti software was used because it is a valid evaluation tool for the organization and analysis of themes (Deterding & Waters, 2021; Saldaña, 2016). Then, commentary from the IYOC Self Review activity was gathered and compared to interview responses, also using ATLAS.ti software; because the Self Review tool used in the IYOC workshop also allows for reflection and justification by the participant and reviewer at the time of training, this qualitative information can be used for comparison to interview questions during this study. Potential changes in teaching practices for online courses can be demonstrated through opportunities for instructor reflection (Ali & Wright, 2017), which were facilitated through interviews and comments from the Self Review exercise during training. Comparing direct responses to previous comments is a form of methodological triangulation used to validate research (Stake, 1995). Improvements to student interactions were the focus of the IYOC workshop. Implementation of those potential improvements in courses, as noted in the interview responses, helped answer Research Question 1. Comparison to the intended improvements noted in the IYOC Self Review activity indicates that transference from training to practice has taken place, helping to answer Research Question 2. See Table D1 for a summary of the participants' perceptions in both the training and this study.

Perceived Changes in Student Interactions (Research Question 1)

Research Question 1 asks how higher education instructors perceive any changes to student interactions within their online courses after participating in the Improving Your Online Course (IYOC) workshop. Responses to interview questions answering this question yielded the following themes: organization of content within the learning management system, utilization of discussion activities, implementation of all three forms of student interaction, and purposeful use

of technology. The results show that course organization was improved to deliver content and promote student interactions through the effective use of technology tools.

When working with adult learners, how the instruction is designed will affect how the learner will interact with the content and delivery (York et al., 2016). All participants use modules to organize their content in a way that makes sense for their teaching styles, with the exception of COMM-1, who uses discussions not only for interaction but also for content organization. ENGL-1 specifically mentioned creating an orientation area labeled “START HERE" and then organizing content into modules. CPSC-3 uses modules to create manageable chunks for students. CPSC-1 creates modules aligned with the online lab supplements from Cisco. Although attempted, ENGL-2 believes the orientation area is confusing to students and is contemplating not using it in the future.

Many instructors also use discussions for student interactions, either with the Blackboard discussion tool, other multimedia tools, optional or required virtual synchronous class meetings, or virtual individual conferences. CPSC-1, NURS-2, and ENGL-2 each commented on their discussion or class interaction requirements each week. HIST-1 adjusts his discussion participation requirements based on age of students, noting that older students do not require as much encouragement to interact. HIST-1 and ENGL-1 also utilize multimedia options, either in addition to or as an alternative to typing discussion posts. NURS-2 requires attending synchronous sessions “to make discussions more lively” and recreate an in-person classroom experience. Both ENGL instructors use discussions as peer review tools for writing assignments. And NURS-2 utilizes a “Cyber Chat Box” discussion for students to ask questions and post announcements for the benefit of the other students.

In addition to interactions between students and content, Community of Inquiry and Constructivist theories view online learning as interactive collaborations between instructor and student and students and each other, where both can act as instructor and student simultaneously (Garrison, 2017; Robinson et al., 2017). So, because of the interactive nature of available learning tools, online learning is distinct from both traditional classroom learning and historical distance and correspondence learning (Garrison, 2017; Joosten & Cusatis, 2019). Participants recognize that activities implemented and tools used can potentially involve all three forms of student interaction, and individual activities can potentially involve more than one. COMM-1, because she uses discussions for organization and interaction, believes her discussion meet all three forms of student interaction because the deliver content and require students postings and comments, which she facilitates. ENGL-1 and HIST-1 use discussion tools external to Blackboard. PHYT-1 uses polling features during synchronous sessions to check for understanding. NURS-1, IDSS-1, and ENGL-1 utilize group projects organized within Blackboard. IDSS-1 promotes using whatever tools students may be comfortable with for completing projects, and then also encourages students to come to class meetings for the in-person sections running concurrently if they need additional support. ENGL-2 records virtual individual conferences so students can have a conversation and not be distracted by the need to take notes. CPSC-3 offers optional synchronous exam review sessions for her asynchronous courses. ENGL-1 unofficially converted his in-person sections to hybrid to accommodate student availability. And HIST-1 purposefully offers content in multiple formats, such as articles and YouTube videos, to specifically address learning preferences.

Higher education faculty who wish to be effective instructors in the online modality need specific professional development, in not only the technology tools but also the pedagogy and

best practices for purposeful course design to provide a quality learning experience in online courses for adult learners (Hodges et al., 2020; Johnson et al., 2019; Joosten & Cusatis, 2019; Kurt, 2016; Luongo, 2018; Nilson, 2003). Technology is purposefully utilized, either through Blackboard's built-in tools or external tools. CPSC-3 purposely chunks videos to 15 minutes or less to address short attention spans, and then edits captions due to the technical content and her foreign accent. SOC-1 attaches audio narrations to her lecture slides so students can read and listen. IDSS-1 gives students the choice to record videos or write assignment submissions. CPSC-2 requires screenshots of coding assignments to deter use of AI. PHYT-1 has developed his own virtual text due to dissatisfaction with publisher materials. NURS-1 requires students to take the institution's Blackboard Student Orientation Course for a grade. NURS-2, PHYT-1, and all three CPSC instructors use technology specific to the professions of their subject areas. And ENGL-2 teaches the ethical use of AI and proper procedures for use on writing assignments. Educators are increasingly expected to utilize technology tools, even though they and their students may not have access, the experience, or training to use them (Brancato, 2003; Johnson et al., 2019; Kurt, 2016), but educators who participate in technology-related professional development perceive that the experience positively impacts their teaching practice (Crews & Wilkinson, 2015; Gregory et al., 2020; Hollowell et al., 2017; Li et al., 2023; Shagrir, 2012).

Transfer of Training to Practice (Research Question 2)

Research Question 2 asks how participants perceived what was transferred from the Improving Your Online Course (IYOC) workshop to regular practice in their online course design and delivery. Responses to interview questions answering this question yielded the following themes: presentation and alignment of learning objectives, intended future improvements to course design and delivery, addressing accessibility, and transfer of training

concepts to other course modalities. The results show that alignment to learning objectives had a direct effect on content organization and development. The results also show that continuing changes in technology and the legal requirements of accessibility will affect future improvements to courses. The results also show that training influenced not only changes in the online courses commented on during training but also other courses in multiple modalities.

Because one of the main goals of the IYOC training was the improvement of alignment to learning objectives in courses, presentation of learning objectives to students is reflected in QM Specific Review Standard (SRS) 2.4, and alignment of content, activities, and assessments is reflected in SRS 4.2. See Appendix A for the descriptions of the QM standards. SRS 2.4 expects that the relationship between learning objectives, learning activities, and assessments is made clear. SRS 4.2 expects the relationship between instructional materials and completion of activities and assessments is clearly explained. Participants showed that they met each of these standards by presenting course learning objectives, usually within the syllabus and course orientation area, and then describing alignment to students within the course content, activities, and assessments.

Instructors had many approaches to demonstrating alignment within their course content based on their personal teaching styles and the learning objectives for their respective programs of study. Each of the participants recognized the value in alignment of content, activities, and assessments to learning objectives within their courses, and in several cases alignment was the basis of course organization. HIST-1 specifically organizes content based on alignment to learning objectives. NURS-1 and CPSC-3 each created a curriculum map. CPSC-1 uses the Cisco online lab environment to mimic how he organizes content within his Blackboard shell because the Cisco exercises are aligned to the course's learning objectives. CPSC-3 and IDSS-1

specifically note alignment in the agendas for each module. And NURS-2 has a reflection assignment at the end of the semester where students have to describe how they believe they met each of the course's learning objectives.

Reflection, as part of the evaluation component in the ADDIE process of instructional design, is a form of evaluation for the effectiveness of what was learned in professional development when applied to an instructor's course design and delivery (Branch, 2009; Hodges & Fowler, 2020). The participants' intentions to make improvements during training were verified through their interviews and within their Blackboard shells. Reflection is also an important part of an educator's active engagement in their own growth process to determine what worked and how to improve (Alsalamah & Callinan, 2021; Branch, 2009; Cahapay, 2021; Cranton & King, 2003; Danielson, 2007; Darling-Hammond et al., 2017; Garrison, 2017; Guskey, 1986; Kennedy et al., 2013; Kurt, 2016). Many of the participants also expressed concerns over future improvements, due to either content needs or technology upgrades being implemented at the institution. These future improvements will affect their course design and student interactions. SOC-1 wants to improve the use of video tools for improving asynchronous lectures. HIST-1 wants students to apply technology tools to real-world scenarios in assignments. PHYT-1 notes the challenge to designing courses when they might be co-taught or taught by others with different teaching styles or experience. And as CPSC-1 noted, when it comes to technology, changes occur frequently, and it is the responsibility of instructors to adapt accordingly.

One of these adaptations expressed by many participants was the legal requirement of accessibility, which was introduced as a concept during training but is now a major concern at this institution due to federal legal requirements by public institutions by April 2026, only a

semester away at the time of this study. HIST-1 and IDSS-1 both expressed that the training reinforced the importance of accessibility and inclusion. HIST-1 and COMM-1 also commented how training reinforced the importance of readability of content. And most participants described how they are offering multiple forms of interaction and content delivery to meet learning preferences.

Another adaptation related to Research Question 2, transference of knowledge was not limited to the courses commented on during training, but also transferred to other online courses, and even other modalities. Even though the IYOC workshop focuses on best teaching practices for the online modality, many pedagogical elements can also be applied to in-person or hybrid courses. Interview responses and verification within Blackboard shells show implementation beyond just the online modality. Many instructors use the same organization within Blackboard because they teach the same course in different modalities. CPSC-1 uses the same organization in Blackboard for each course he teaches regardless of modality. NURS-2 uses a curriculum map in all the courses she teaches, both online and in-person, and also uses the same tools in all modalities, such as the Blackboard journal tool or online texts, including Open Educational Resource (OER) texts. HIST-1 uses discussions heavily in both online and in-person courses. ENGL-1 unofficially implemented hybrid delivery options in his in-person sections based on student availability. IDSS-1 encourages online students to come to the in-person section for additional interaction and support, and also uses group projects in Blackboard for all modalities taught. Because IYOC training showed benefits beyond the intended audience of online instructors, the results of this study also reflect the possible benefits of future professional development efforts. This can affect instructor trainings, student achievement, and institutional leadership choices.

Interpretations and Implications

Kirkpatrick (2006) suggests the following guidelines for evaluating the results of training:

1. Use a control group if practical.
2. Allow time for results to be achieved.
3. Measure both before and after program if practical.
4. Repeat the measurement at appropriate times.
5. Consider cost versus benefits.
6. Be satisfied with evidence if proof is not possible.

For number 1, using a control group was not practical for this study due to the low number of potential participants and the background of the possible control group. A significant number of the IYOC participants overall were College of Education faculty, and it can be assumed that this group has had formal training in teaching pedagogy already. Even though the IYOC workshop focuses on the pedagogical aspects of online course design and delivery, some of the IYOC subject matter might be considered a review for this group. It can, therefore, be assumed that this group may already have been implementing best teaching practices in their courses, regardless of modality, so this group was eliminated as potential participants in this study. A future study could use the College of Education faculty as a control group to compare with participants in this study or additional future studies.

An additional control group for future studies could be all the participants in this study. This study used the 6th Edition of the QM Higher Education Rubric for consistency of training application, but QM has since updated the rubric to a 7th Edition. Future studies at this institution could be a comparison of perceptions from 6th Edition training participants to 7th

Edition training participants. Training workshops have continued at this institution since the adoption of the QM 7th Edition Rubric, and after enough time has passed, there may be a large enough number to create another pool of potential participants.

The purpose of this study was specifically designed to address items 2 and 3. The IYOC trainings as part of this study took place between March 2021 and October 2023, between two and four-and-a-half years before this study, which is a significant period of time to allow changes in behavior to take place. Evaluation specifically compared qualitative data from before and after training. The Self Review responses were the culminating activity in the IYOC workshop, and those responses were compared to the interviews of this study, along with verification in the Blackboard learning management system of course sections from before and after the training took place. Future studies would follow this same model, allowing for some time to elapse before gathering reflections from a new participant pool. QM training workshops using the 7th Edition have been administered at this institution since February 2024, so at the time of this study, more time is necessary to gather a larger number of potential study participants to be able to address item number 4.

For item number 6, proof in this study that the training was effective was confirmed in the implementation by participants by comparing participant reflections during training to interview responses, and then verified by tool usage within Blackboard shells. Effective professional development utilizes critical self-reflection to help instructors decide how or why to implement what was learned during training (Alsalamah & Callinan, 2021; Cahapay, 2021; Cranton & King, 2003; Kennedy et al., 2013; Kurt, 2016; Meyer & Marsick, 2003). Instructor reflections on professional development can help guide educator practice when embracing new pedagogical ideas and available technology tools for online course design and delivery and help

correct previous misconceptions, making new knowledge meaningful to their own experiences (Darling-Hammond et al., 2017; Kurt, 2016; Luongo, 2018; Merseth, 1994; Picciano, 2017; Robinson et al., 2017; York et al., 2016). Reflection is actively used in the IYOC workshop where participants use the Self Review tool, a digital version of the QM Rubric, to reflect on what is working and what might need improvement in their online course design.

Quotes were identified in participant interviews based on themes determined in the researcher's journal, and when compared to the Self Review comments and verified with tool usage in Blackboard shells, implementation of training was confirmed. The categories identified in interview responses show how participants reflected on perceived changes to student interactions within their courses after training, answering Research Question 1. The categories identified in interview responses show how participants reflected on perceived changes to their course design processes after training, answering Research Question 2. Reflection is considered part of the standard instructional design process through which better teaching practices are achieved for any modality (Hodges & Fowler, 2020). See Table D1 for a summary of the participants' reflections in both the training and this study.

Item number 5, cost effectiveness, is a challenge for institutions. QM requires an annual subscription by institutions to have access to training materials (Quality Matters™, 2023). Participant reflections show that transference of training from the IYOC workshop to regular practice has taken place, as has been shown by the examples of implementation and perceptions of effectiveness. The practical implications relate to understanding how faculty participants are influenced by the IYOC workshop, as well as QM's role as a provider of professional development. This study is considered heuristic because it explains whether an innovation worked or failed to work (Merriam, 1998). This institution's administration decided to begin a

subscription to Quality Matters™ in February 2020, before there was even a thought of a possible forced nationwide shutdown due to COVID. Administration already saw a potential value in utilizing QM as a training resource to help improve faculty understanding of best practices for teaching online courses. Through the evaluation of QM training at this institution with this study, the training has proven to be successful in helping faculty apply online course design best practices. A recommendation for leadership will be to continue to subscribe to QM and encourage future offerings of QM training at this institution, knowing it is meaningful, even though there is an annual expense.

There is justification for using the Kirkpatrick Model to evaluate training workshops provided by QM (Budzick, 2014; Chisum, 2019; Oberfoell et al., 2019; Simunich, 2023). Level 3 of the Kirkpatrick Model specifically evaluates the effectiveness of training by analyzing changes in behavior after training. Evaluation of training involves assessing a participant's change in behavior (Alsalamah & Callinan, 2021; Cahapay, 2021; Judd, 2015; Kennedy et al., 2013; Kurt, 2016). Effective professional development is not a one-off event, but requires sufficient time for learning, application, reflection, and iterative improvement (Darling-Hammond et al., 2017). Several studies note that data from Level 1: Reaction and Level 2: Learning are easily gathered at the very end of training but do not provide much information beyond motivation to participate, satisfaction with the delivery modality or the facilitator, assessment of activity completion for participants, and the possible intention of implementing what was learned (Budzick, 2014; Cahapay, 2021; Chisum, 2019; Judd, 2015; Kennedy et al., 2013; Maina et al., 2022; Oberfoell et al., 2019; Sowcik, 2018). As a result, there is a call for more studies that specifically address changes in behavior and the potential to contribute more thoroughly to the overall results that can benefit an institution (Alsalamah & Callinan, 2021;

Budzik, 2014; Cahapay, 2021; Chisum, 2019; Kennedy et al., 2013; Kurt, 2016; Maina et al., 2022; Oberfoell et al., 2019). This study contributes knowledge to the education field because it demonstrates changes in instructors' behavior, implementing what was learned as a result of training.

Impact

The results have shown that instructors who participated in this training found value in applying what was learned. Participants perceive that organization of materials and designing purposeful student interactions beyond correspondence enhance learning in the online modality. Topics introduced in training including content organization, student interactions, alignment to learning objectives, and purposeful use of technology tools have been applied not only to online courses but also other modalities, showing that training in best teaching practices can have a perceived benefit for all instructors.

The topic of accessibility and inclusion is also shown to be affected by this training. Participants in this training were introduced to not only being reactive to address the legal requirements of accessibility, but also how being proactively inclusive shows a benefit for all students, which is emphasized in Universal Design for Learning (UDL) (CAST, 2024). By embracing best practices of utilizing accessibility guidelines of properly designated headings and paragraphs, alt text on images, and captions on videos, as well as the inclusive practices of providing multiple modes for content delivery and alternatives for assignment submission, as a result of training and implementing what was learned, participants are better meeting the needs of all students while also addressing legal requirements as a matter of habit.

Limitations

Limitations of this study include the pool of potential participants. The 21 faculty in the sample pool represent only 8.71% of the total 241 full-time faculty population at this institution (Univstats, 2023). Therefore, this sample pool might not entirely represent the total faculty population, but the methods to be used in this study could be used in future studies to build upon the results from this study, which could include cross-case analysis of implementation levels across different participant groups. Embedded single-case studies have multiple groups for analysis (Yin, 2018).

Because of the small sample size, there are only a few similar themes between members of these sub-groups worth noting, but not enough similarities between this set of sub-groups to do a full embedded analysis. The small sample size and lack of richness of data from the sub-groups do not make a true embedded case study design feasible (Yin, 2018). The breakdown of these sub-groups possibly presents an opportunity to analyze larger numbers from distinct subject areas in a future study.

Also, 15 training participants were not included in the potential sample pool for this study because they are College of Education faculty, and it is assumed that these participants may have had pedagogical training in teaching as well as designing courses for one or more modalities. Because QM assumes participants in QM workshops have not had training in how to be course designers or teachers (Simunich, 2022), it can also be assumed that higher education faculty in non-education disciplines might never have been trained in how to be teachers, regardless of modality (Budzick, 2014), hence the exclusion of College of Education faculty. A potential for future studies can compare the College of Education faculty's results based on this assumption, using these faculty as a control group, as defined by Kirkpatrick (2006) for Level 3, to compare to the results for non-education faculty as a whole, or by individual subject areas. For Level 4,

institution-wide results are needed to get a more well-rounded report on changes in behavior beyond a single case study. Recommendations to leadership will include additional case studies at this institution to continually evaluate results of training, and the effects on both instructors and students.

Time constraints were a limitation of this study. The sample pool was made up of previous IYOC participants employed at this Midwest regional mid-sized institution from March 2021 to June 2023. February 2020 is when this institution first started subscribing to QM, and afterward, the researcher was trained as an official QM workshop facilitator and course reviewer. March 2021 was the first IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. June 2023 was the last IYOC facilitated by the researcher using the 6th Edition of the QM Higher Education Rubric. After July 2023, the QM Higher Education Rubric was updated to the 7th Edition, and terminology and numbering were changed from the 6th Edition (Quality Matters™, 2023). March 2021 through June 2023 was the time frame chosen for this study for the sake of consistency, using the same QM Rubric and Self Review tool used in the IYOC workshops, and as a result, the participant pool was from those same IYOC workshops previous to July 2023. This study was longitudinal in that it studied the before and after of a phenomenon, in this case participation in training, by members of the same case (Yin, 2018). Future longitudinal case studies could be performed at this institution with later training participants and use this study as a model. Future studies at this institution could also include participants in the workshops based on the 7th Edition; as of the time of this study, just under two years has passed since the first IYOC using the 7th edition was offered in February 2024, so more time is necessary to provide a potential participant pool for a future study updated for the new training material.

It is important to acknowledge potential bias on the part of this researcher because this study is based on professional development that this researcher was the facilitator of. As such, the researcher is not independent of the qualitative data – the material learned during training was delivered by the researcher, and the reflections and comments by the participants within that training may be directly affected by the researcher's participation. To practice reflexivity and mitigate bias, this researcher used a research journal to reflect on the process after each interview. Active reflection during the research process can help a researcher continually question their thought process and consider how their own perspective may influence the conclusions drawn (Burnam, 2025). A research journal can capture both the practical aspects of conducting research and the intellectual and emotional responses to the process (Turner, 2020; Lincoln & Guba, 1982). Especially because this researcher was the facilitator of the professional development being evaluated in this study, using a journal can help recognize how background, assumptions, and biases might influence the research, which is essential for maintaining research integrity and transparency (Hurst, 2023; Turner, 2020; Lincoln & Guba, 1982).

It is also possible that the participants made the comments they did during training to "please the teacher", and they may have also participated in this study because they are professional peers of the researcher. Interviews and training comments could be considered forms of self-reporting because the participants are expressing their own perceptions of reality, but self-reporting is mitigated by trying to corroborate the interview responses with data from other sources (Yin, 2018). In this study, interview data was corroborated by comparing responses to comments made during training and verification of implementation within Blackboard shells.

Another limitation could be the single institution context of this study. Several studies of training evaluation using the Kirkpatrick Model note that data from Levels 1 and 2 are easily gathered but do not provide much information beyond motivation to participate, satisfaction with the delivery modality or the facilitator, assessment of activity completion for participants, and the possible intention of implementing what was learned (Budzick, 2014; Cahapay, 2021; Chisum, 2019; Judd, 2015; Kennedy et al., 2013; Maina et al., 2022; Oberfoell et al., 2019; Sowcik, 2018). As a result, there is a call for more studies that specifically address changes in behavior and the potential to contribute more thoroughly to the overall results that can benefit an institution (Alsalamah & Callinan, 2021; Budzik, 2014; Cahapay, 2021; Chisum, 2019; Kennedy et al., 2013; Kurt, 2016; Maina et al., 2022; Oberfoell et al., 2019). This study provides an analysis of changes in behavior after training using Kirkpatrick's Level 3 at a single institution, but more institutions may wish to consider using this model to further this research as well.

Recommendations for Leadership

The first recommendation to leadership is to increase training efforts for faculty in online course design and delivery. Training is necessary and expected for professionals in several career fields to gain new skills, which in this case is best teaching practices and technology tools for the online modality. Improving current practices or learning new and evolving technologies are important goals of professional development in specific industries (Kendall & Kendall, 2014; King, 2003; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016; Meyer & Marsick, 2003; Nilson, 2003), including education (Alsalamah & Callinan, 2021; Borko, 2004; Cahapay, 2021; Darling-Hammond et al., 2017; Kennedy et al., 2013; Kurt, 2016). Historically, professional development for teaching online focused on the technology tools but not necessarily on teaching pedagogy with those tools (Johnson et al., 2019; Robinson et al., 2017). Prior to the

COVID-19 pandemic, many educators were not fully aware of the influence the lack of effective purposeful online learning was already having on education; online learning was already on the rise and predicted to become mainstream before the pandemic (Picciano, 2017; Simunich, 2022; Zimmerman et al., 2020), and the number of faculty teaching online was already increasing but many did not have proper training or preparation to do so (Johnson et al., 2019). Therefore, instructors must continue to receive professional development for best practices in online course design to be effective. The emergency remote teaching scenario that arose out of the pandemic has shown that institutions need to scale up their approach to online learning to meet the needs of students, and it is essential for future professional development in online teaching to be guided by evidence-based needs (Li et al., 2023). The results of this study show the potential benefits of increasing the QM training initiative at this institution.

Currently, at this institution, professional development is not mandated by the administration, but efforts can be made with faculty labor groups to further encourage trainings for online course design and delivery. Imposing a mandate could mean that organizational conflicts will increase due to the need to coordinate resources and possible employee pushback, but the results could mean increased student achievement. Professional development in education may look different at different levels and in different disciplines, such as in PreK-12 where mandated training is expected and regularly offered. But in higher education, training tends to be specific to subject areas, climate and culture, or compliance training, and not always on best teaching practices. Regardless of level, professional development in education is an ongoing and systematic process involving discussion, reflection, experimentation with new methods, expansion of knowledge, and acquiring new skills to implement change that improves practice (Alsalamah & Callinan, 2021; Borko, 2004; Cahapay, 2021; Darling-Hammond et al., 2017;

Guskey, 1986; Kennedy et al., 2013; Kurt, 2016; Shagrir, 2012). Educators, at all levels, are expected to be lifelong learners, so professional development, including reflection on practices, is encouraged and expected (Burridge & Carpenter, 2013; Danielson, 2007; Davies & Preston, 2002; King, 2003; Shagrir, 2012). As a result of social restrictions from the COVID-19 pandemic, there is increased student and instructor demand for flexibility in the modality of course delivery (Bozkurt et al., 2022; Hodges & Fowler, 2020; Hodges et al., 2020). With this demand, recommending professional development for instructors to be effective in the online modality for students to succeed is recommended.

Kirkpatrick's (2016) Level 4: Results expects continual reporting on training outcomes, which in this single-case study has been done, but then leadership can use the results to promote an ongoing system of support. To be successful in training initiatives, the intangibles of training completion and participant satisfaction (Levels 1 and 2), must be leveraged into tangible behaviors and results (Levels 3 and 4) (Kirkpatrick & Kirkpatrick, 2005). Early success stories, such as the participants in this study, can be shared by the participants themselves to help bolster this training initiative. Early adopters are the ones with the can-do attitude who can be transformed from learners into leaders (Kirkpatrick & Kirkpatrick, 2005). Trainers and other innovation leaders deliver training by modeling, usually by peers and sometimes by early adopters of an innovation, to potential future adopters via organized demonstrations (King & Lawler, 2003; Rogers, 2003). Educators who participate in technology-related professional development perceive that the experience positively impacts their teaching practice and helps create a culture of support (Crews & Wilkinson, 2015; Gregory et al., 2020; Hollowell et al., 2017; Li et al., 2023; Shagrir, 2012). Support for training efforts creates a climate for action and

employee satisfaction (Kirkpatrick & Kirkpatrick, 2005), which leaders can foster by incentivizing training participation with stipends or decreased teaching load.

Future Research Recommendations

Because of the single institution context of this study, additional studies need to be conducted at other institutions. Several studies of training evaluation using the Kirkpatrick Model note that data from Levels 1 and 2 are easily gathered during training but do not provide much information about actual implementation of what was learned after training (Budzick, 2014; Cahapay, 2021; Chisum, 2019; Judd, 2015; Kennedy et al., 2013; Maina et al., 2022; Oberfoell et al., 2019; Sowcik, 2018). More institutions may wish to consider using Kirkpatrick's Level 3 to further research in the behavior of implementation. This study can be used as a model for future studies evaluating Kirkpatrick's Level 3.

Kirkpatrick (2005; 2006; 2007; 2016) recommends using control groups for analyzing Level 3: Behavior. In this study, the College of Education faculty were eliminated for analysis due to their familiarity of the subject matter before training, so it was assumed they were likely already implementing much of what was presented during training, but future studies might wish to compare Education faculty participants as a control group to other subject area participants to see to what degree the subject matter was implemented between groups. Holistic single-case studies have a single unit of analysis, and embedded single-case studies have multiple units of analysis (Yin, 2018). This study was a holistic context due to the small sample size, but if large enough numbers of participants warrant the creation of large sub-groups, or if there is a large sub-group that can act as a control group, future studies may be able to justify an embedded analysis. Performance in previous case studies may serve as a prediction for performance in future studies; pattern matching compares findings from one case study with a prediction from a

previous one (Yin, 2018). At institutions such as in this study where there has been transition from the QM 6th Edition Rubric to the 7th Edition, 6th Edition training participants could be the control group for 7th Edition participants to be compared to, and because this study showed transference from training to practice for these participants in multiple course modalities, that might predict a similar outcome for future participants.

To broaden the scope of results for Kirkpatrick Level 4, additional future studies could be conducted to show any possible relationship between QM training for faculty and actual student performance in online courses such as course grades and evaluations on individual assignments. In this study, Kirkpatrick's Level 3 provided a model to evaluate faculty perceptions of student performance and satisfaction in this training. Suggested future studies include analyzing student performance as well as perceptions of their learning experiences from both the students themselves and their instructors. It is essential for future professional development in online teaching to be guided by evidence-based needs (Li et al., 2023). The perceptions of instructors from this study suggest that improved student interactions in courses may lead to increased student achievement, or at least improved student perceptions of their own interactions within courses, which is not limited to the online modality.

Conclusion

Using Kirkpatrick's Level 3: Behavior as a model for analysis, the Quality Matters™ (QM) Improving Your Online Course (IYOC) training workshop offered at this institution showed that after training, instructors perceived what was learned in training was implemented into regular practice. Transfer of training to performance is one of the most critical concerns for trainers and administrators (Dick, Carey, & Carey, 2015; Kirkpatrick & Kirkpatrick, 2006; Kirkpatrick & Kirkpatrick, 2016). The training goals of improving student interactions and

alignment of course content to learning objectives helped develop the research questions for this study to determine if and to what degree transference from training to practice had taken place. Results showed that instructors have applied what was learned in training beyond the intended online modality into multiple courses in different learning modalities.

Themes yielded during interviews were compared to training comments and implementations within the learning management system. Research Question 1 asks how higher education instructors perceive any changes to student interactions within their online courses after participating in training. Results show that course organization was improved to deliver content and promote student interactions through the effective use of technology tools. Research Question 2 asks how participants perceived what was transferred from training to regular practice in their online course design and delivery. Results show that alignment to learning objectives had a direct effect on content organization and development, changes in technology and the legal requirements of accessibility will affect future improvements to courses, and that training influenced not only changes in courses commented on during training but also other courses in multiple modalities.

Limitations of this study include the small pool of potential participants, time constraints, potential biases on the part of both the researcher and the participants, and the single institution context. The small pool of participants does not entirely represent the total faculty population, but the methods used in this study could be used in future studies to build upon the results, which could include cross-case analysis of implementation levels across different participant groups. There was a potential for an embedded analysis in this study, but the small sample size and lack of data within subgroups did not make an embedded analysis feasible. The time frame chosen for this study was based on consistency of training materials; this institution started offering

IYOC workshops in March 2021, and when QM updated its higher education rubric and training materials from the 6th Edition to the 7th Edition in July 2023. Longitudinal studies show the before and after of a phenomenon for a specific case (Yin, 2018), and this study could be used as a model for future studies. Future studies at this institution could include larger participant pools to allow for the formation of larger sub-groups based on subject areas for embedded analysis, and can now include participants in workshops using the newer training materials.

The recommendation to leadership is to increase training efforts for faculty in online course design and delivery. Although professional development is not mandated at this institution, efforts can be made with faculty labor groups to further encourage participation in training for online course design and delivery. Kirkpatrick's (2016) Level 4: Results expects continual reporting on training outcomes, which leadership can use the results from this study and possible future studies to promote an ongoing system of support. The success of the early adopters in this study can be shared to help bolster this training initiative. Early adopters are the ones with the can-do attitude who can be transformed from learners into leaders, and for training efforts, create a climate for action and employee satisfaction (Kirkpatrick & Kirkpatrick, 2005), which leaders can also foster with other incentives.

Additional studies need to be conducted at other institutions to analyze Kirkpatrick's Level 3: Behavior. Levels 1 and 2 provide information during training, but Level 3 provides information about implementation after training. Future studies can have an embedded context if sample sizes are large enough to have multiple sub-groups for effective comparison. Comparing control groups to other sub-groups of participants is also a possibility, such as using Education faculty or participants who received training under the QM 6th Edition Rubric compared to those who received training under the 7th Edition. This study only analyzed the perceptions of faculty,

but additional studies could also include comparing perceptions of faculty to perceptions of students. This study can be also used as a model for future studies evaluating Kirkpatrick's Level 3. To broaden the scope of results for Kirkpatrick's Level 4, additional future studies could be conducted to show any possible relationship between QM training for faculty and actual student performance in online courses, such as course grades and evaluations on individual assignments.

It is the hope of this researcher, as an educator of both students and instructors, and devoted to the improvement of best practices for technology integration into course design, that the results of this study will assist others in the field in professional development. It is further hoped that this body of knowledge will continue to be developed as eductors continue to affirm the commitment to improving teaching and learning for students to succeed in higher education and beyond, regardless of learning modality.

REFERENCES

Alamri, W. A. (2019, MAY). Effectiveness of Qualitative Research Methods: Interviews and Diaries. *International Journal of English and Cultural Studies, 2*(1), 65-70.

Alexander, L. K., Lopes, B., Ricchetti-Masterson, K., & Yeatts, K. B. (2014). *Randomized Controlled Trials (Experimental Studies).* Retrieved June 12, 2025, from ERIC Notebook: https://sph.unc.edu/epid/eric/

Ali, R., & Wright, J. (2017). Examination of the QM process: Making a case for the transformative professional development model. *Faculty Publications, 4150*.

Alsalamah, A., & Callinan, C. (2021). Adaptation of Kirkpatrick's Four-Level Model of Training Criteria to Evaluate Training Programmes for Head Teachers. *Education Sciences, 11*(116), 1-25.

Barton, P. (2020). Elementary School Teacher Characteristics Associated with Self-Efficacy for Using Technology to Implement Constructivist Content Instruction: An Explanatory Correlational Study.

Beggs, R. (2018). *Technology for the Sake of It, I Don't Think So!* Retrieved October 5, 2025, from Ulster University: https://altc.alt.ac.uk/2018/sessions/technology-for-the-sake-of-it-i-dont-think-so-18-176/#gref

Berkovic, D. (2023). Researcher Positionality. In D. Ayton, T. Tsindos, & D. Berkovic, *Qualitative Research – a practical guide for health and social care researchers and practitioners.* Open Educational Resources Collective.

Berry, D. C., Neal, M., Hall, E. G., McMurray, R. G., Schwartz, T. A., Skelly, A. H., & Smith-Miller, C. (2013, January). Recruitment and retention strategies for a community-based weight management study for multi-ethnic elementary school children and their parents. *Public Health Nursing, 30*(1), pp. 80-86.

Bogdan, R. C., & Biklen, S. K. (2006). *Qualitative research for education: An introduction to theory and methods* (5th ed.). Boston, MA: Allyn and Bacon.

Borko, H. (2004). Professional Development and Teacher Learning: Mapping the Terrain. *Educational Researcher, 33*(8).

Bozkurt, A., Karakaya, K., Turk, M., Karakaya, O., & Castellanos-Reyes, D. (2022, September). The impact of COVID-19 on education: A meta-narrative review. *TechTrends, 66*(5), 883-896.

Brancato, V. C. (2003). Professional development in higher education. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 59-65). San Francisco: Jossey-Bass.

Branch, R. M. (2009). *Instructional design: The ADDIE approach.* New York: Springer.

Buchanan, E. A., & Zimmer, M. (2021). Internet research ethics (revised). In E. N. Zalta (Ed.), *The Stanford Encyclopedia of Philosophy* (Spring ed.).

Budzick, D. (2014, October). *Examining the relationship between the quality matters rubric workshop and faculty knowledge and perceptions of course design: An action research study*. Retrieved September 2023, from ProQuest Dissertations & Theses Global: http://proxy.govst.edu:2048/login?url=https://www.proquest.com/dissertations-theses/examining-relationship-between-quality-matters/docview/1626380108/se-2

Burnam, L. (2025, January 7). *Reflexivity in Qualitative Research: Why You'll Never Be An Objective Observer*. Retrieved June 15, 2025, from User Interviews: https://www.userinterviews.com/blog/reflexivity-in-qualitative-research

Burridge, P., & Carpenter, C. (2013, September). Expanding pedagogical horizons: A case study of teacher professional development. *Australian Journal of Teacher Education, 38*(9).

Cahapay, M. B. (2021). Kirkpatrick Model: Its Limitations as Used in Higher Education Evaluation. *International Journal of Assessment Tools in Education, 8*(1), 135-144.

Casa-Todd, J. (2025). *The Most Compelling Argument For and Against Educational Technology in Schools*. Retrieved October 5, 2025, from Jennifer Casa-Todd: https://jcasatodd.com/the-most-compelling-argument-for-and-against-educational-technology-in-schools/

CAST. (2024). *CAST Universal Design for Learning Guidelines version 3.0.* Retrieved October 29, 2025, from CAST Universal Design for Learning Guidelines: https://udlguidelines.cast.org/static/udlg3-graphicorganizer-digital-numbers-a11y.pdf

Chisum, R. (2019, April). An evidence-based faculty development program for online teaching In higher education. *All Theses And Dissertations, 269*. Retrieved November 2023, from DUNE: DigitalUNE: https://dune.une.edu/cgi/viewcontent.cgi?article=1268&context=theses

Cranton, P., & King, K. P. (2003). Transformative learning as a professional development goal. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 31-37). San Francisco: Jossey-Bass.

Crews, T., & Wilkinson, K. (2015, Spring/Summer). Online quality course design vs. quality teaching: Aligning Quality Matters standards to principles of good teaching. *The Journal of Research in Business Education, 57*(1), 47-63.

Daley, B. J. (2003). A case for learner-centered teaching and learning. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 23-30). San Francisco: Jossey-Bass.

Danielson, C. (2007). *Enhancing professional practice : A framework for teaching* (2nd ed.). Alexandria, VA: Association for Supervision and Curriculum Development.

Darling-Hammond, L., Hyler, M. E., & Gardner, M. (2017, June 5). *Effective Teacher Professional Development.* Retrieved July 19, 2025, from Learning Policy Institute: https://learningpolicyinstitute.org/product/effective-teacher-professional-development-report

Davies, R., & Preston, M. (2002). An evaluation of the impact of continuing professional development on personal and professional lives. *Journal of In-service Education, 28*(2), 231-254.

Deterding, N. M., & Waters, M. C. (2021). Flexible Coding of In-depth Interviews: A Twenty-first-century Approach. *Sociological Methods and Research, 50*(2), 708-739.

Dick, W., Carey, L., & Carey, J. O. (2015). *The systematic design of instruction* (Vol. Eighth edition). Boston: Pearson.

Fisher, L. (2019). *Exploring use of the Bridge21 model as a 21st century method of continuing professional development (CPD) in computer science (CS) for teachers in Ireland.* Retrieved October 2023, from 1Library: https://1library.net/article/comparison-of-guskey-and-kirkpatrick-evaluation-models.zpn9247y

Forbes, D. (2013, July). Blinding: an essential component in decreasing risk of bias in experimental designs. *Evidence Based Nursing, 16*(3), pp. 70-71.

Garrison, D. R. (2017). *E-learning in the 21st century : A framework for research and practice* (Vol. 3rd Ed.). New York: Routledge.

Gibson, C. C. (1998). *Distance learners in higher education : Institutional responses for quality outcomes.* Madison, WI: Atwood Pub.

Governors State University. (2021). *About*. Retrieved from Governors State University: https://www.govst.edu/about/

Goyes, D. R., & Sandberg, S. (2024). Trust, nuance, and care: Advantages and challenges of repeat qualitative interviews. *Qualitative Research, 25*(2), 330-349.

Gravel, R. J. (2014, May). *The effects of pre-kindergarten through twelfth grade in-service teachers' participation in a professional development course aligned with the TPACK framework.* Retrieved from ProQuest Dissertations & Theses Global: http://proxy.govst.edu:2048/login?url=https://www.proquest.com/dissertations-theses/effects-pre-kindergarten-through-twelfth-grade/docview/1554012509/se-2

Gregory, R. L., Rockinson-Szapkiw, A. J., & Cook, V. S. (2020, June). Community college faculty perceptions of the Quality Matters™ Rubric. *Online Learning Journal, 24*(2), 128-141.

Guskey, T. R. (1986, May). Staff development and the process of teacher change. *Education Researcher, 15*(5), 5-12.

Guskey, T. R. (2002, March 1). *Does It Make a Difference? Evaluating Professional Development.* Retrieved October 2023, from Association for Supervision and Curriculum Development: https://www.ascd.org/el/articles/does-it-make-a-difference-evaluating-professional-development

Hess, D. R. (2004, October). Retrospective Studies and Chart Reviews. *Respiratory Care, 49*(10), 1171-1174.

Hodges, C. B., & Fowler, D. J. (2020). The COVID-19 crisis and faculty members in higher education: From emergency remote teaching to better teaching through reflection. *International Journal of Multidisciplinary Perspectives in Higher Education, 5*(1), 118-122.

Hodges, C., Moore, S., Lockee, B., Trust, T., & Bond, A. (2020, March 27). *The difference between emergency remote teaching and online learning*. Retrieved February 2023, from Educause Review: https://er.educause.edu/articles/2020/3/the-difference-between-emergency-remote-teaching-and-online-learning

Hollowell, G. P., Brooks, R. M., & Anderson, Y. B. (2017). Course design, Quality Matters training, and student outcomes. *American Journal of Distance Education, 31*(3), 207-216.

Horton, W. (2001). *Evaluating e-learning.* Alexandria, VA: American Society for Training & Development.

Hurst, A. (2023). Reflexivity. In *Introduction to Qualitative Research Methods.* Oregon State University.

Johnson, K. R., Hewapathirana, G. I., & Bowen, M. M. (2019). Faculty development for online teaching. In J. S. Keengwe, *Handbook of research on virtual training and mentoring of online instructors* (pp. 40-55). IGI Global.

Joosten, T., & Cusatis, R. (2019, December). A cross-institutional study of instructional characteristics and student outcomes: Are quality indicators of online courses able to predict student success? *Online Learning Journal, 23*(4), 354-378.

Judd, J. D. (2015). CTO mentor program: examining the effectiveness of the CTO mentor program and its impact on the K-12 technology leader's career. *Theses and Dissertations, 553*. Retrieved from Pepperdine University Digital Commons: https://digitalcommons.pepperdine.edu/etd/553

Kaman, Z. K., & Othman, Z. (2016). Validity, Reliability and Triangulation in Case Study Method: An Experience. *Proceedings of the 2nd UUM International QUALITATIVE RESEARCH CONFERENCE*, 350-356.

Kendall, K. E., & Kendall, J. E. (2014). *Systems Analysis and Design.* Upper Saddle River, NJ: Pearson.

Kennedy, P. E., Chyung, S. Y., Winieki, D. J., & Brinkerhoff, R. O. (2013). Training professionals' usage and understanding of Kirkpatrick's Level 3 and Level 4 evaluations. *International Journal of Training and Development, 18*(1), 1-21.

King, K. P. (2003). Learning the new technologies: Strategies for success. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 49-57). San Francisco: Jossey-Bass.

King, K. P., & Lawler, P. A. (2003). Trends and issues in the professional development of teachers of adults. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 5-14). San Francisco: Jossey-Bass.

Kirkpatrick Partners, LLC. (2023). *About Dr. Don Kirkpatrick*. Retrieved November 2023, from Kirkpatrick Partners: https://www.kirkpatrickpartners.com/About-Us/Don-Kirkpatrick/

Kirkpatrick, D. L., & Kirkpatrick, J. D. (2005). *Transferring Learning to Behavior: Using the Four Levels to Improve Performance.* San Francisco: Berrett-Koehler.

Kirkpatrick, D. L., & Kirkpatrick, J. D. (2006). *Evaluating training programs : The four levels* (3rd ed.). San Francisco: Berrett-Koehler Publishers, Inc.

Kirkpatrick, D. L., & Kirkpatrick, J. D. (2007). *Implementing the Four Levels: A Practical Guide for Effective Evaluation of Training Programs.* San Francisco: Berrett-Koehler Publishers, Inc.

Kirkpatrick, J. D., & Kirkpatrick, W. K. (2016). *Kirkpatrick's four levels of training evaluation.* Alexandria, VA: Association for Talent Development Press.

Kratochwill, T. R. (1978). *Single Subject Research: Strategies for Evaluating Change.* New York: Academic Press.

Kratochwill, T. R., Hitchcock, J., Horner, R. H., Levin, J. R., Odom, S. L., Rindskopf, D. M., & Shadish, W. R. (2010, June). *Single-case designs technical documentation.* Retrieved June 12, 2025, from What Works Clearinghouse: http://ies.ed.gov/ncee/wwc/pdf/wwc_scd.pdf

Kurt, S. (2016, October 24). *Kirkpatrick Model: Four Levels of Learning Evaluation.* Retrieved June 2025, from Educational Technology: https://educationaltechnology.net/kirkpatrick-model-four-levels-learning-evaluation/

Lakens, D. (2022). Sample Size Justification. *Collabra: Psychology, 8*(1).

Lawler, P. A. (2003). Teachers as adult learners: A new perspective. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 15-22). San Francisco: Jossey-Bass.

Lawler, P. A., & King, K. P. (2003). Changes, challenges, and the future. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 83-91). San Francisco: Jossey-Bass.

Li, S., Craig, S. D., & Schroeder, N. L. (2023, January). Lessons learned from online learning at scale: A study of exemplar learning organizations. *TechTrends, 67*(1), 84-97.

Lincoln, Y. S., & Guba, E. G. (1982, March). *Establishing Dependability and Confirmability in Naturalistic Inquiry through an Audit.* Retrieved July 20, 2025, from ERIC Institute of Education Sciences: https://files.eric.ed.gov/fulltext/ED216019.pdf?ref=quirkos.com

Lowell, V. L., & Yang, M. (2023, January). Authentic learning experiences to improve online instructor's performance and self-efficay: The design of an online mentoring program. *TechTrends, 67*(1), 112-123.

Luongo, N. (2018, July 31). *An examination of distance learning faculty satisfaction levels and self-perceived barriers.* Retrieved February 2023, from Journal of Educators Online: https://www.thejeo.com/archive/archive/2018_152/luangopdf

Maina, M., Guardia, L., Cabrera, N., Fernandez-Ferrer, M., Malerba, M. L., & Garcia, C. (2022). Evaluation of a Study Programme for the implementation of blended learning at the Maldives National University. *Proceedings of the Central European Conference on Information and Intelligent Systems. 33*, pp. 249-253. Dubrovnik, Croatia: CECIIS.

Manolov, R., Moeyaert, M., & Fingerhut, J. E. (2021, March 25). A Priori Justification for Effect Measures in Single-Case Experimental Designs. *Perspectives on Behavior Science, 45*, 153-186.

Marceau, G. (2003). Professional development in adult basic education. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 67-74). San Francisco: Jossey-BAss.

McKim, C. (2023). Meaningful Member-Checking: A Structured Approach to Member- Checking. *American Journal of Qualitative Research, 7*(2), 41-52.

McLeod, S. (2024, December 20). *Peer Debriefing In Qualitative Research*. Retrieved June 15, 2025, from Simply Psychology: https://www.simplypsychology.org/peer-debriefing-in-qualitative-research.html

Merriam, S. B. (1998). *Qualitative Research and Case Study Applications in Education.* San Francisco: Jossey-Bass.

Merseth, K. K. (1994, November). *Cases, case methods, and the professional development of educators.* Retrieved November 2022, from ERIC Digest: https://files.eric.ed.gov/fulltext/ED401272.pdf

Meyer, S. R., & Marsick, V. J. (2003). Professional development in corporate training. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 75-82). San Francisco: Jossey-Bass.

Moss, A. (2025). *4 Strategies to Improve Participant Retention In Online Longitudinal Research Studies*. Retrieved June 15, 2025, from CloudResearch: https://www.cloudresearch.com/resources/blog/strategies-retain-participants-longitudinal-studies/

Nilson, L. B. (2003). *Teaching at its best : A research-based resource for college instructors* (Vol. 2nd ed.). Bolton, MA: Anker Pub. Co.

Oberfoell, A., Dao, D., Martinez-Marrero, I., & Bailey, S. (2019). Design4Online-A unique faculty training model at the University of Iowa: A best practices report. *AECT 2019. 16*, pp. 425-432. Association for Educational Communications and Technology.

Ozcan, P., Han, S., & Graebner, M. (2017). Single Cases: The What, Why and How. In R. Mir, & S. Jain (Eds.), *The Routledge Companion to Qualitative Research in Organization Studies* (pp. 92-112). Routledge.

Picciano, A. G. (2017). Theories and frameworks for online education: Seeking an integrated model. *Online Learning, 21*(3), 166-190.

Quality Matters™. (2023, October 23). *Applying the QM Rubric*. Retrieved from Quality Matters: https://www.qualitymatters.org/index.php/professional-development/workshops/higher-ed-appqmr

Quality Matters™. (2023). *Improving Your Online Course (IYOC)*. Retrieved May 9, 2023, from Quality Matters™: https://www.qualitymatters.org/professional-development/workshops/higher-ed-iyoc

Quality Matters™. (2023, October 23). *QM Members Already Benefiting from New Higher Ed Rubric* . Retrieved from Quality Matters: https://www.qualitymatters.org/qa-resources/resource-center/articles-resources/higher-ed-seventh-edition-course-reviews

Quality Matters™. (2025). *Higher Ed. Options for QM Certification Course Reviews*. Retrieved October 2025, from Quality Matters™: https://www.qualitymatters.org/reviews-certifications/course-design-reviews/higher-education-official-review-options

Quintao, C., & Andrade, P. (2020). How to Improve the Validity and Reliability of a Case Study Approach. *Open Journals in Education, 9*(2), 264-275.

Ritzhaupt, A. D., Kumar, S., & Martin, F. (2021). The Competencies for Instructional Designers in Higher Education. In J. E. Stefaniak, S. Conklin, B. Oyarzun, & R. M. Reese, *A Practitioner's Guide to Instructional Design in Higher Education.* EdTech Books.

Robinson, H. A., Kilgore, W., & Warren, S. J. (2017, December). Care, communication, learner support: Designing meaningful online collaborative learning. *Online Learning Journal, 21*(4), 29-51.

Rogers, E. M. (2003). *Diffusion of innovations* (Vol. 5th ed.). New York: Free Press.

Rudra, S. (2025, June 13). *A Guide to the ADA Title II Accessibility Rule*. Retrieved October 5, 2025, from EdTech: Focus on Higher Education: https://edtechmagazine.com/higher/article/2025/06/guide-ada-title-ii-accessibility-rule-perfcon

Saldaña, J. (2014). Coding and Analysis Strategies. In P. Leavy, *The Oxford Handbook of Qualitative Research* (pp. 581-605). Oxford University Press.

Saldaña, J. (2016). *The Coding Manual for Qualitative Researchers* (3rd ed.). London: SAGE.

Shagrir, L. (2012, February). How evaluation processes affect the professional development of teachers in higher education. *Journal of the Scholarship of Teaching and Learning, 12*(1), 23-35.

Simunich, B. (2022, November 8). Quality Matters Coordinators Meeting. *QM Connect Conference 2022*. Tuscon, AZ.

Simunich, B. (2023, September 6). Vice President, Innovation and Research. (S. Thesen, Interviewer)

Smith, J. A., & Nizza, I. E. (2022). *Essentials of interpretative phenomenological analysis.* American Psychological Association.

Sowcik, M., Benge, M., & Niewoehner-Green, J. (2018). A practical solution to developing county extension director's leadership skills: Exploring the design, delivery and evaluation of an online leadership development program. *Journal of Agricultural Education, 59*(3), 139-153.

Stake, R. E. (1995). *The Art of Case Study Research.* SAGE.

Turner, D. (2020, April 2). *Reflexive journals in qualitative research*. Retrieved July 19, 2025, from Quirkos: https://www.quirkos.com/blog/post/reflexive-journals-in-qualitative-research/

U.S. Department of Justice, Civil Rights Division, Disability Rights Section. (2024, April 8). *Fact Sheet: New Rule on the Accessibility of Web Content and Mobile Apps Provided by State and Local Governments*. Retrieved October 29, 2025, from ADA.gov: https://www.ada.gov/resources/2024-03-08-web-rule/

Univstats. (2023). *GSU Faculty Salary by Rank*. Retrieved May 9, 2023, from Univstats: https://www.univstats.com/salary/governors-state-university/faculty/

Van Wart, M., Ni, A., Medina, P., Canelon, J., Kordrostami, M., Zhang, J., & Liu, Y. (2020). Integrating students' perspectives about online learning: A hierarchy of factors. *International Journal of Educational Technology in Higher Education, 17*(53). Retrieved from Springer Open: https://doi.org/10.1186/s41239-020-00229-8

Webster, W. (2025). *What is data saturation in qualitative research?* Retrieved June 15, 2025, from Qualtrics: https://www.qualtrics.com/experience-management/research/data-saturation-in-qualitative-research/

Wlodkowski, R. J. (2003). Forstering motivation in professional development programs. In K. P. King, & P. A. Lawler, *New perspectives on designing and implementing professional development of teachers of adults* (pp. 39-48). San Francisco: Jossey-Bass.

Yin, R. K. (2018). *Case study research and applications: Design and methods* (6th ed.). Sage.

York, C. S., Yamagata-Lynch, L. C., & Smaldino, S. E. (2016). Adult reflection in a graduate-level online distance education course. *Reflective Practice, 17*(1), 40-58.

Zimmerman, W., Altman, B., Simunich, B., Shattuk, K., & Burch, B. (2020, December). Evaluating online course quality: A study on implementation of course quality standards. *Online Learning, 24*(4), 147-163.

APPENDICES

APPENDIX A: The Quality Matters™ Higher Education Rubric, 6th Edition

HE Specific Review Standards from the QM Higher Education Rubric, Sixth Edition

General Standards	Specific Review Standards	Points
Course Overview and Introduction	1.1 Instructions make clear how to get started and where to find various course components.	3
	1.2 Learners are introduced to the purpose and structure of the course.	3
	1.3 Communication expectations for online discussions, email, and other forms of interaction are clearly stated.	2
	1.4 Course and institutional policies with which the learner is expected to comply are clearly stated within the course, or a link to current policies is provided.	2
	1.5 Minimum technology requirements for the course are clearly stated, and information on how to obtain the technologies is provided.	2
	1.6 Computer skills and digital information literacy skills expected of the learner are clearly stated.	1
	1.7 Expectations for prerequisite knowledge in the discipline and/or any required competencies are clearly stated.	1
	1.8 The self-introduction by the instructor is professional and is available online.	1
	1.9 Learners are asked to introduce themselves to the class.	1
Learning Objectives (Competencies)	2.1 The course learning objectives, or course/program competencies, describe outcomes that are measurable.	3
	2.2 The module/unit-level learning objectives or competencies describe outcomes that are measurable and consistent with the course-level objectives or competencies.	3
	2.3 Learning objectives or competencies are stated clearly, are written from the learner's perspective, and are prominently located in the course.	3
	2.4 The relationship between learning objectives or competencies and learning activities is clearly stated.	3
	2.5 The learning objectives or competencies are suited to the level of the course.	3
Assessment and Measurement	3.1 The assessments measure the achievement of the stated learning objectives or competencies.	3
	3.2 The course grading policy is stated clearly at the beginning of the course.	3
	3.3 Specific and descriptive criteria are provided for the evaluation of learners' work, and their connection to the course grading policy is clearly explained.	3
	3.4 The assessments used are sequenced, varied, and suited to the level of the course.	2
	3.5 The course provides learners with multiple opportunities to track their learning progress with timely feedback.	2
Instructional Materials	4.1 The instructional materials contribute to the achievement of the stated learning objectives or competencies.	3
	4.2 The relationship between the use of instructional materials in the course and completing learning activities is clearly explained.	3
	4.3 The course models the academic integrity expected of learners by providing both source references and permissions for use of instructional materials.	2
	4.4 The instructional materials represent up-to-date theory and practice in the discipline.	2
	4.5 A variety of instructional materials is used in the course.	2
Learning Activities and Learner Interaction	5.1 The learning activities promote the achievement of the stated learning objectives or competencies.	3
	5.2 Learning activities provide opportunities for interaction that support active learning.	3
	5.3 The instructor's plan for interacting with learners during the course is clearly stated.	3
	5.4 The requirements for learner interaction are clearly stated.	2
Course Technology	6.1 The tools used in the course support the learning objectives or competencies.	3
	6.2 Course tools promote learner engagement and active learning.	3
	6.3 A variety of technology is used in the course.	1
	6.4 The course provides learners with information on protecting their data and privacy.	1
Learner Support	7.1 The course instructions articulate or link to a clear description of the technical support offered and how to obtain it.	3
	7.2 Course instructions articulate or link to the institution's accessibility policies and services.	3
	7.3 Course instructions articulate or link to the institution's academic support services and resources that can help learners succeed in the course.	3
	7.4 Course instructions articulate or link to the institution's student services and resources that can help learners succeed.	1
Accessibility* and Usability	8.1 Course navigation facilitates ease of use.	3
	8.2 The course design facilitates readability.	3
	8.3 The course provides accessible text and images in files, documents, LMS pages, and web pages to meet the needs of diverse learners.	3
	8.4 The course provides alternative means of access to multimedia content in formats that meet the needs of diverse learners.	2
	8.5 Course multimedia facilitate ease of use.	2
	8.6 Vendor accessibility statements are provided for all technologies required in the course.	2

* Meeting QM Specific Review Standards regarding accessibility does not guarantee or imply that the specific accessibility regulations of any country are met. Consult with an accessibility specialist to ensure that accessibility regulations are met.

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For more information or access to the fully annotated QM Rubric, visit qualitymatters.org or email info@qualitymatters.org

QUALITY MATTERS QM

APPENDIX B: Pool of Faculty Participants (Anonymized)

Last	First	College(s)	Division(s)	Program(s)	Training	Participant Code	Consent Signed	Interview
		CHHS	DON	NURS	March 2021		8/11/25	Not available
		COB	DAFME	BAN, MIS	March 2021			
		CAS	DAL	HIST, SOSC	March 2021	HIST-1	8/12/25	8/14/25
		CAS, COB	DSMT, DAFME	CPSC, MIS, IT	March 2021	CPSC-1	8/19/25	9/2/25
		CAS	DAL	ART	March 2021			
		CAS	DSMT	CPSC, IT	March 2021	CPSC-2	8/12/25	8/26/25
		CHHS	DHA	HLAD	March 2021		On med leave	
		CAS	DAL	CJUS	March 2021			
		CHHS	DON	DNP, HLSC, NURS	March 2021	NURS-1	8/19/25	9/3/25
		CAS	DAL	ENGL	June 2021	ENGL-1	8/13/25	9/5/25
		CAS	DAL	CMP, COMM, GNSX	June 2021	COMM-1	9/3/25	9/9/25
		CAS	DSMT	BIOL, CPSC	July 2021	CPSC-3	9/3/25	9/4/25
		COB	DMME	MGMT	July 2021		Opted out	
		CHHS	DSW	SOCW	Nov 2021			
		CAS	DAL	ENGL	Nov 2021	ENGL-2	9/10/25	9/15/25
		CHHS	DON	DNP, HLSC, NURS	Nov 2021	NURS-2	8/19/25	8/25/25
		CHHS	DPT	PHYT	March 2022	PHYT-1	8/12/25	8/20/25
		CHHS	DON	NURS	March 2022			
		CAS	DAL	IDSS	July 2022	IDSS-1	8/22/25	8/28/25
		COB	DAFME	MIS	June 2023			
		CAS	DAL	SOC	June 2023	SOC-1	8/19/25	9/4/25

Potential** non-COE GSU faculty who have completed the official QM Improving Your Online Course (IYOC) workshop as of 6/6/23	21
Potential Academic Programs Represented	22
TOTAL STUDY PARTICIPANTS	12
ACTUAL PROGRAMS REPRESENTED	16

Total faculty by program who completed IYOC? *	
ART	1
BAN	1
BIOL	1
CJUS	1
CMP	1
COMM	1
CPSC	3
DNP	2
ENGL	2
GNSX	1
HIST	1
HLAD	1
HLSC	2
IDSS	1
IT	2
MGMT	1
MIS	3
NURS	4
PHYT	1
SOC	1
SOCW	1
SOSC	1

Faculty participating in this study by program? *	
ART	0
BAN	0
BIOL	1
CJUS	0
CMP	1
COMM	1
CPSC	3
DNP	2
ENGL	2
GNSX	1
HIST	1
HLAD	0
HLSC	2
IDSS	1
IT	2
MGMT	0
MIS	1
NURS	3
PHYT	1
SOC	1
SOCW	0
SOSC	1

Program Key	
ART	*Art*
BAN	*Business Analytics*
BIOL	*Biology*
CJUS	*Criminal Justice*
CMP	*Communication, Media, and Performance*
COMM	*Communication*
CPSC	*Computer Science*
DNP	*Doctorate of Nursing Practice*
ENGL	*English*
GNSX	*Gender and Sexuality Studies*
HIST	*History*
HLAD	*Health Administration*
HLSC	*Health Sciences*
IDSS	*Interdisciplinary Studies*
IT	*Information Technology*
MGMT	*Management*
MIS	*Management Information Systems*
NURS	*Nursing*
PHYT	*Physical Therapy*
SOC	*Sociology*
SOCW	*Social Work*
SOSC	*Social Sciences*

How many faculty have completed IYOC by college? *	
CAS	11
COB	4
CHHS	7

How many faculty have participated in this study by college? *	
CAS	9
COB	1
CHHS	4

College Key	
CAS	*College of Arts and Sciences*
COB	*College of Business*
CHHS	*College of Health and Human Services*

How many faculty have completed IYOC by division? *	
DAL	8
DSMT	3
DAFME	3
DMME	1
DHA	1
DON	4
DPT	1
DSW	1

How many faculty have participated in this study by division? *	
DAL	6
DSMT	3
DAFME	1
DMME	0
DHA	0
DON	3
DPT	1
DSW	0

Division Key	
DAL	*Div. of Arts and Letters*
DSMT	*Div. of Science, Math, and Technology*
DAFME	*Div. of Accounting, Finance, MIS and Economics*
DMME	*Div. of Management, Marketing, and Entrepreneurship*
DHA	*Div. of Health Administration*
DON	*Div. of Nursing*
DPT	*Div. of Physical Therapy*
DSW	*Div. of Social Work*

**** = Some faculty teach in multiple colleges, divisions, and/or programs, so the individual college, division, and program totals for each might add up to more than the actual overall total.***

***** = Any College of Education faculty who participated in IYOC were purposely excluded from this pool of participants.***

APPENDIX C: Interview Script and Questions

Welcome to PARTICIPANT NAME. Today is DAY DATE and we are conducting an interview for Scott Thesen's capstone research project evaluating the transference of training to practice in online course design.

Some disclaimers before we continue:

- Participation in this study is voluntary. Whether or not you continue to participate is up to you. You can choose not to participate at any time during this process. Your decision will not be held against you or result in penalty.

- This interview is being recorded, and transcripts will be used as part of this study. Additionally, previous courses in Blackboard and materials from previous training may be accessed as part of this study. Your personal information will be kept private in the results, and recordings will be securely stored in a private, password-protected account.

- You can ask all the questions you want before making a decision. Feel free to ask any questions now.

AFTER ASKING/ANSWERING QUESTIONS – Do you wish to continue with this interview?

IF YES – Please introduce yourself. What is your teaching position, and what courses do you teach?

In MONTH YEAR, you participated in the official Quality Matters "Improving Your Online Course" professional development workshop. The culminating activity in that workshop was a Self Review of the course COURSE ID. Do you still teach that course? Please describe that course.

How do you show students that your course activities are aligned with learning objectives? (QM SRS 2.4)

How do your instructional materials prepare students for the activities they are asked to do? (QM SRS 4.2)

How do your learning activities provide opportunities for student interaction? How do your activities meet each of the three forms of student interaction (student-content, student-instructor, student-student)? Do any of your activities meet more than one, or all three? (QM SRS 5.2)

How do you personally interact with students in your online courses? Where can students find your policy and/or expectations for this? (QM SRS 5.3)

What specific technology tools within Blackboard or from any external sources are you using to provide all three forms of student interaction? Please provide examples of each. (QM SRS 6.2)

Have you applied any of these IYOC concepts to your other online courses?

Have you applied any of these IYOC concepts to your in-person or hybrid courses?

Do you have any additional questions for me?

This concludes this interview. Thank you for your time.

APPENDIX D: Table D1 (Summary of Perceived Implementations of Training)

Table D1

Summary of perceived implementations of training by IYOC participants

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
COMM-1	COMM-3200: Strategic Writing (Undergraduate)	• Presents learning objectives in syllabus and orientation materials • Believes there is a number of active learning opportunities • Explains why activities are important and provides instructions how to meet learning objectives • Communicates via email • Provides feedback on graded assignments • Sends class communications through Announcements • Main tool used is discussion board	• Needs to present learning objectives within individual assignments • Improving assignment guidelines within discussion boards • Needs to clarify how long grading should take • Doesn't always respond within all discussion boards	• Has pursued and passed an official QM course review • Uses discussions for both student interactions and organizing content instead of learning modules • Content includes modeling examples for assignment submission • Interactions include students posting drafts and students and instructor providing feedback and potential redirection • Shows alignment to learning objectives in assignment descriptions, the syllabus, and in the Blackboard orientation area • Posts lecture videos to YouTube and also links to relevant video supplements to text	• Addressing accessibility to make sure proper heading and paragraph designations are used in text descriptions	• Online • In-person
CPSC-1	CPSC-4580: Information Security (Undergraduate)	• Presents learning objectives in unit overview • Actively moderates discussions • Uses announcements to summarize or stress main focus of unit's discussions • Each week has an instruction document • Uses grading comments in first week to guide student expectations • Uses text and links to supplemental articles for content • Students participate in online lab activities	• Needs to present learning objectives and purpose within individual assignments • Needs to set student policies and clarify interactions better • Adding alignment to learning objectives within instructions	• Alignment is shown in Cisco text, not in Blackboard, so instructor will direct students to text if there are alignment questions • Student interactions include use of weekly discussions, but participation is required throughout the week and not just on weekends to ensure active participation • Instructor responses in discussions are used for guidance and feedback based on content • Instructor-created videos on YouTube supplement the content • Virtualization software is recommended for use on students' personal workstations to execute programming content • Netlabs software is used for ethical hacking exercises • Cisco provides another virtual learning environment aligned to text for lab exercises • Assignment submissions and grading feedback are on Blackboard	• Adapting to changes in technology, especially when relevant to subject area	• Online • In-person • Hybrid

Table D1 – *continued*

Faculty ID	Course ID and Title	• Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
CPSC-2	IT-4537: Advanced Linux Administration (Undergraduate)	• Online labs are used for content delivery, and scaffolded to encourage mastery before moving on to advanced material • Uses bi-weekly discussions for student interactions including instructor moderation • Interactions for the semester are explained in welcome announcement	• Needs to expand weekly agenda showing how material covered in the chapters and labs relate to learning objectives	• Learning objectives are in the syllabus and introduced to students in first week • Specific learning objectives are introduced each week based on content • Synchronous sessions were only used during COVID, so all courses are now either asynchronous or hybrid meeting in person every other week • Records videos of PowerPoints with assignment demonstrations from student point of view • Assignments are Linux command-line programming to be completed on students' personal workstations • Instructor interacts with students daily within the weekly discussion assignments • NetAcad and Cisco provide online lab supplements for content interaction • Cisco online lab includes an online text • VMWare is used for workstation simulation • Quizzes are taken online but are proctored virtually using Respondus • Students submit screenshots of assignment completion in online labs to Blackboard assignments for grading and feedback (screenshots include student name and personal approaches to programming problems to eliminate use of AI) • Sends announcements weekly • Responds to emails within 24 hours	• Intends to take Panopto training for video hosting instead of YouTube.	• Online • In-person • Hybrid

Table D1 – *continued*

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
CPSC-3	CPSC-3310: Introduction to Object-oriented Programming (Undergraduate)	• Topics are presented in learning modules • Each learning module is clearly marked with learning objectives it is specifically aligned to • Lecture instructions also show alignment to learning objectives • VoiceThread videos and discussion board provide opportunities for student interaction	• None stated	• Uses modules to organize content starting with 3-4 learning objectives for that module aligned to the overall course objectives • Intentionally designs structure to map all learning objectives to overall course objectives • Uses asynchronous videos for lectures and demonstration of assignments • Chunks videos into 10-15 minute segments to hold student attention • Quizzes are discussion questions posted by instructor based on video lectures, and students respond to questions but do not have to interact with each other (unless they want to) • Optional virtual office hours are available by appointment • Scheduled virtual sessions are used for group conferences when students can ask questions about previous or upcoming assessments like Midterm or Final • Respondus is used for remote proctoring of exams • Uses YouTube instead of Collaborate for captioning recordings, although they need to be edited due to technical terms and foreign accent • Uses IDLE scripting tool for that comes with Python coding download for coding assignments	• None stated	• Online • In-person • Hybrid

Table D1 – *continued*

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
ENGL-1	ENGL-4550: Shakespeare's Plays (Undergraduate)	• General relationship between learning objectives and materials/activities is described during synchronous meetings • Student interactions are required in an active response section within lectures and in discussions with analytical questions and required responses to other students	• Needs to be more systematic in approach to showing alignment to learning objectives in item descriptions, and possibly with a curriculum map • Intends to include more regular, active student interactions • Wants to better facilitate group work during synchronous sessions	• Syllabus is present in an orientation area for reference • Welcome video is sent at beginning of course explaining resources, policies for communication, and learning objectives • Learning objectives are present in the syllabus, the orientation area, listed at start of weekly modules, and explained within assignment prompts • Alignment to learning objectives has helped with organization of course materials • Teaches both in-person and online versions of classes, but has purposefully (and unofficially) converted in-person sections to hybrid, including creating virtual meeting options and moving his class to hyflex classrooms to accommodate students who cannot attend scheduled in-person sessions • Uses Collaborate Ultra for virtual sessions • Uses both Blackboard discussion tool and FlipGrid for online discussions • Class-wide and small groups discussions are student-led based on writing assignments including drafts for peer feedback	• Will reluctantly convert to Teams for virtual sessions since this is the new tool at the institution, and Collaborate Ultra is being deprecated	• Online • In-person • Hybrid

Table D1 – *continued*

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
ENGL-2	ENGL-3330: Studies in Literature (Undergraduate)	• Learning objectives are listed in each module • Announcements are posted and emailed detailing the expectations and work for the week • Discussions and interaction with learning materials are required throughout the week – participation cannot be pushed back to the weekend • Virtual office hours are held in Collaborate, and individual appointments can be scheduled in Webex via email (recorded upon request) • Essay assignments have a student peer review component • Professional communication discourse is expected by students • Emails will be responded to within 24 hours • Grade feedback on discussions and essays will be posted within 7 days of the due date • Uses Norton Digital Learning Assignments as a supplement to text chapters	• Needs to link learning outcomes to explanations of individual activities and materials • Needs to update communication and feedback policies • Needs to update descriptions of Norton assignments to show alignment to textbook chapter learning objectives	• Weekly modules are organized by learning objectives covered, with activities specifically aligned to activities with a logical progression through the module • Student interactions include discussion assignments that function as peer reviews of drafts for four essay assignments, and partners are assigned based on participation in discussions where they have ongoing dialogue throughout the semester • Instructor also interacts in discussions as well as weekly announcements, responding to emails, and recorded one-on-one conferences by appointment (but available weekly) in Webex (previously) or Teams (currently) for future student reference so they can focus on conversation instead of worring about taking notes • ClaudeAI is purposefully used to help organize writing per assignment rubrics and help students with research and eliminating bias • Instructor provides previous student examples of assignments to give students a guide for what is expected	• Continue to evaluate instructional approaches for what is working and what isn't, and then adapt accordingly • Wants to provide better descriptions of how activities meet learning objectives • Believes students became confused by having an orientation area that included the syllabus and specific policy items, so is considering revising or even removing the orientation area	• Online • In-person
HIST-1	FYS-1001: First Year Seminar (Undergraduate)	• Students are expected to regularly participate in Blackboard discussions in preparation for future writing assignments (30% of grade is participation) • Provides regular feedback in grading • All assignments directly reference learning outcomes • Uses Blackboard discussions, VoiceThread, and Films on Demand for content delivery and student interactions	• Needs to update assignment and activity instructions with a better explicit connection to learning objectives • Needs to have a better defined plan for feedback and communication in syllabus and orientation materials • Wants to vary use of technology tools to promote student engagement, possibly with increased use of VoiceThread for recording lectures and encouraging active learning for collaborative analysis of texts	• Orientation is where students are introduced to learning objectives • Specific assignments are aligned to learning objectives • Uses a variety of online resources to present content to students • Relies heavily on discussions for student interaction • Uses VoiceThread for multimedia discussions	• Possible conversion of current in-person sections to online • Improve instructor presence in online sections, including posting weekly videos, and surveys about progress of semester-long projects • Wants to integrate a media campaign into focused, semester-long projects • Introduce debates into online sections, but unsure how to approach this, possibly with videos • Continuing to improve accessibility for all students, including the visual and hearing accommodations seen most often	• Online • In-person

Table D1 – *continued*

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
IDSS-1	IDSS-3100: Interdisciplinary Theories (Undergraduate)	• Relationship between learning objectives, materials and activities is explained in written instructions, regular announcements, and instructor videos • Students can use the Course Messaging tool in Blackboard for communication with instructor • Active learning is primarily through student interactions with content via recorded lectures or online supplements, discussions with other students (including peer feedback on assignments) or instructor, or with instructor for grading feedback • Weekly announcements on Sundays outline expectations for the coming week • Module objectives are aligned to course objectives • Progress through course is on student's timetable as long as deadlines are being met as shown in assignment descriptions • A welcome video outlines the interaction policies from the syllabus • Communication with instructor for answering questions is encouraged • Collaborate Ultra is used for virtual session and student-created presentations	• Needs to be more intentional about alignment of learning objectives to materials and activities beyond just the syllabus	• In asynchronous online courses, encourages students to attend in-person sections to get live interaction • Student introductions are required to use a video tool • Collaborate Ultra is used for four full-class synchronous sessions and also group work on their own schedule • Goes over learning objectives in opening presentation for each module • Addresses accessibility by allowing students to choose between written or video submissions for assignments • Uses group projects in all online and in-person courses	• None stated	• Online • In-person • Hybrid

Table D1 – *continued*

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
NURS-1	NURS-8107: Gerontology Acute Care Alterations (Graduate)	• Alignment between learning objectives and assigned learning activities is clearly explained during the first class session and the course schedule outlining the weekly assignments, including the competency check out sheet • Assignments include readings, a PowerPoint, and an online quiz • Course practicum competency is administered by a site preceptor and review by a validation tool by the preceptor and student at the clinical site outside of online course • Utilizes an learning agreement assignment so students acknowledge semester expectations and policies for interactions • Group presentations, peer critiques, and discussions for sharing networking opportunities and clinical experiences are used for student interactions • Instructor provides regular announcement reminders for weekly materials and meetings, feedback on assignments, encourages email for questions, hosts synchronous sessions, and virtual office hours by appointment • Instructor only responds to discussion after all students have responded to at least two other student initial posts first • Collaborate Ultra is used for conducting synchronous sessions and virtual office hours	• Set up weekly introduction messages via Collaborate Ultra • Wants to create a curriculum map to share with students to better show alignment between learning objectives, materials and activities • Needs to update the course orientation area	• Students have to sign a learning agreement in the orientation area as a graded assignment, acknowledging that they understand all course policies, learning objectives, and expectations • Students are also required to take the Blackboard Student Orientation Course through CTL as a graded assignment • Assignments have rubrics that show alignment to learning objectives • Collaborate Ultra rooms are set up for individual student groups every semester to have virtual sessions throughout to work on group projects • Group projects are partially graded by peer review of participation • Provides specific feedback when grading discussions on how to improve substance of posts, and then general feedback about trends in weekly announcements • Weekly virtual synchronous sessions are held via Webex (formerly) or Collaborate Ultra (currently) and recorded	• Continue to make improvements based on feedback from department evaluation committee every semester • Continue to make accommodations for meeting with students due to non-traditional work hours in the profession • Wants to take training on upcoming changes in Blackboard due to the interface upgrade • Needs to learn Teams and Panopto because Collaborate Ultra is being deprecated • Needs to learn Blackboard Ultra due to the campus-wide LMS upgrade • May be interested in learning Google Forms or Microsoft Forms (within Teams) for interactive feedback during synchronous sessions	• Online • In-person • Hybrid
NURS-2	NURS-7109: Informatics for Nurse Executives (Graduate)	• Learning contract assignment summarizes interaction and assignment submission expectations and policies • Discussions are used for content delivery and student interactions with each other and the instructor • Discussion rubrics are shared with students so they are aware of grading requirements • Discussion directions provide submission and response deadlines • Official institution email is noted as required method of communication for students with instructor, and instructor commits to responding within 24 hours M-F or 48 hours over a weekend • Instructor commits to having an active presence in discussions • Students will participate in group projects	• None stated	• Specifically teaches technologies used in the profession • Interactions suggest logging in at least three times per week for two hours at a time • Requires synchronous sessions to make discussions more "lively" like in-person classes • Purposefully includes technical support in orientation resources • Uses discussions for students to post professional resources to share with each other • Applies alignment to learning objectives in all modalities taught • Students submit reflection statement at end of semester stating how they believe they met the learning objectives • Uses Blackboard tools for journals and content delivery in in-person courses	• Addressing accessibility requirements in all courses, including learning new Ally plugin for Blackboard coming out soon on campus to help evaluate course compliance	• Online • In-person • Hybrid

Table D1 – *continued*

Faculty ID	Course ID and Title	Perceived Implementations before Training	Intended Improvements after Training	Actual Implementations after Training	Possible Future Improvements	Modalities Implemented after Training
PHYT-1	PHYT-6743: Research in Physical Therapy Practice (Graduate)	• Chapter summaries are provided at the start of each module, which show alignment of learning objectives to readings and assignments • Individual activities also include alignment information in the descriptions • Tools for interactions used discussions, Google docs for collaborative projects, online research databases, Endnote for citations, and Collaborate Ultra for video conferencing • Communication through email will be responded to within 24 hours • Grading feedback will be posted within 7 days of due date	• Add clearer explanation for links between learning objectives and activities • State specific learning materials to be used/referenced in specific assignments	• Explains how activities will meet learning objectives, and alignment has helped with departmental accreditation • Makes assessments specifically tied to learning objectives • Many assignments are discussions that include participation expectations in the instructions • Has written their own virtual interactive textbook • Utilizes online interactive MRI and CT imaging tools as used in the profession • Made an introductory video • Records synchronous sessions for future student review • Has posted communication policies • Uses polling in quiz question review sessions to determine why majority of students may have gotten a specific question wrong • Has one open synchronous session per semester for asking questions	• Continue to focus on alignment each semester as materials are created or updated • Wants to improve application of training concepts to in-person sections, but this is difficult when many courses are co-taught and other instructors have different ideas and approaches	• Online • In-person
SOC-1	SOC-1100: Introduction to Sociology (Undergraduate)	• Modules contain a weekly content description of topics and key concepts with a list of learning activities • Email is best way for communication, and students can expect a response within 24 hours • Students can schedule virtual office hours using Collaborate Ultra by appointment • Instructor contact info and availability are listed in Blackboard Instructor Info tab • Weekly announcements outlining content and other pertinent information will be emailed • Student interactions include text readings and videos, discussions for reading and commenting on other student posts, and grading feedback on papers and discussions • Test answers are posted with explanations 2 weeks after the due date	• Needs to add bullet points to weekly descriptions showing alignment to learning objectives, and list learning materials needed to complete the activities each week • Arrows could be used in instructional materials to show alignment to learning objectives • Wants to have more collaborative discussion assignments beyond the current post/comment dynamic	• Learning objectives are listed in syllabus, orientation area of Blackboard shells, and in weekly units • Alignment to learning objectives is present in description of assignments • Students can download PowerPoint slides with audio narrations attached, and PDF versions are also available • Videos of lectures and assignment demos or YouTube supplements to text content are included • Utilizes discussions for student interactions • Provides feedback on assignments • Announcements are posted weekly	• Needs to convert PowerPoints with audio narrations to videos using Panopto	• Online • In-person

Subject-area Key		Response Key	
COMM	Communications	**Perceived Implementations before Training:**	Participants stated during training that they believed they were already implementing these training tropics.
CPSC	Computer Science or Information Technology	**Intended Improvements after Training:**	Participants stated during training in the Self Review activity that they intended to implement these training topics.
ENGL	English and Literature	**Actual Implementations after Training:**	Training topics implemented after training as stated in interviews and verified within Blackboard shells.
HIST	History	**Possible Future Improvements:**	Participants stated during interviews that they need or want to implement these training topics in the future.
IDSS	Interdisciplinary Studies	**Modalities Implemented after Training:**	Course modalities training topics were implemented in after training, as stated in interviews and verified within Blackboard shells.
NURS	Nursing		
PHYT	Physical Therapy		
SOC	Sociology		

APPENDIX E: Institutional Review Board

Governors State UNIVERSITY

Institutional Review Board
Room G353
1 University Parkway
University Park, IL 60484
www.govst.edu/irb

Aug 7, 2025 11:55:50 AM CDT

Deborah King, Scott Thesen
Education

Re: Exempt - Initial - IRB-FY2026-3 Perceptions of Implementation from Higher Education Participants in Online Course Design Professional Development

Dear Dr. Deborah King and Scott Thesen:

The Governors State University Institutional Review Board has granted exempt approval for your project titled "Perceptions of Implementation from Higher Education Participants in Online Course Design Professional Development." You may begin your research.

- All research related to this project **must** be conducted **exactly** as stated in the approved research protocol.
- If you would like to make any changes to research personnel or to the way that this research is conducted, you must complete a **Modification submission** and **wait for IRB approval** to enact any changes.
- Please be advised that you may **only** distribute the **GSU IRB approved** text, forms, documents, and materials to all participants. Your approved consent form and recruitment materials can be found under the "Attachments" tab on the "Study Details" page.
- After you have completed the project, please complete a **Project Closure submission**.
- Please include the assigned IRB project number, PI name, and exact title of your project in any correspondence about this project.

Decision: Exempt

Selected Category: Exemption Category Category 2.(ii). Research that only includes interactions involving educational tests (cognitive, diagnostic, aptitude, achievement), survey procedures, interview procedures, or observation of public behavior (including visual or auditory recording) if at least one of the following criteria is met:
Any disclosure of the human subjects' responses outside the research would not reasonably place the subjects at risk of criminal or civil liability or be damaging to the subjects' financial standing, employability, educational advancement, or reputation; or

Sincerely,
Governors State University Institutional Review Board
irb@govst.edu

APPENDIX F: Recruitment Email

Capstone Recruitment Email Draft • sthesen@govst.edu

Send | Discard | Attach File | Signature | Polls | Viva Insights | ··· | Copilot

From: Thesen, Scott (sthesen@govst.edu)

To:

Cc:

Bcc:

Subject: Capstone Recruitment Email Draft

Importance

Aptos 10

Hello _____.

In MONTH 202#, you completed the official QM Improving Your Online Course (IYOC) workshop with me as your facilitator. I am now working on the capstone project for my EdD in the College of Education and Human Development Interdisciplinary Leadership program, analyzing transference from training to practice. Now that some time has passed for you to implement what was learned during the workshop, I would like to know if you are interested in participating in my study.

Participation would require about an hour of your time during a recorded interview. Interviews will be conducted in August and September, and we can meet based on your availability. Your information will be kept confidential, and recordings will be set to "private" and stored securely in my password-protected GSU Panopto account. If you are willing to participate, I will reach out with an Informed Consent form for you to review and sign.

I hope to hear back from you soon. Thanks in advance for your consideration.

scott

Scott B. Thesen, M.A.Ed. & M.A.T.
Senior Online Instruction Specialist / Adjunct Faculty
Quality Matters™ Coordinator, Reviewer & Facilitator
708.534.4106 Office • sthesen@govst.edu

Draft saved just now